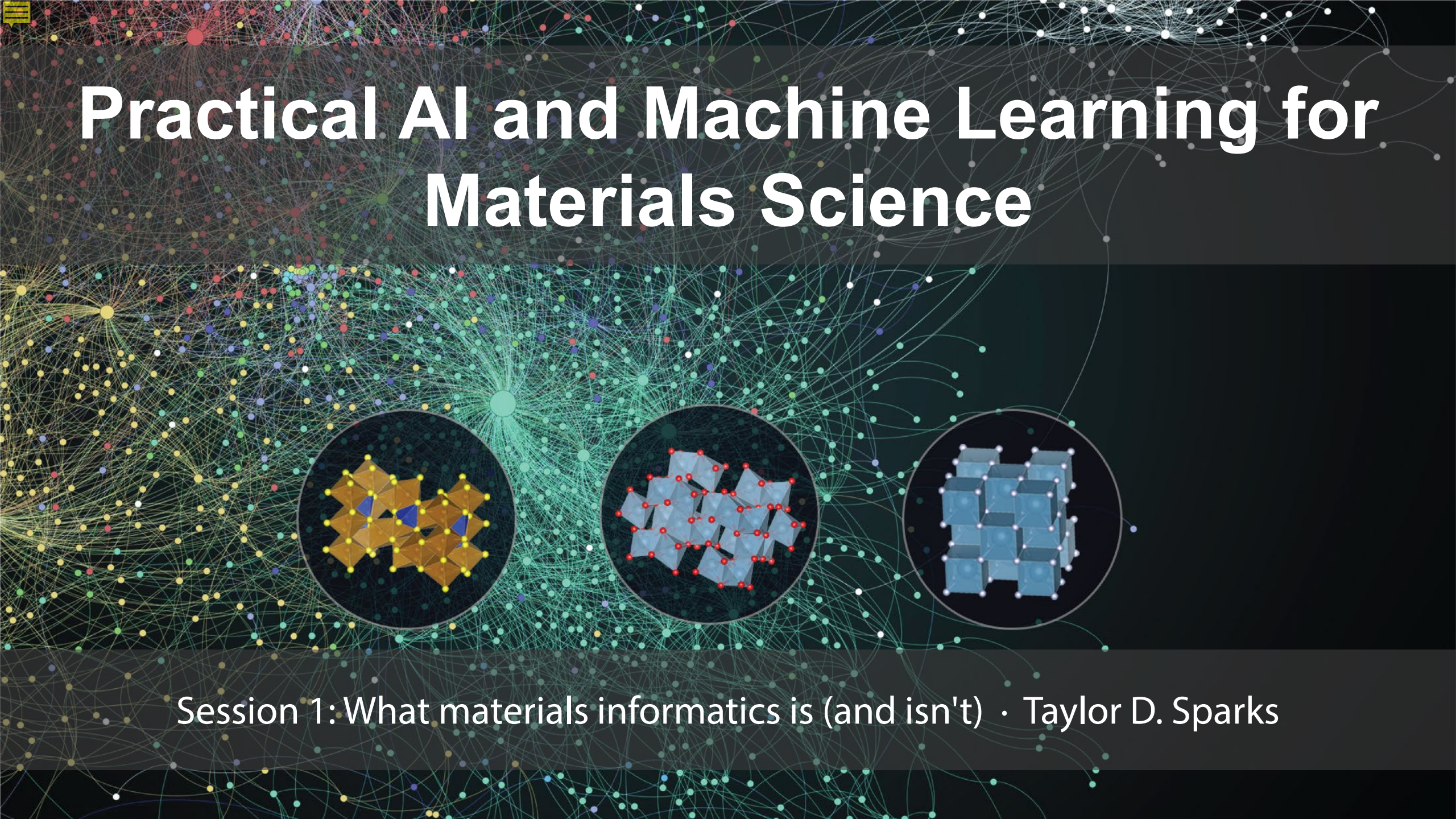
Practical AI and Machine Learning for Materials Science



Session 1: What materials informatics is (and isn't) · Taylor D. Sparks

Today we go from pattern finding to self-driving labs

Pattern finding
Structure-processing-property: the original data science

→

Four types of ML
Supervised, unsupervised, semi-supervised, reinforcement

→

ML for ceramics
Regression, classification, images, generation

→

Closing the loop
Automation, LLMs and agents

→

Hands-on
30 min with real datasets in Colab

90 minutes of lecture + 30 minutes exploring datasets and notebooks

Six sessions take you from data to decisions

1 · Tue 10/13
What materials informatics is (and isn't)

2 · Wed 10/14
Feature engineering and composition-based features

3 · Thu 10/15
Supervised learning, validation, metrics

4 · Tue 10/20
Interpretation, uncertainty and trust

5 · Wed 10/21
Bayesian optimization and active learning

6 · Thu 10/22
LLMs, agents, vibe coding and the capstone

github.com/sp8rks/ACerS_AI_ML_Workshop · every notebook opens in Google Colab

Materials discovery needs to be accelerated!

"The temperature gauge was stuck on 900 degrees,
and I thought I had ruined the furnace …

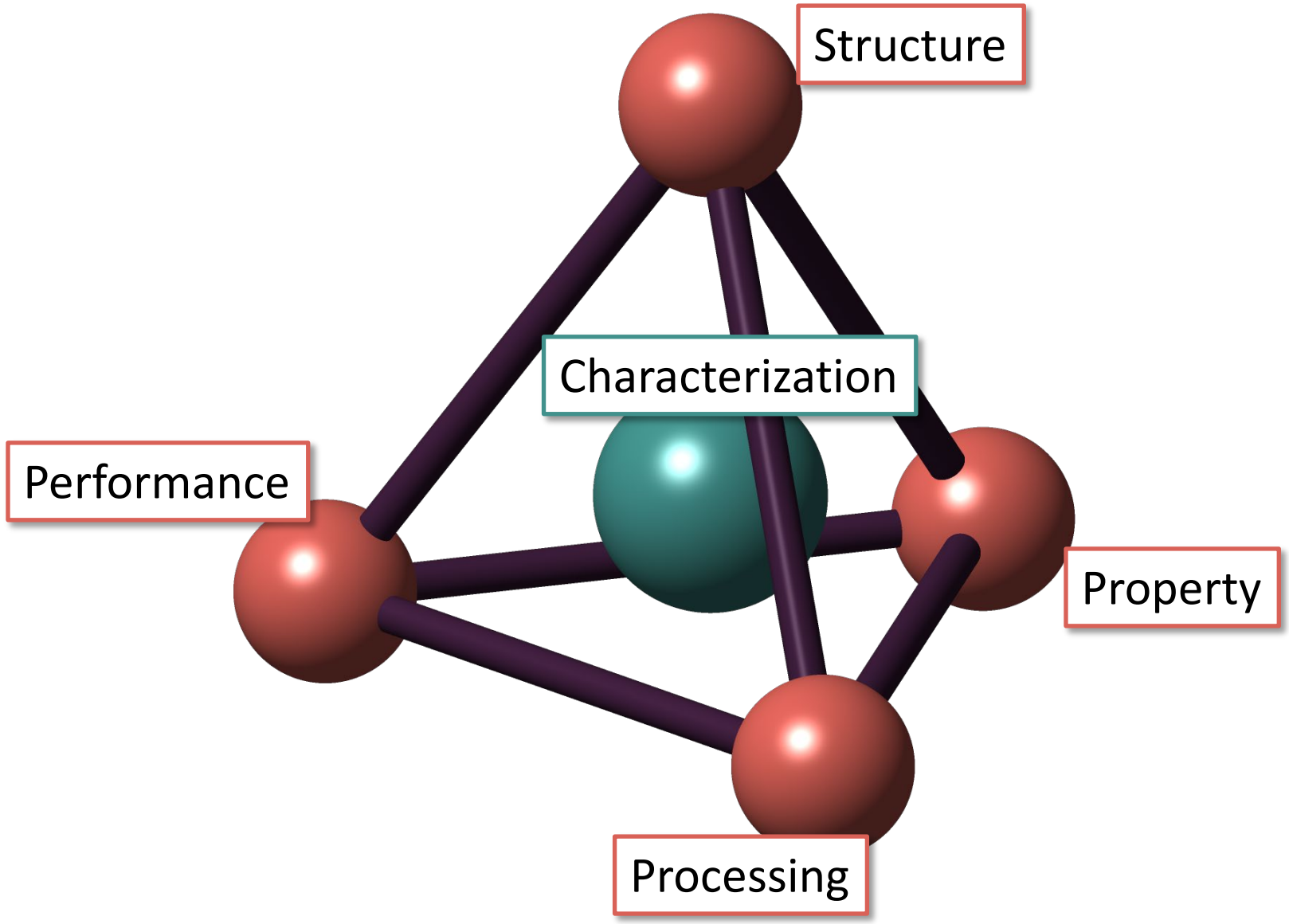
I grabbed some tongs to get it out as fast as I could,
but the glass slipped out of the tongs and fell to the floor.

The thing bounced and didn't break."

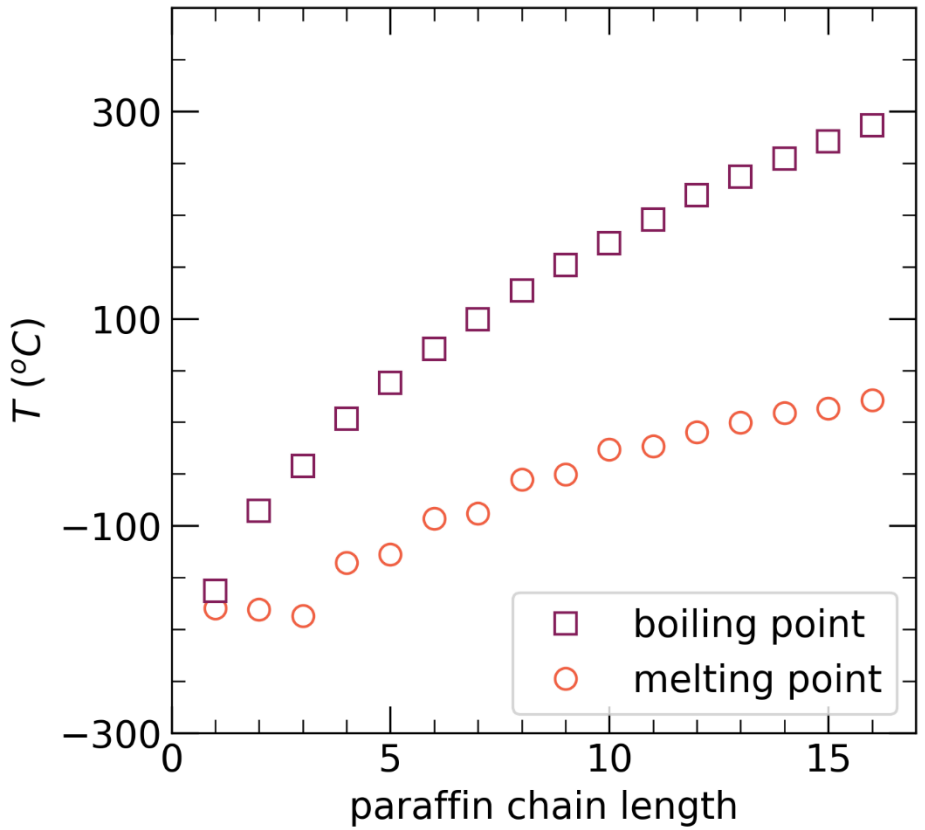
Donald Stookey (1915-2014)

AI for materials has kicked off a new materials gold rush!

Materials science is the study of structure-processing-property links

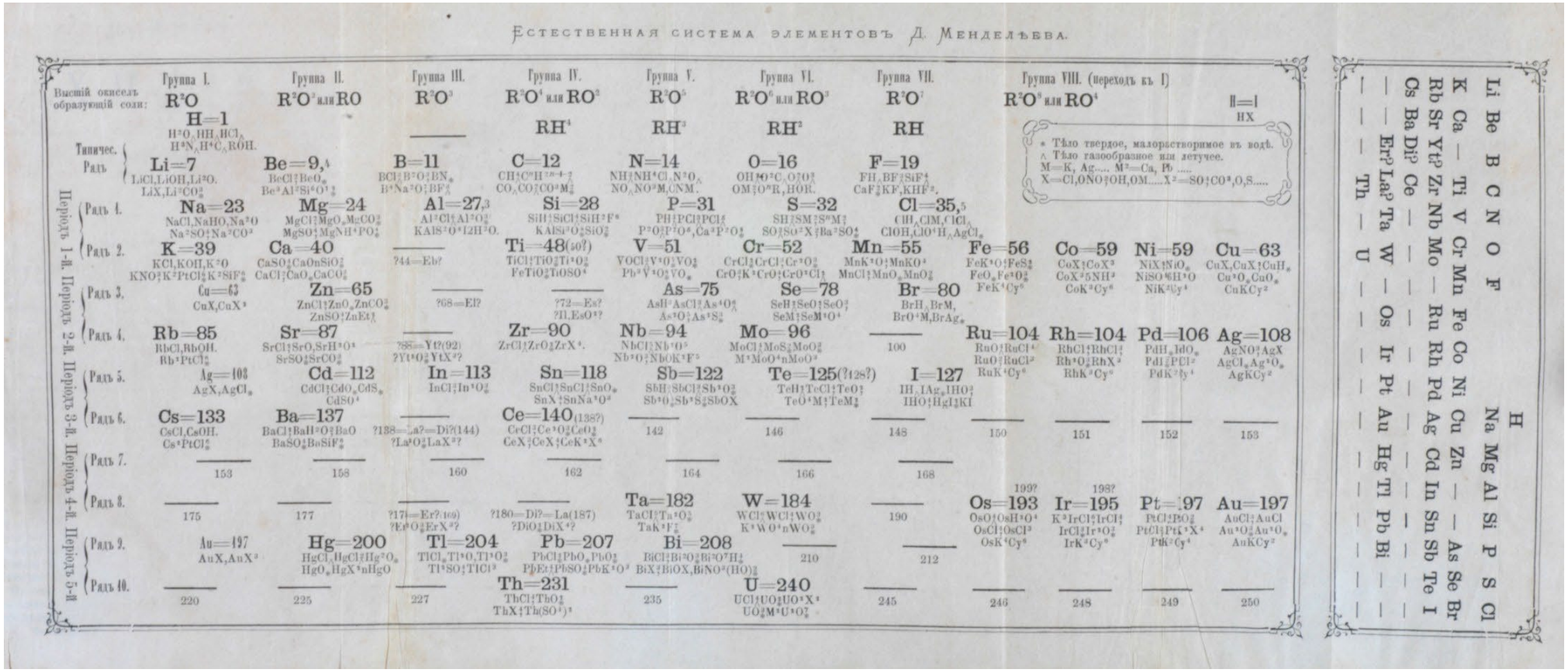


Materials scientists have noticed patterns in data forever!



Mendeleev found the periodic table by spotting patterns in data

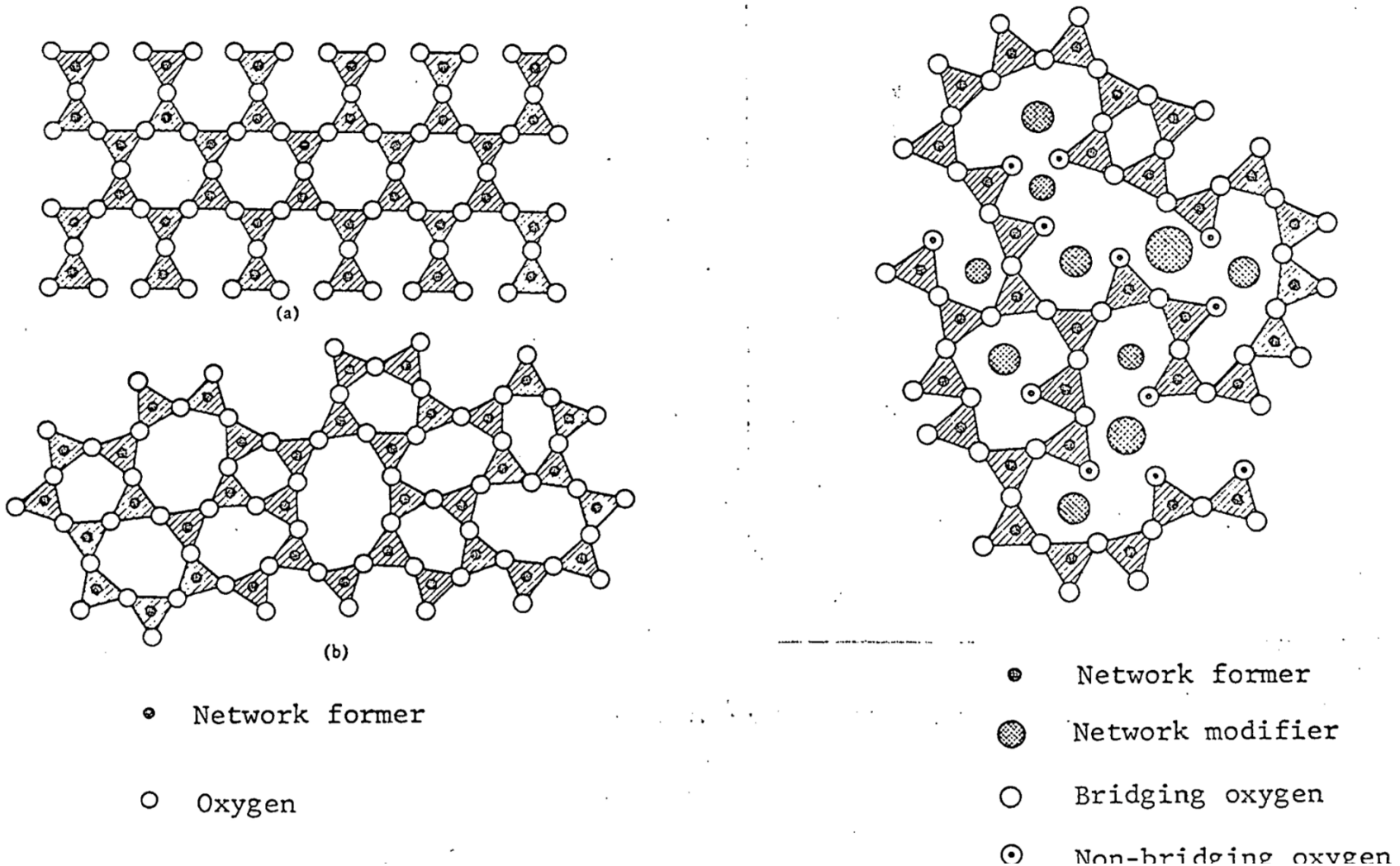
Each column is headed by its oxide formula: the first great materials pattern was an oxide pattern



D. I. Mendeleev, Osnovy Khimii, 1869-1871 (public domain; Science History Institute via Wikimedia Commons) https://commons.wikimedia.org/wiki/File:Dmitry_Mendeleyev_Osnovy_Khimii_1869-1871_first_periodic_table.jpg

Zachariasen's glass-forming rules came from patterns in structure

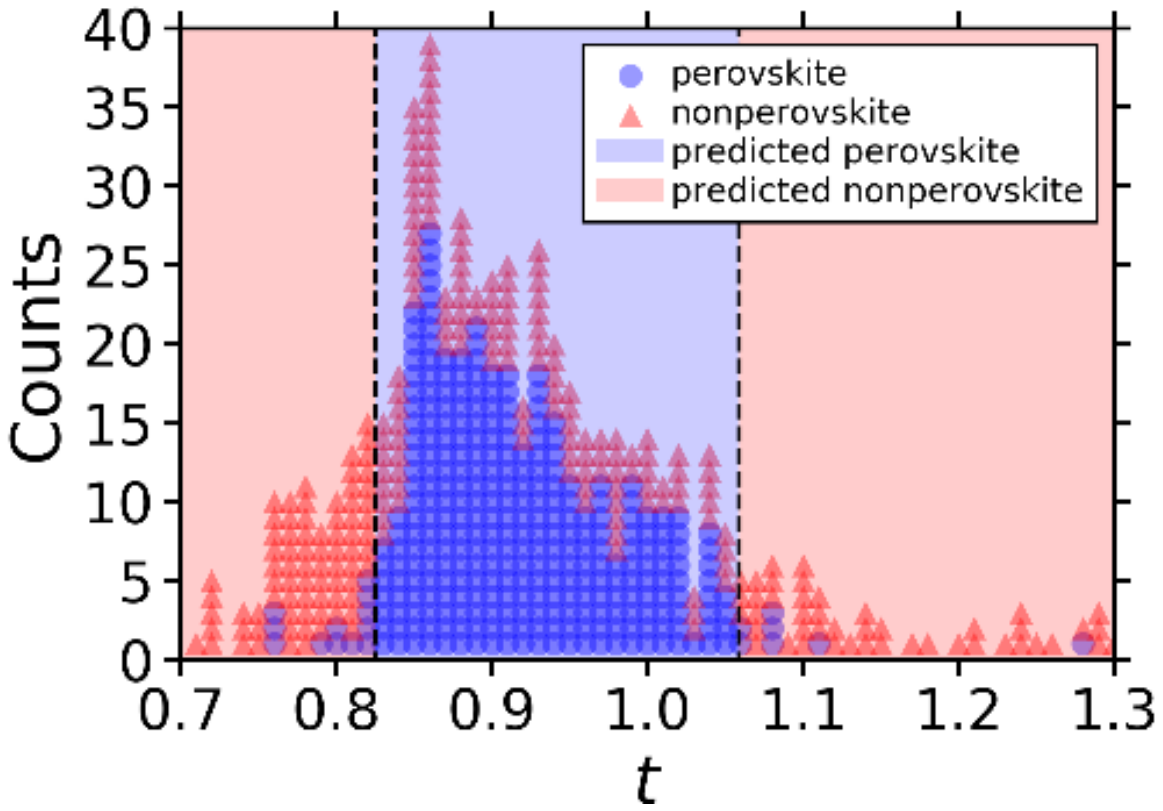
Network formers, modifiers and bridging oxygens are still the features glass ML models use

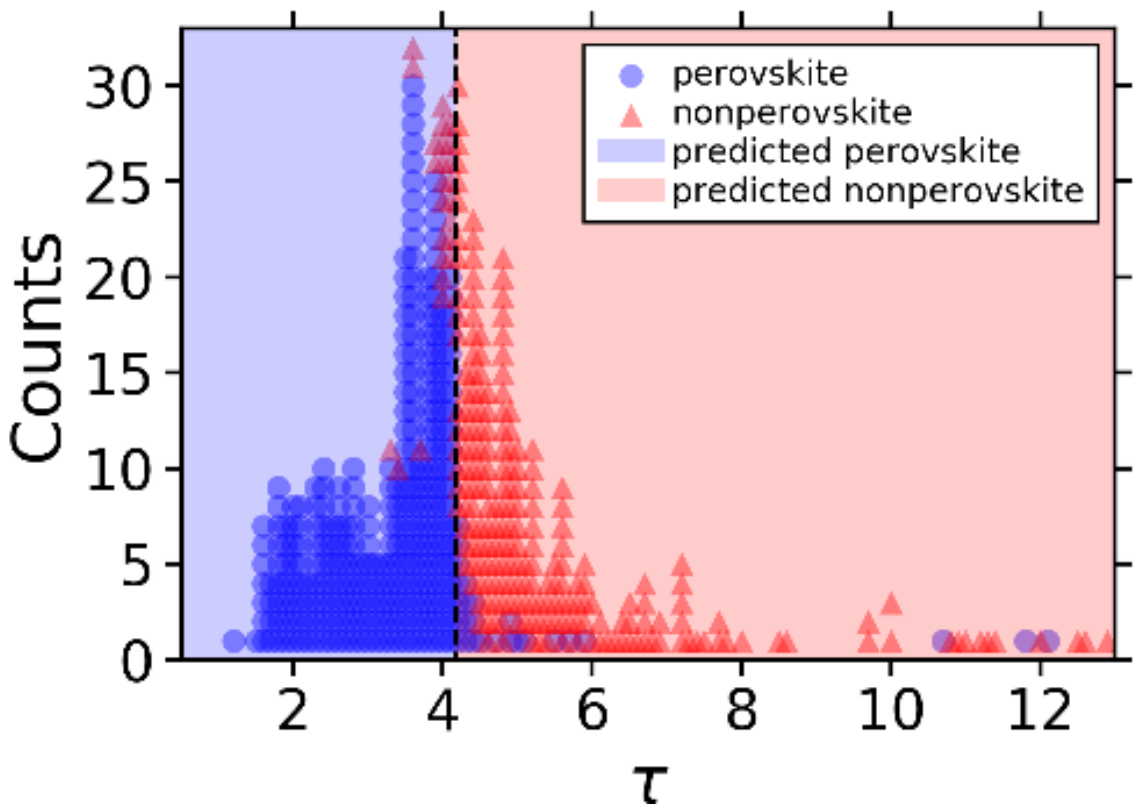


After Zachariasen, J. Am. Chem. Soc. 1932 and Warren, J. Am. Ceram. Soc. 1941 (drawing: Pohl, Love & Stephens, 1973) https://commons.wikimedia.org/wiki/File:Two-dimensional_schematic_diagram_of_quartz,_silica,_and_silica-based_glass.png (source: https://www.osti.gov/servlets/purl/4607018)

A data-mined tolerance factor beats Goldschmidt's 1926 rule of thumb

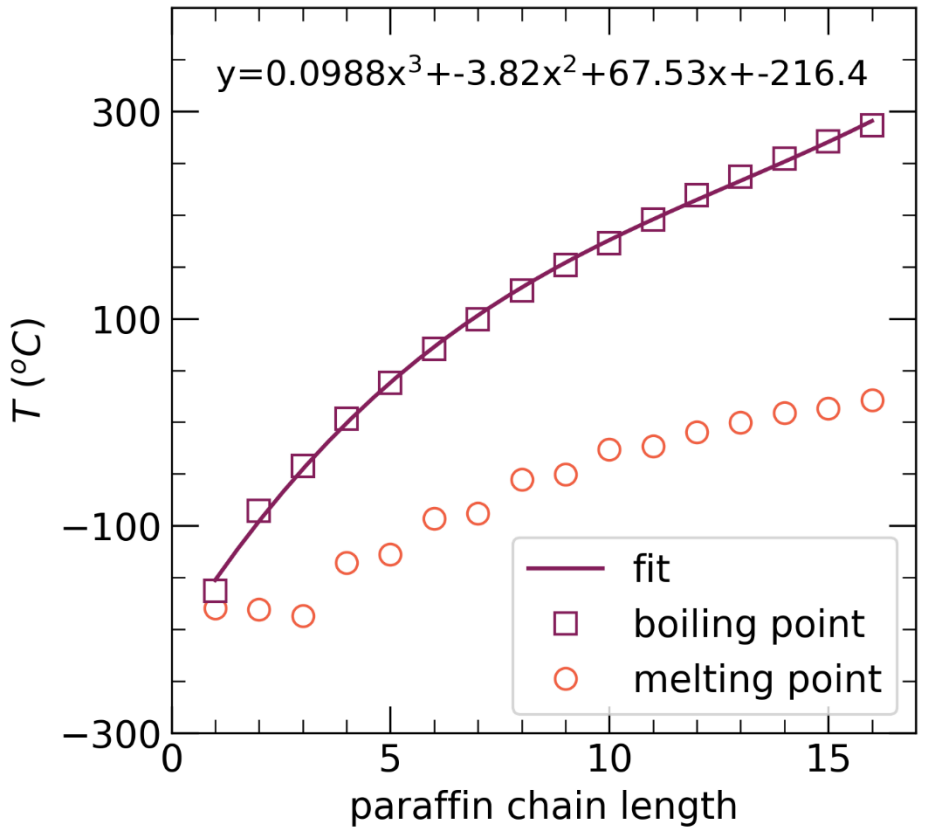
Accuracy rises from 74% (t) to 92% (tau) on 576 known ABX_3 compounds.





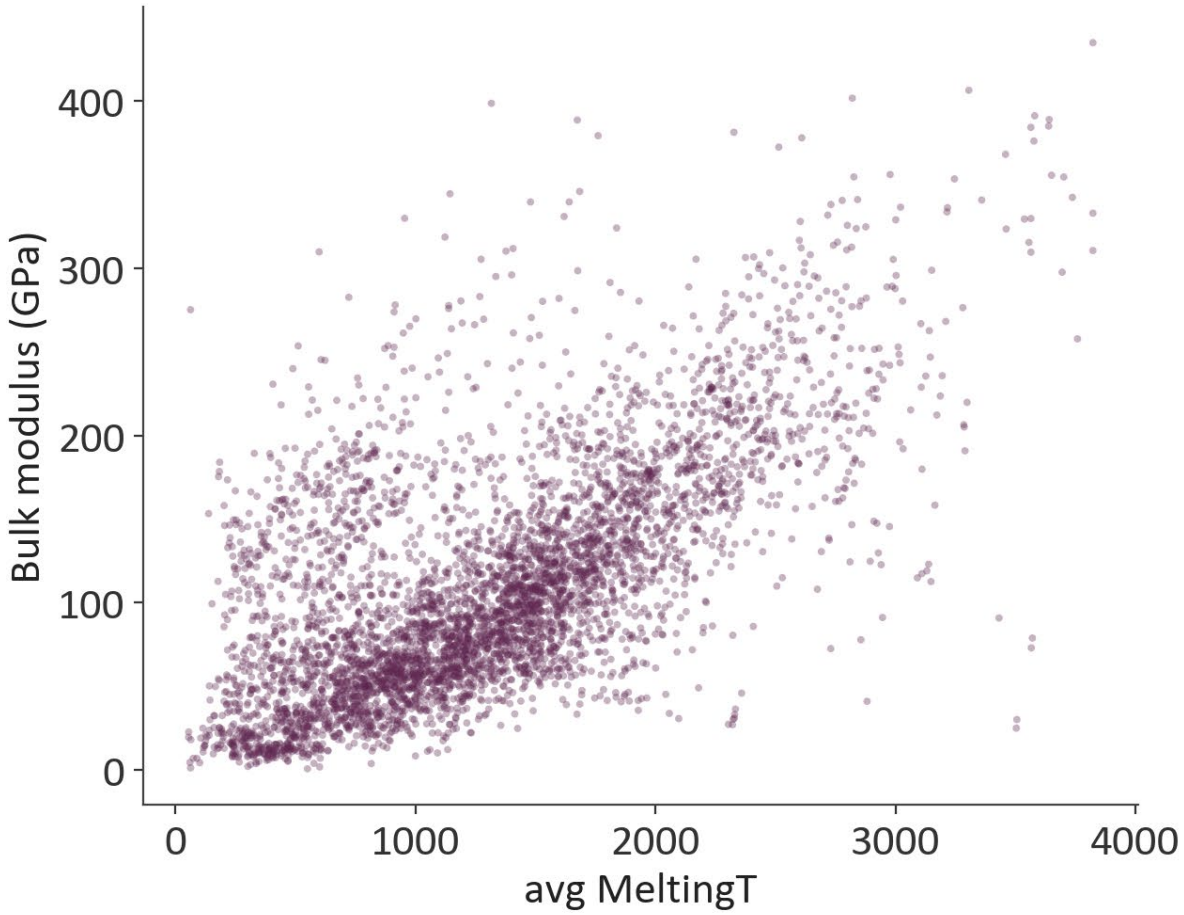
Bartel et al., Sci. Adv. 2019 https://doi.org/10.1126/sciadv.aav0693 (arXiv:1801.07700)

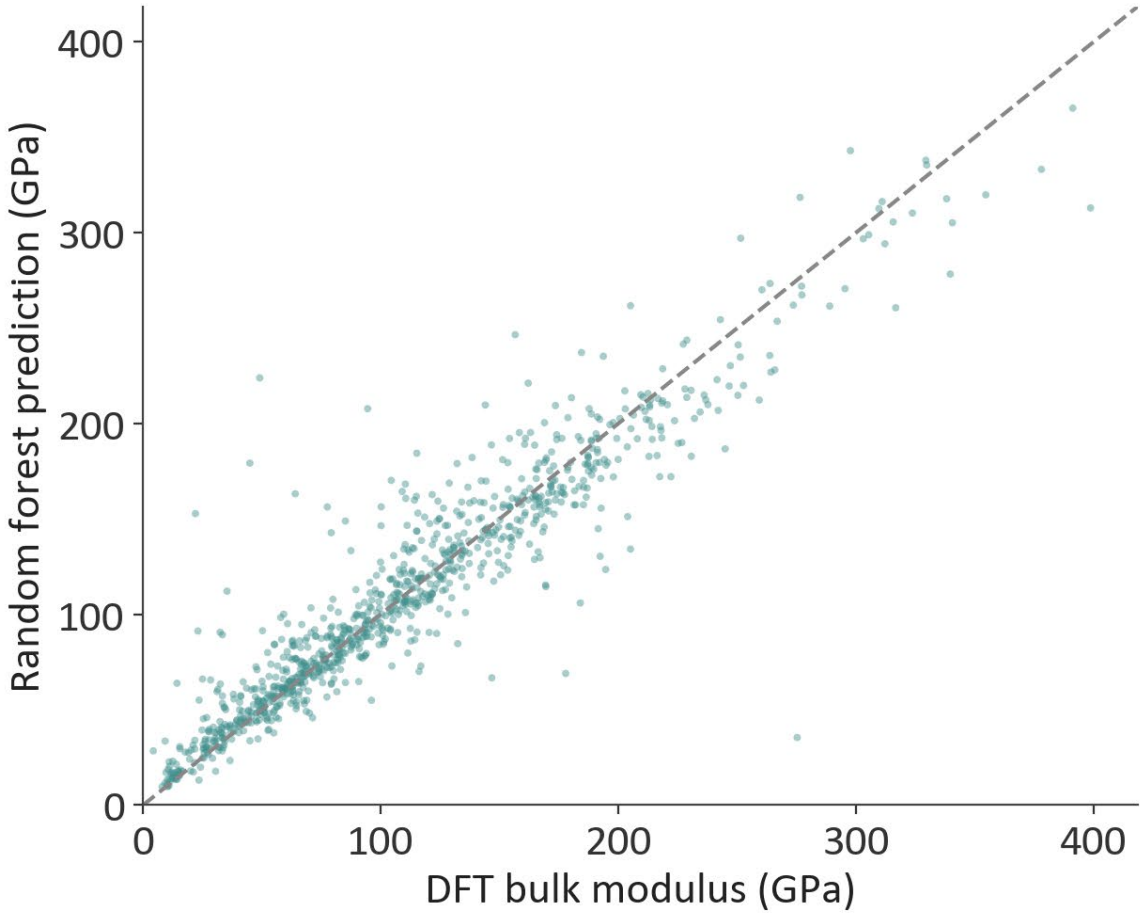
Empirical trends may not be supported by theory



Humans see one or two variables at a time; ML sees them all

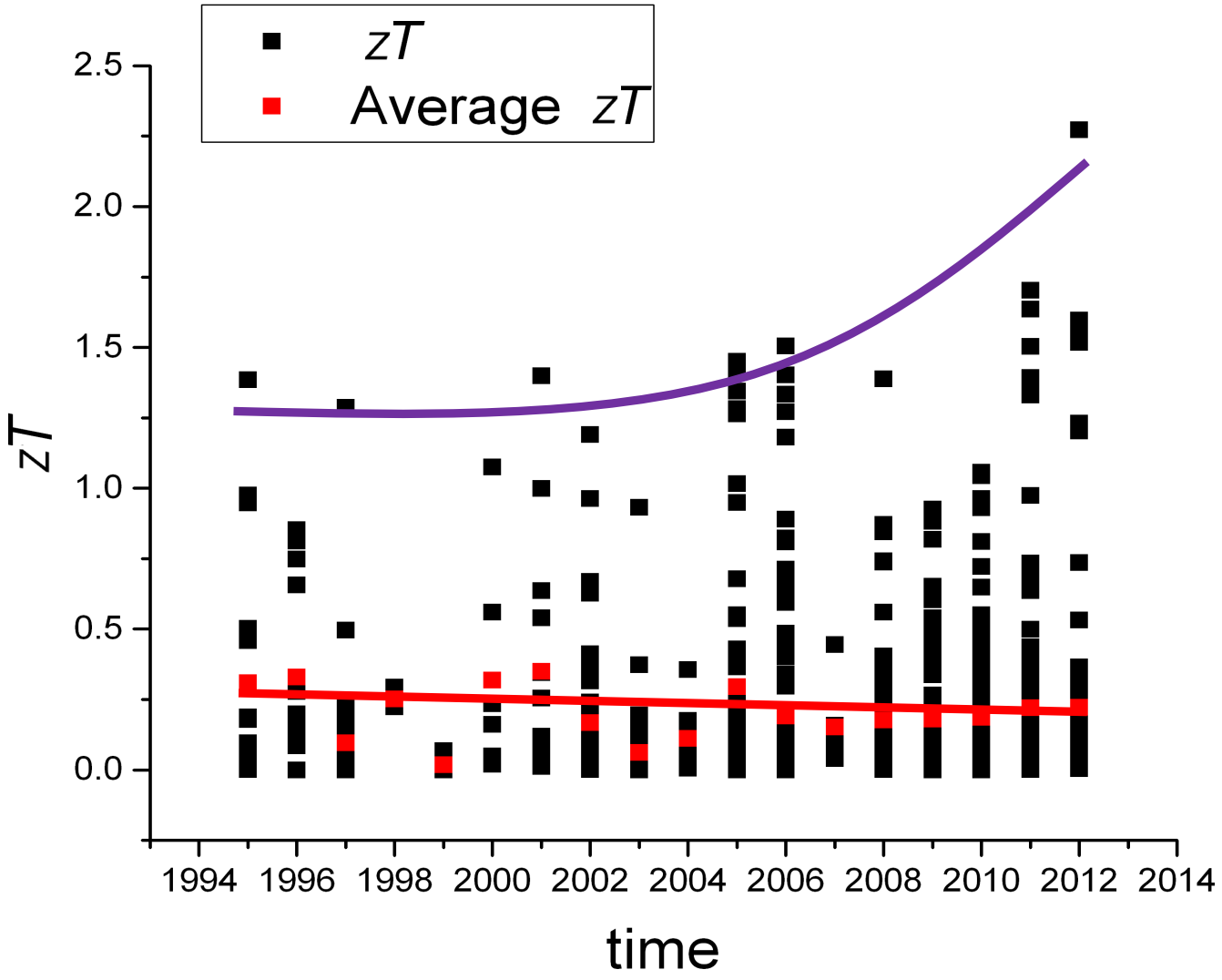
Bulk modulus of 4,905 compounds: melting point alone is a smear, 132 features together give the pattern





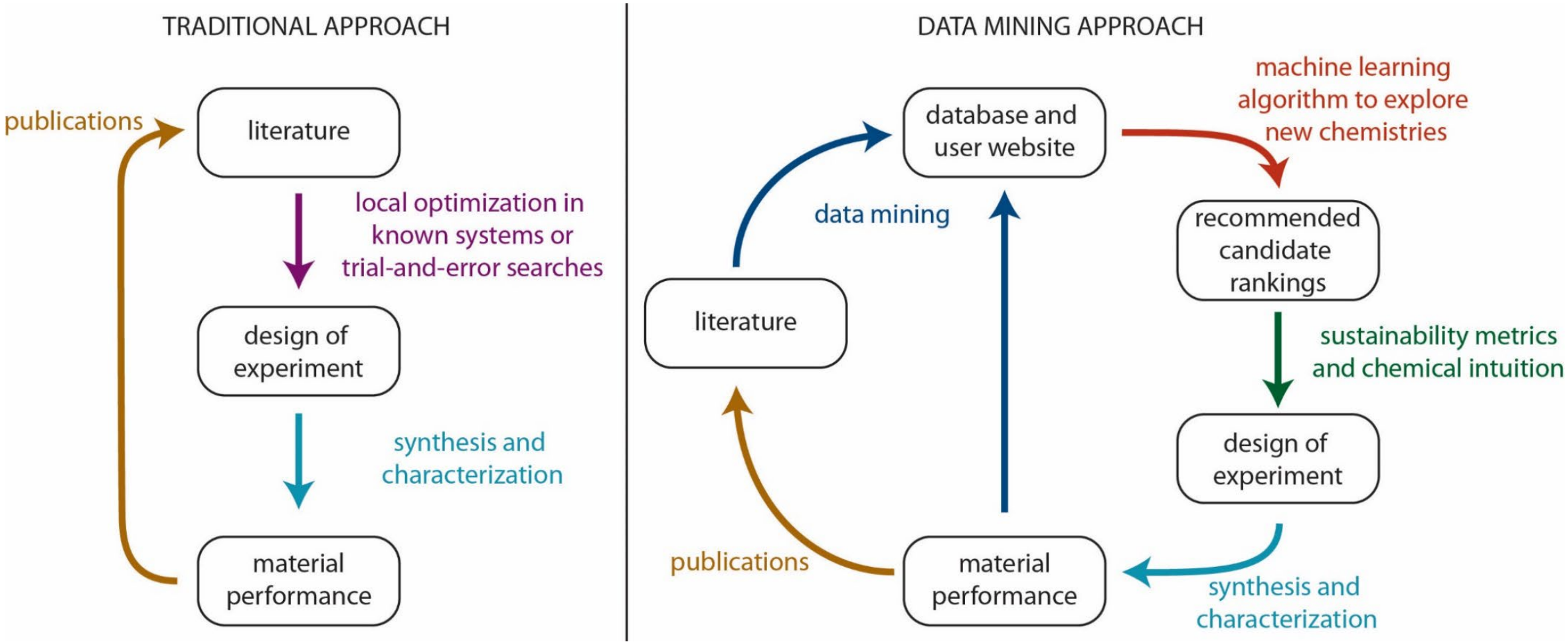
Course dataset: AFLOW AEL-AGL bulk moduli (Wang et al., Chem. Mater. 2020). Random forest, 80/20 random split, Magpie features

Most materials improvements are incremental



Breakthroughs are the uncommon exception

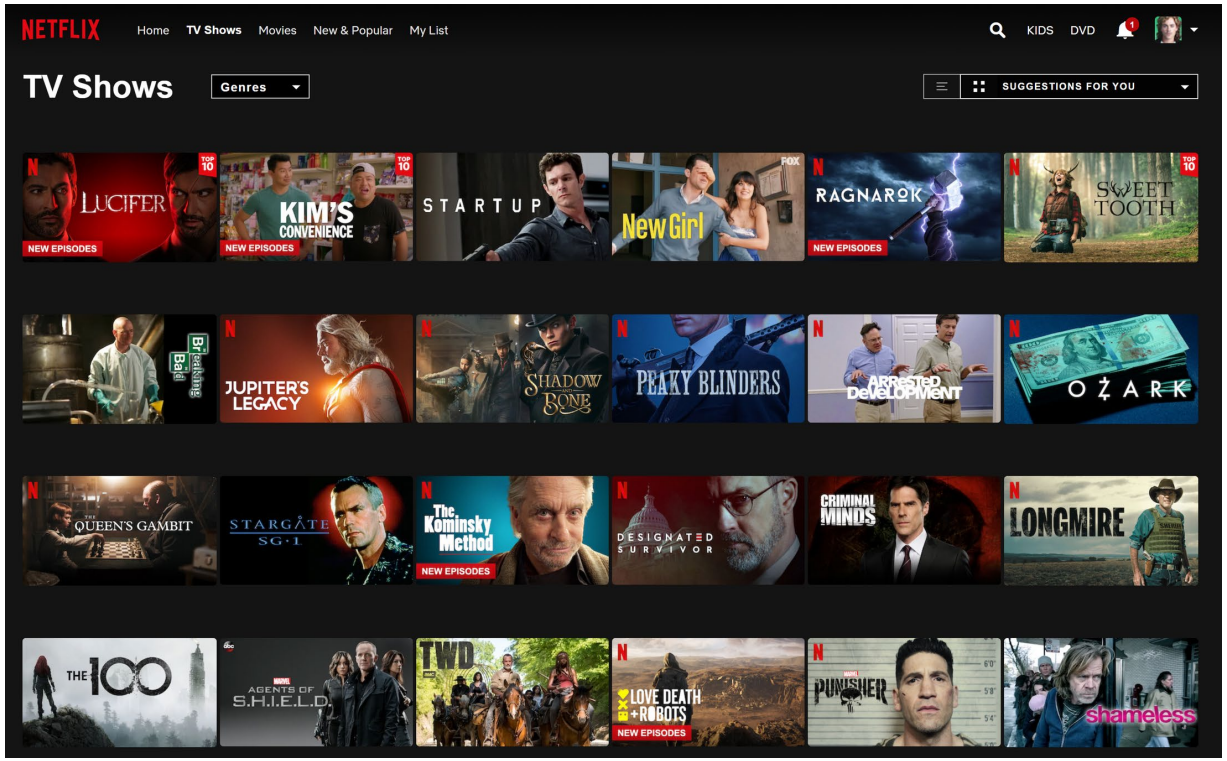
New tools of discovery are needed for “chemical whitespace”

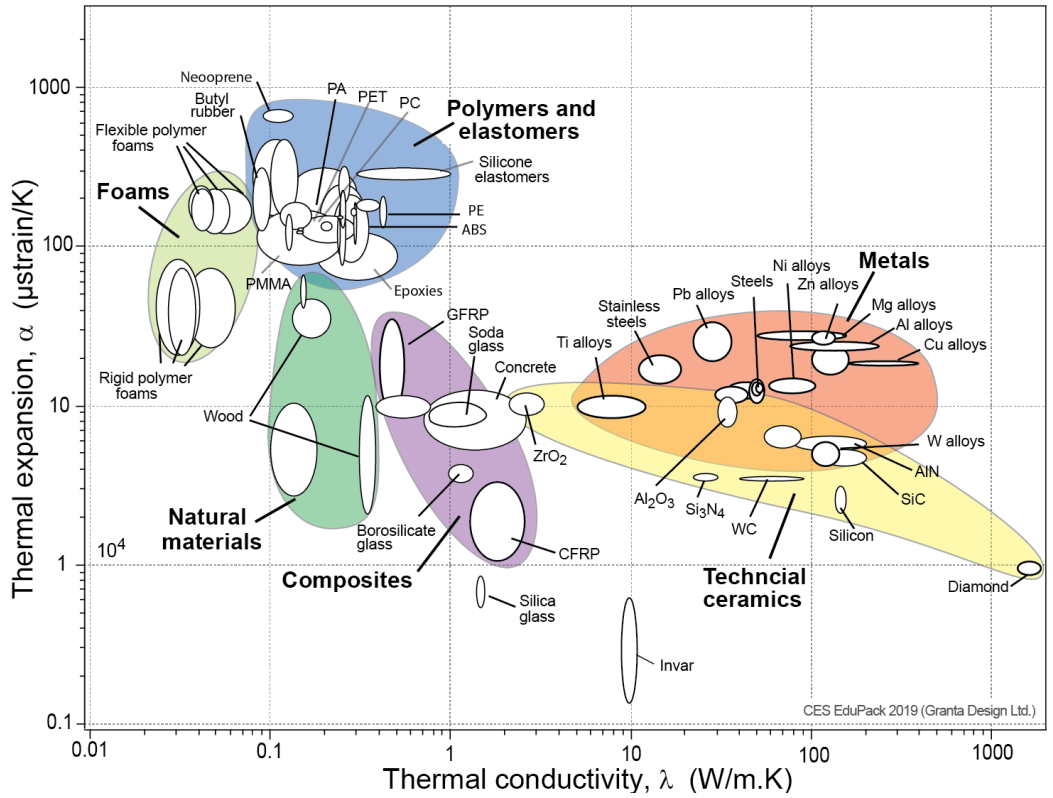


Machine learning and materials informatics differ substantially!

Aspect	Traditional Machine Learning	Materials Informatics
Data Availability (number of data records available)	Very large, typically $>10^6$ and data *down sampling* is common for computational tractability	Small, typically $<10^3$
Task	Identify value for given input	Identify extremes (large or small) for given input
Data heterogeneity	Possible to achieve high homogeneity	Typically multimodal with wide heterogeneity
Uncertainty quantification	Not important	Important
Features	Not important with deep learning from large data	Very important due to limited data
Interpretability	Typically less important	Very important

Materials informatics is more interested in outliers and extrapolation

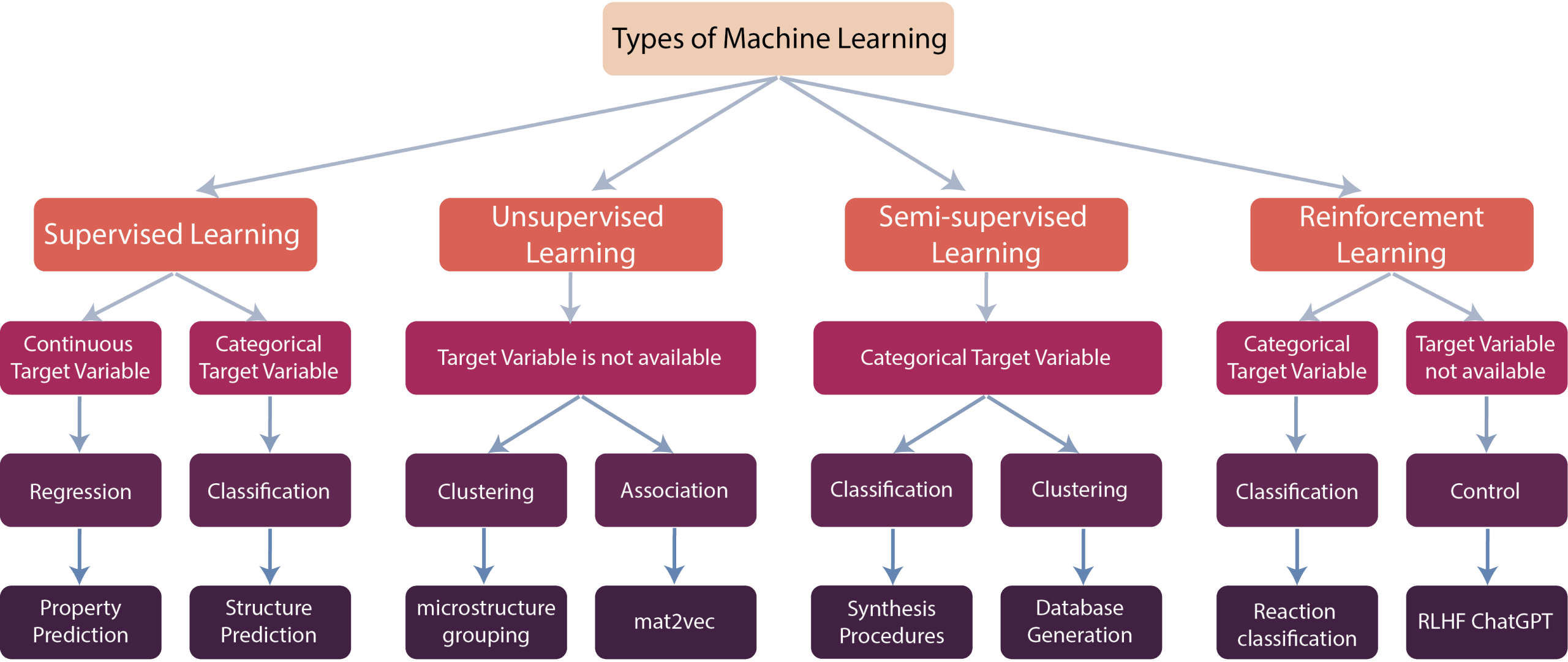




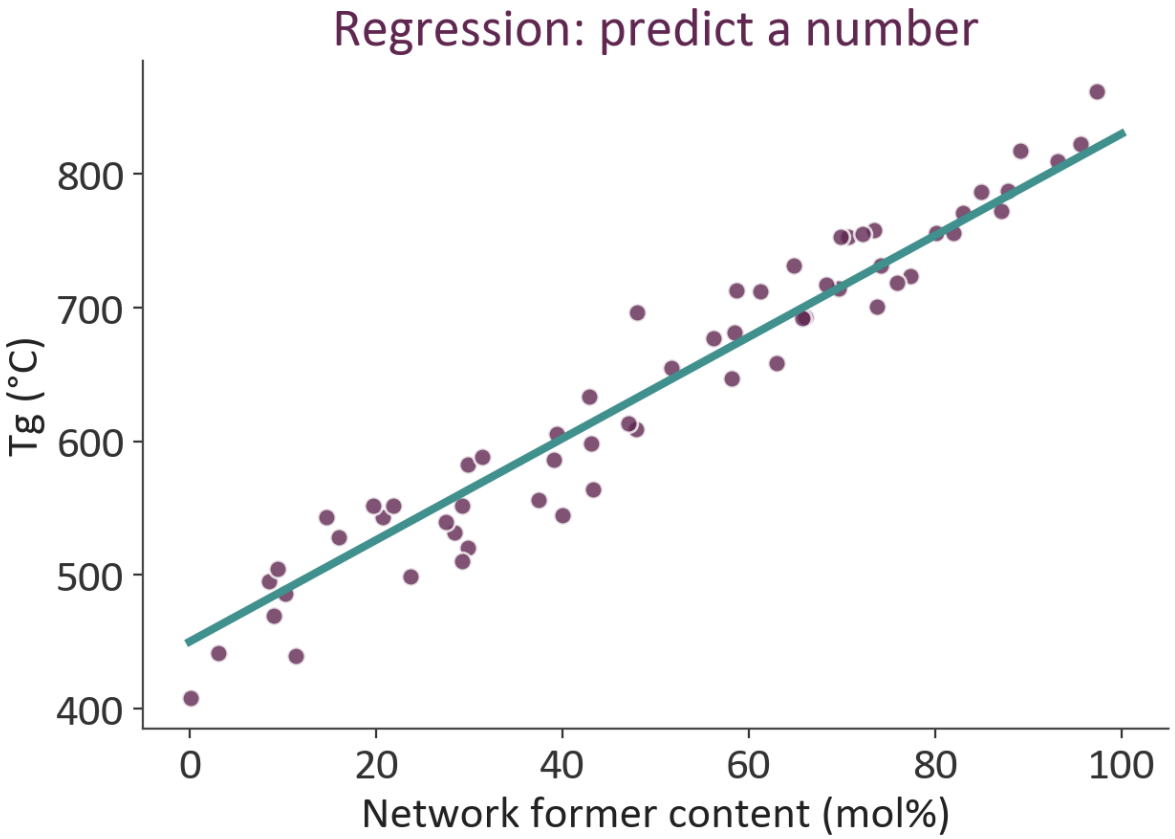
Materials informatics is a tool for pattern finding, not magic

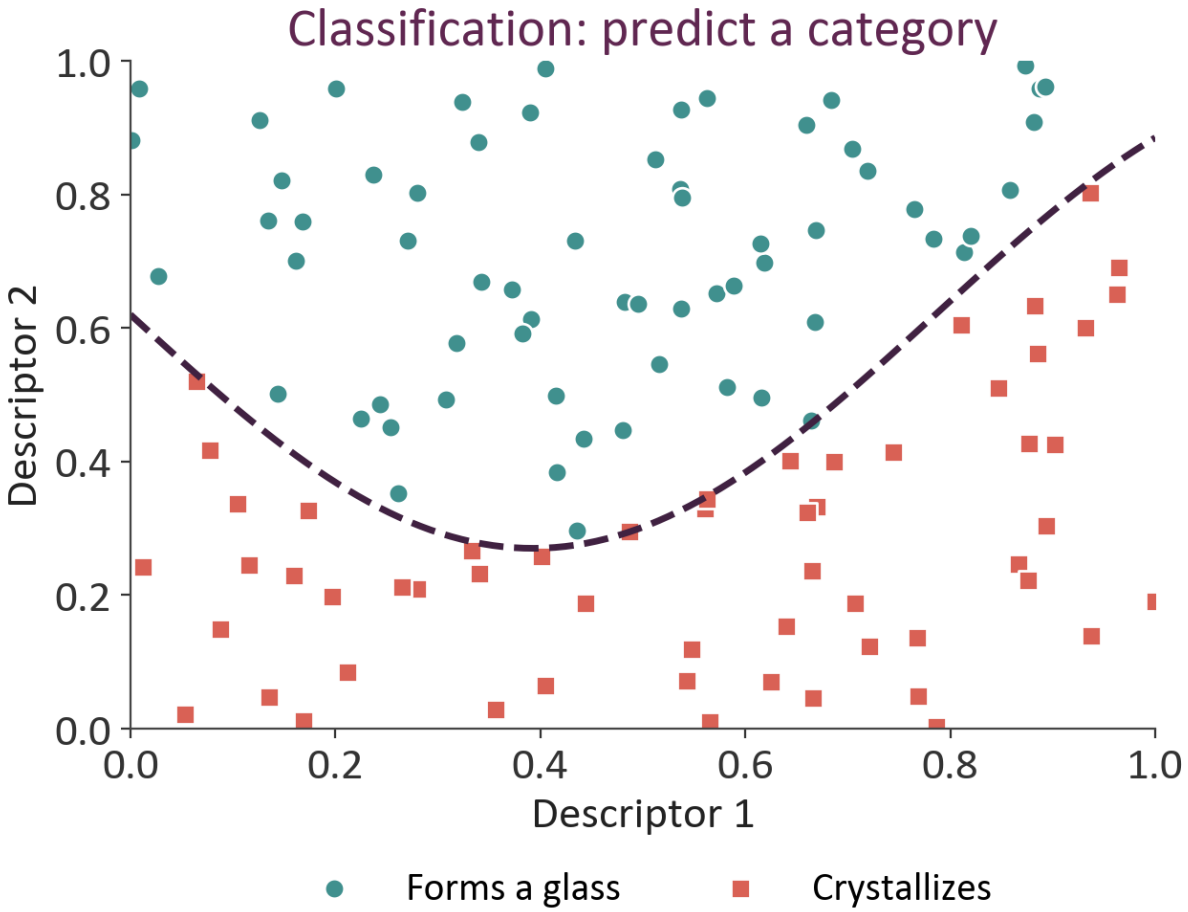
It is…	It isn't…
Learning composition, processing and structure → property links from data	A replacement for domain knowledge or experiments
Fast screening of huge design spaces	Something that works without good data
Quantifying images and spectra you already collect	Only deep learning (random forests and GPs do most of the work)
Choosing the next experiment when data are expensive	A guarantee of extrapolation to new chemistries

Machine learning is often grouped in four categories



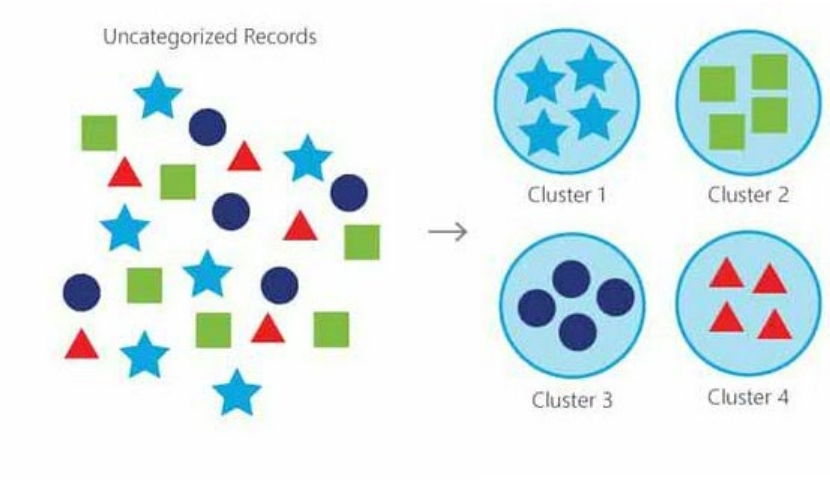
Supervised learning predicts labels you already have examples of



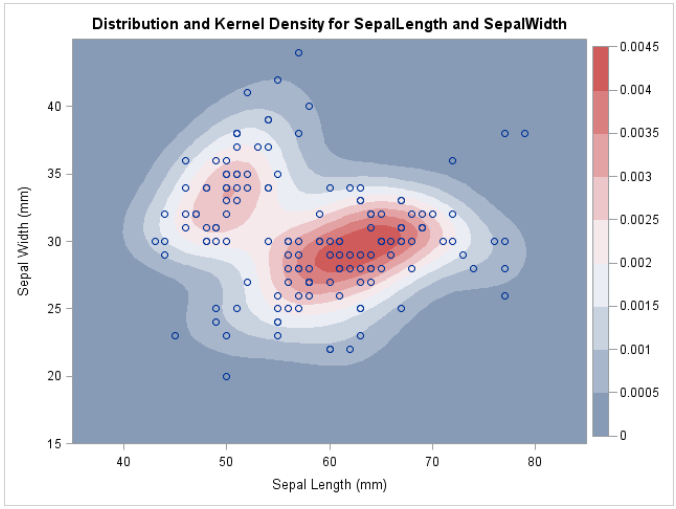


Unsupervised is typically clustering, density estimation, projection

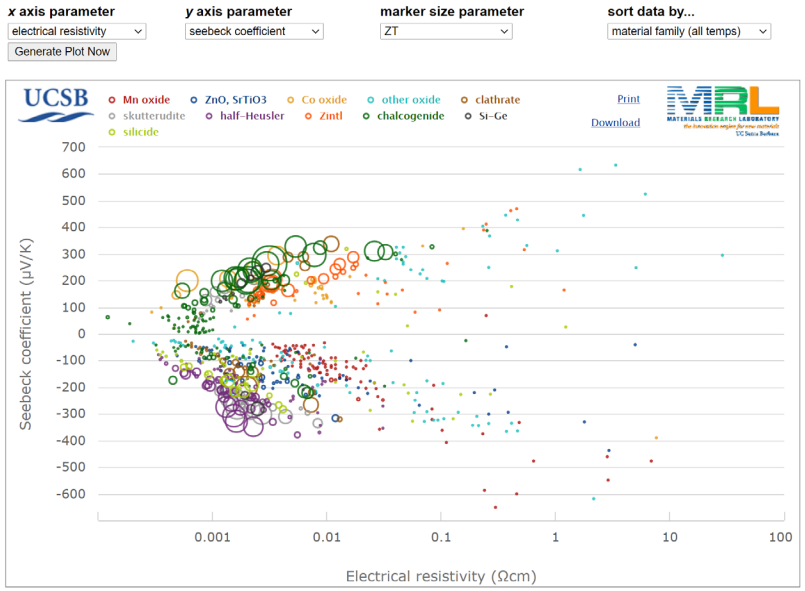
Clustering



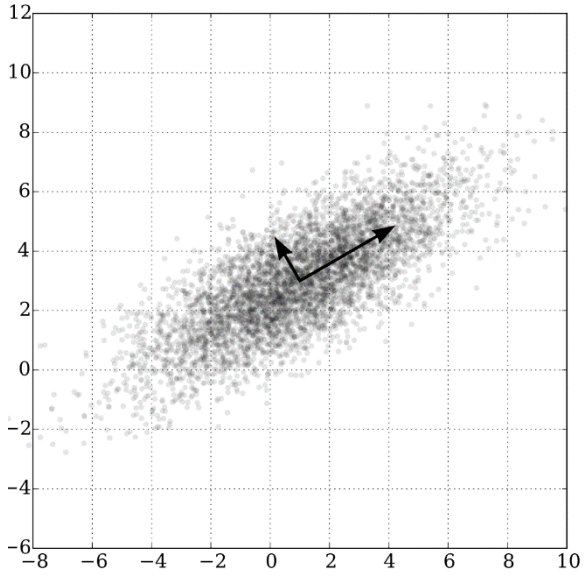
Density Estimation



Visualization

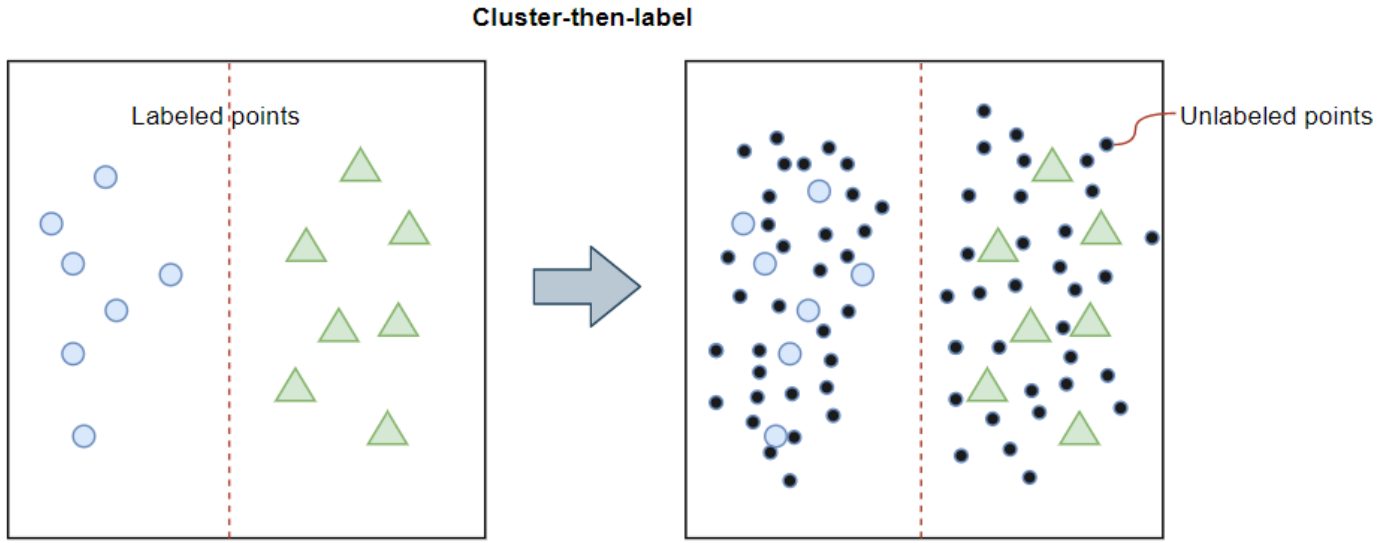


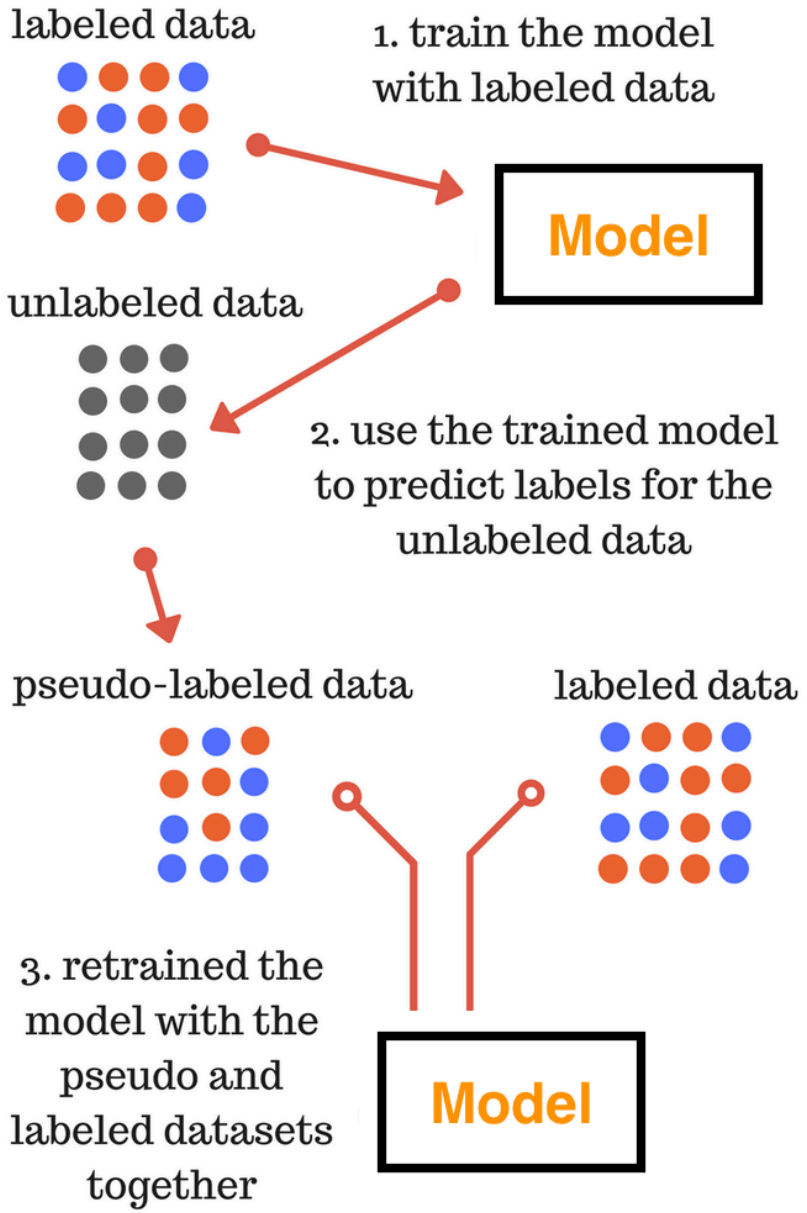
Projection



Semi-supervised makes the most of partially labeled data

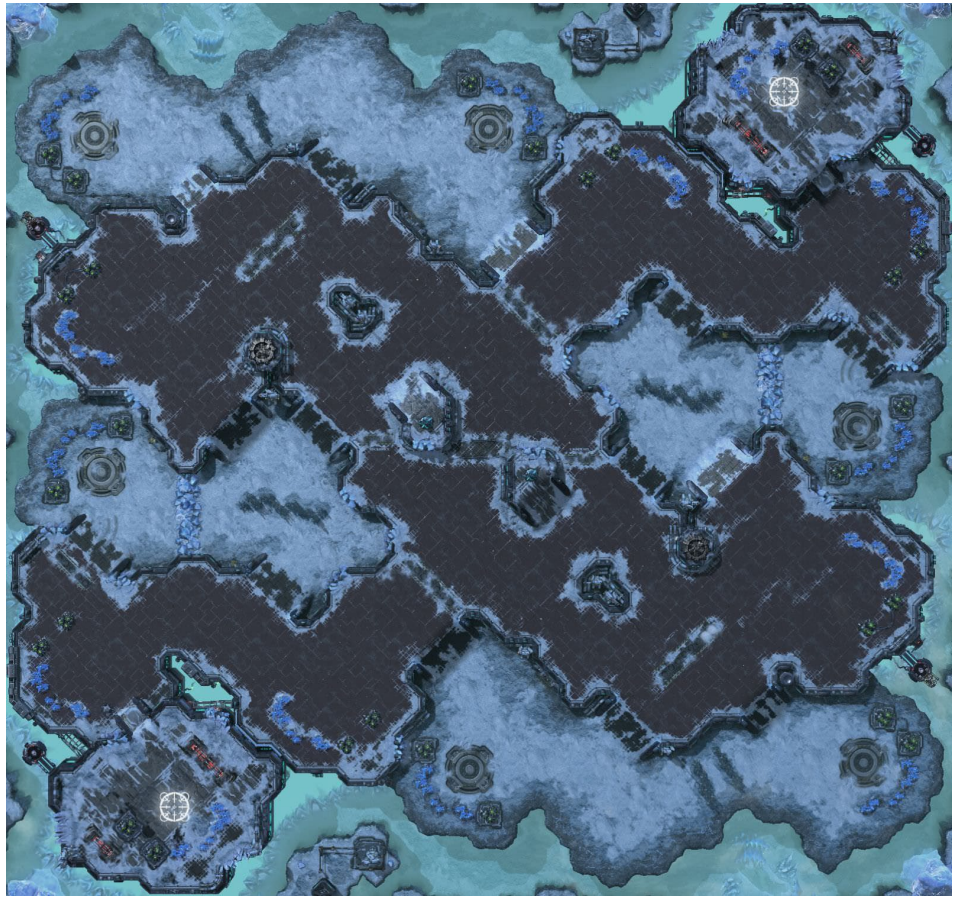
Algorithms are used to learn relationships between labeled and unlabeled data to then use all the data

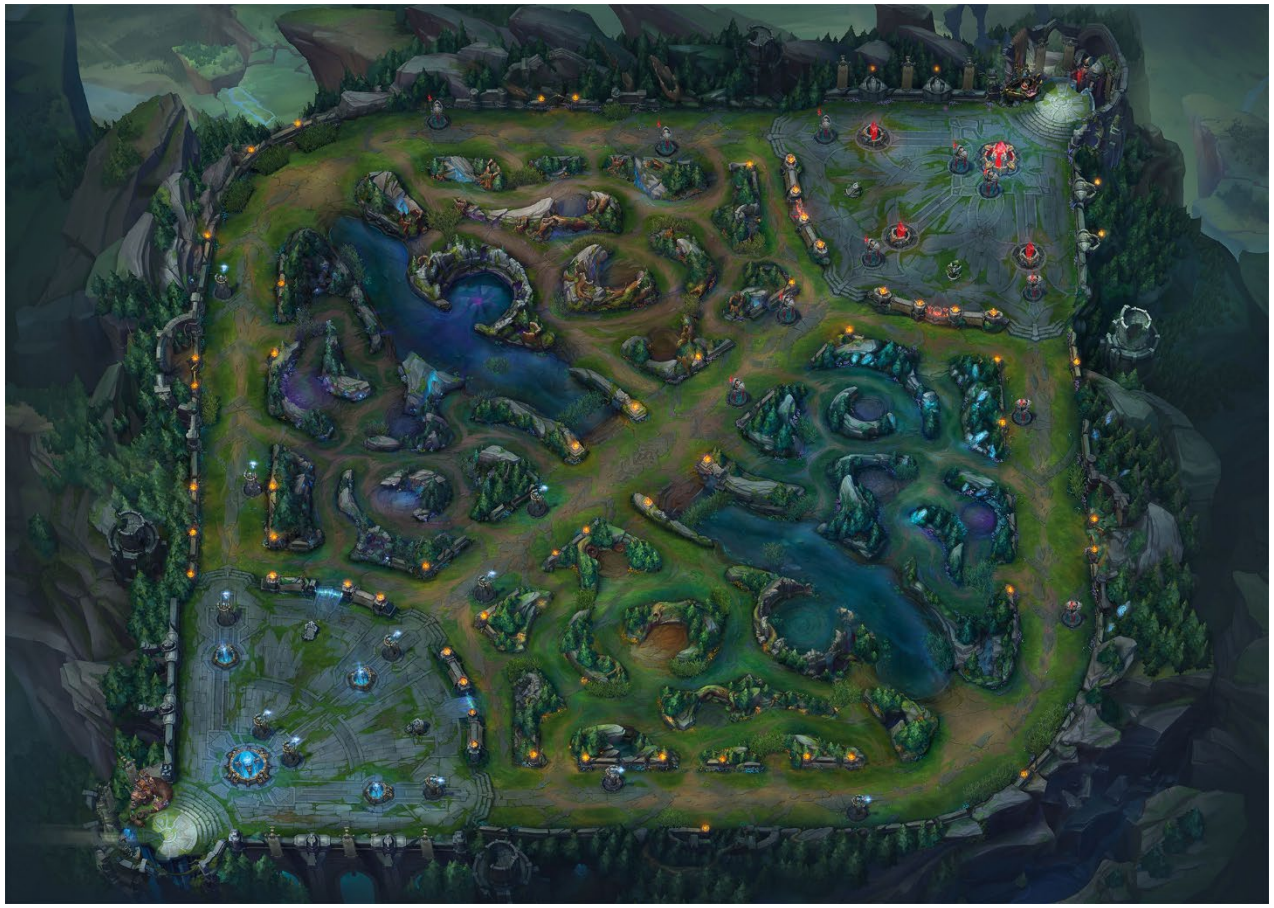




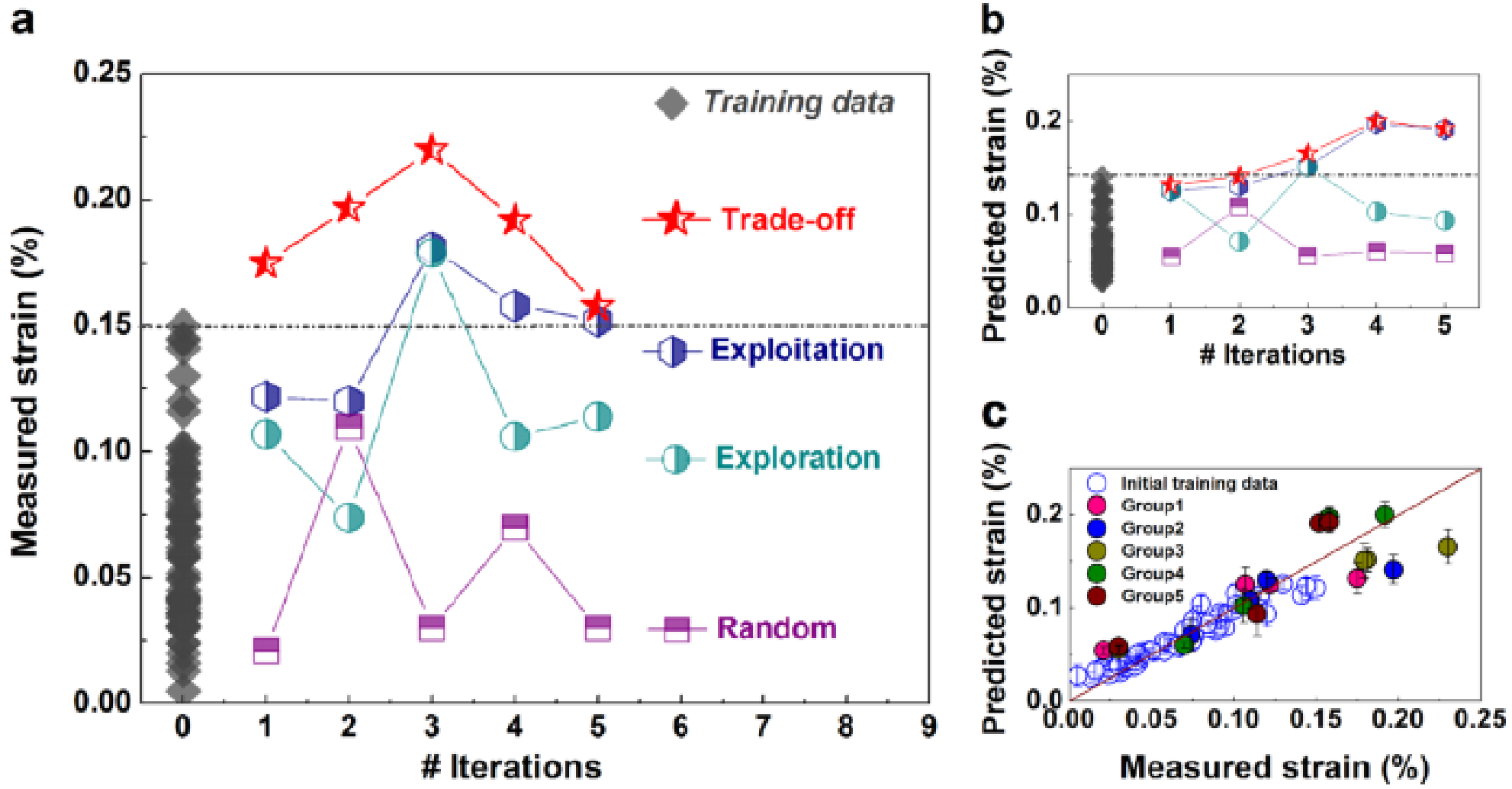
Reinforcement learning differs from supervised learning

An agent operates in an environment and must learn to operate using feedback
- No fixed dataset and the feedback may be delayed or noisy





Active learning is a model that retrains constantly with new info



We have many ML algorithms to choose from

Ensemble Techniques:
- Random forest
- Gradient boosted
- Adaboost
- Extra Trees

Bayesian:
- Kriging or GP
- Gaussian RF
- Bayesian NN

Neural Networks:
- ANN
- GAN
- CNN

Support Vector Machine:
- SVR
- Linear SVR

Linear Models:
- Lasso
- Ridge
- K nearest neighbors

Today's ceramics examples map onto these learning types

Regression

Predict Tg, modulus, permittivity

— Supervised —

Classification

Single phase? Glass former? Defect?

— Supervised —

Maps & clusters

Phase maps, embeddings

— Unsupervised —

Segmentation

Grains, pores, cracks in micrographs

— Supervised (per pixel) —

Generation

New crystals and glass compositions

— Self-/unsupervised —

Automation

Closed-loop synthesis and BO

— Active / reinforcement —

LLMs & agents

Read papers, plan, call tools

— Self-supervised + RL —

Let’s consider a materials informatics success story!

Superhard discovery was stuck between two design rules

Light-element covalent

Diamond, c-BN, B_6O, c-BC_2N

\+ Cheap, abundant elements

– Needs extreme pressure and temperature

Transition metal + light element

ReB_2, WB_4

\+ Arc-melts at ambient pressure

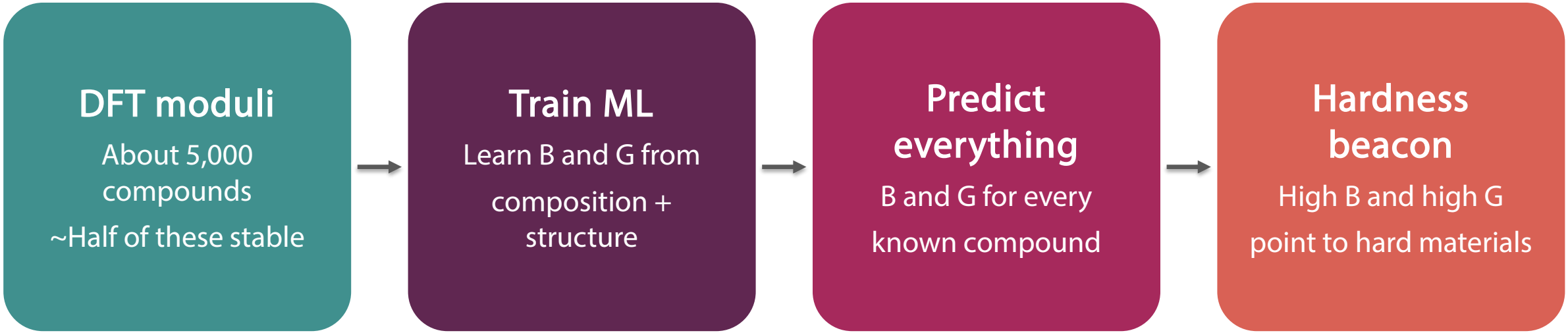
– Scarce, expensive metals

Millenia of trial-and-error and rules of thumb. Could data point somewhere new?

Mansouri Tehrani, Oliynyk, Parry, Rizvi, Couper, Lin, Miyagi, Sparks & Brgoch, J. Am. Chem. Soc. 2018 https://doi.org/10.1021/jacs.8b02717

Predict a proxy: moduli are computable, hardness is not

Bulk (B) and shear (G) modulus both track hardness, and DFT can compute them



Lesson for your own problems:

if the property you care about can't be computed or labeled, find a proxy that scales with it.

Mansouri Tehrani, Oliynyk, Parry, Rizvi, Couper, Lin, Miyagi, Sparks & Brgoch, J. Am. Chem. Soc. 2018 https://doi.org/10.1021/jacs.8b02717

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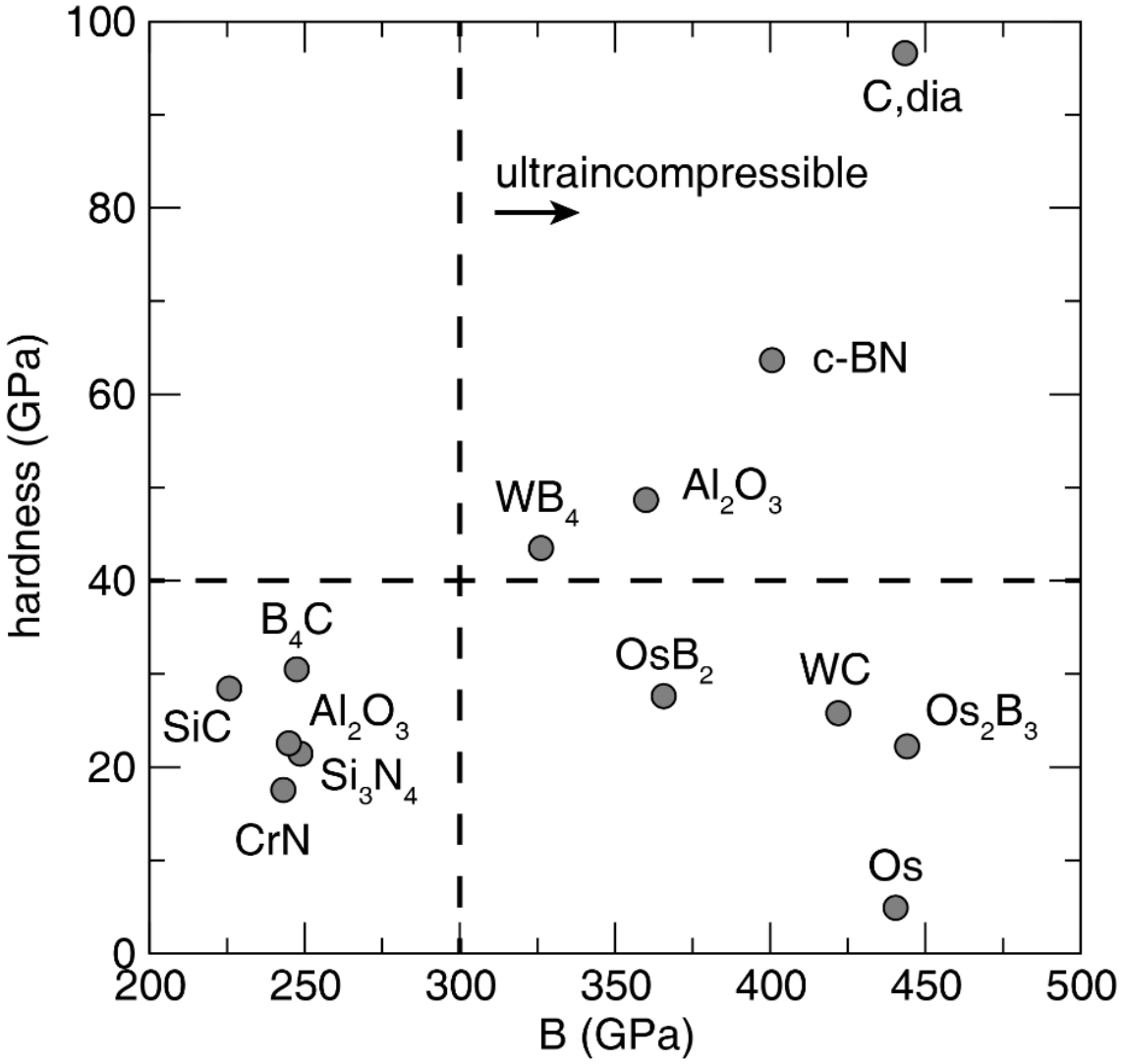
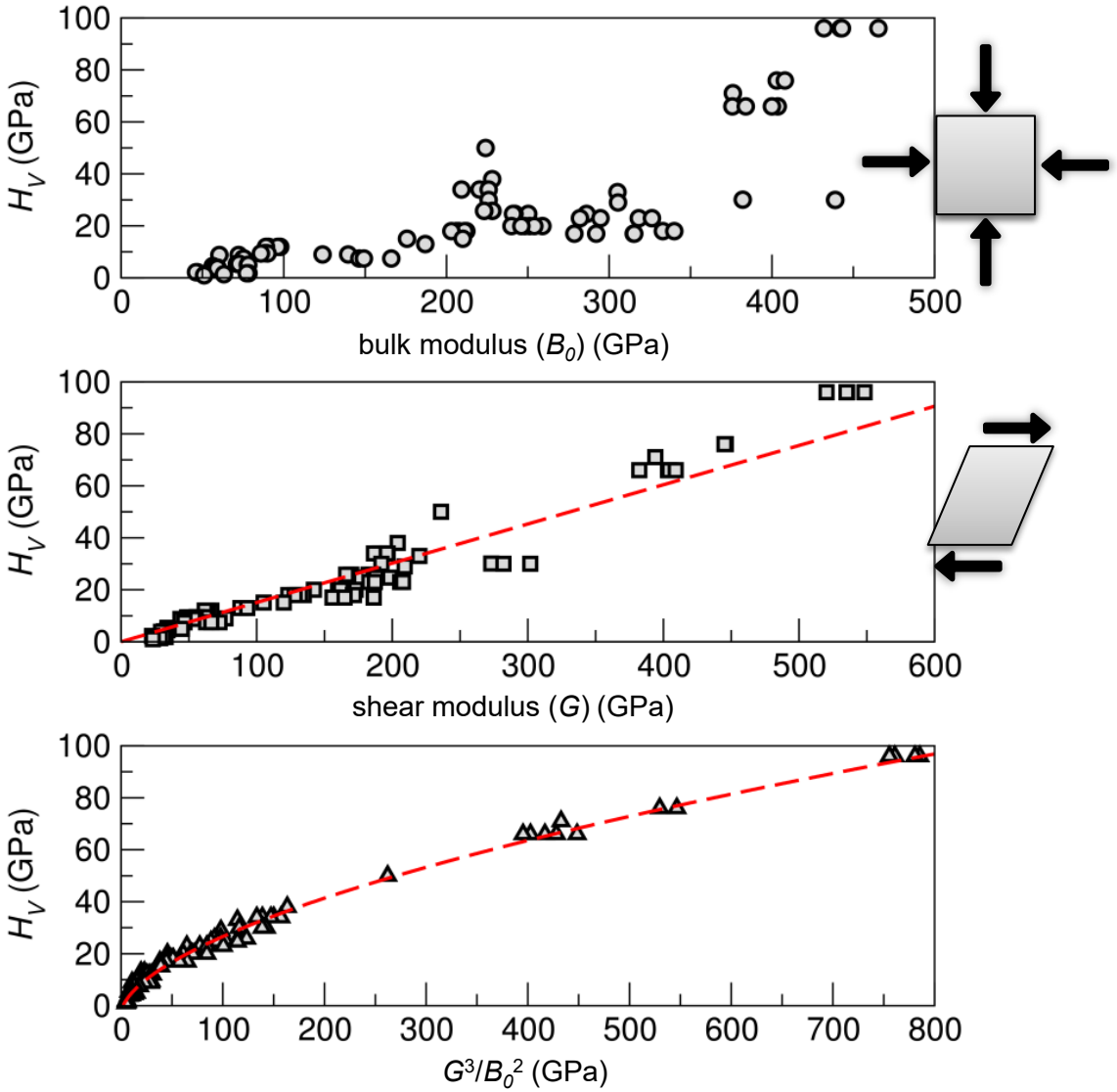




Garbage in, garbage out, so clean your data!

Materials Project elastic moduli (de Jong et al., Sci. Data 2015), cleaned for training

3,248 → **2,572**

raw DFT moduli — training compounds

Data curation is the method

Every filter is a scientific judgment about what the model should learn.

What was removed

- High-pressure-only phases (checked against Pearson's)
- Unphysical values, such as negative moduli
- Anything beyond quaternaries
- Noble gases, H, Tc and Z > 83 (except U, Th)
- Thin films and foils

Mansouri Tehrani, Oliynyk, Parry, Rizvi, Couper, Lin, Miyagi, Sparks & Brgoch, J. Am. Chem. Soc. 2018 https://doi.org/10.1021/jacs.8b02717

A support vector model on 150 descriptors, trained on a desktop

SVR

RBF kernel

10-fold cross-validation

150 descriptors

34 element properties × 4 stats

+ 14 structural

Feature selection

Remove unhelpful features

52% for B, 44% for G

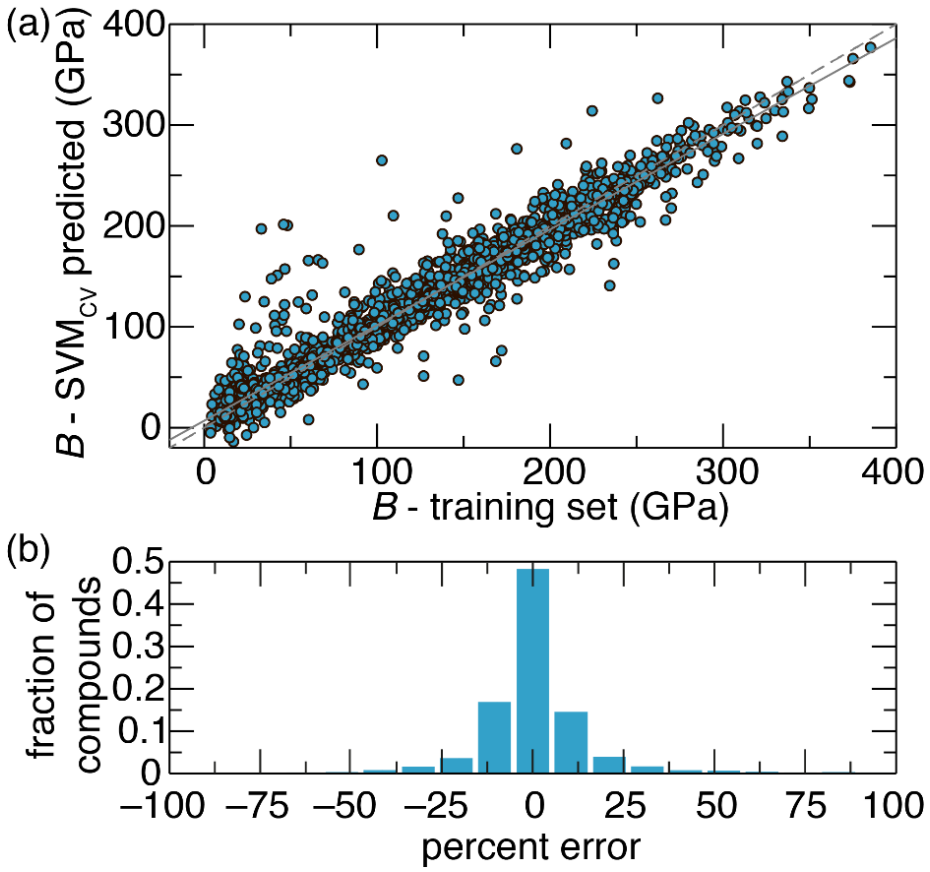
Predict B & G for ICSD

Simple Intel Core i5 laptop for inference

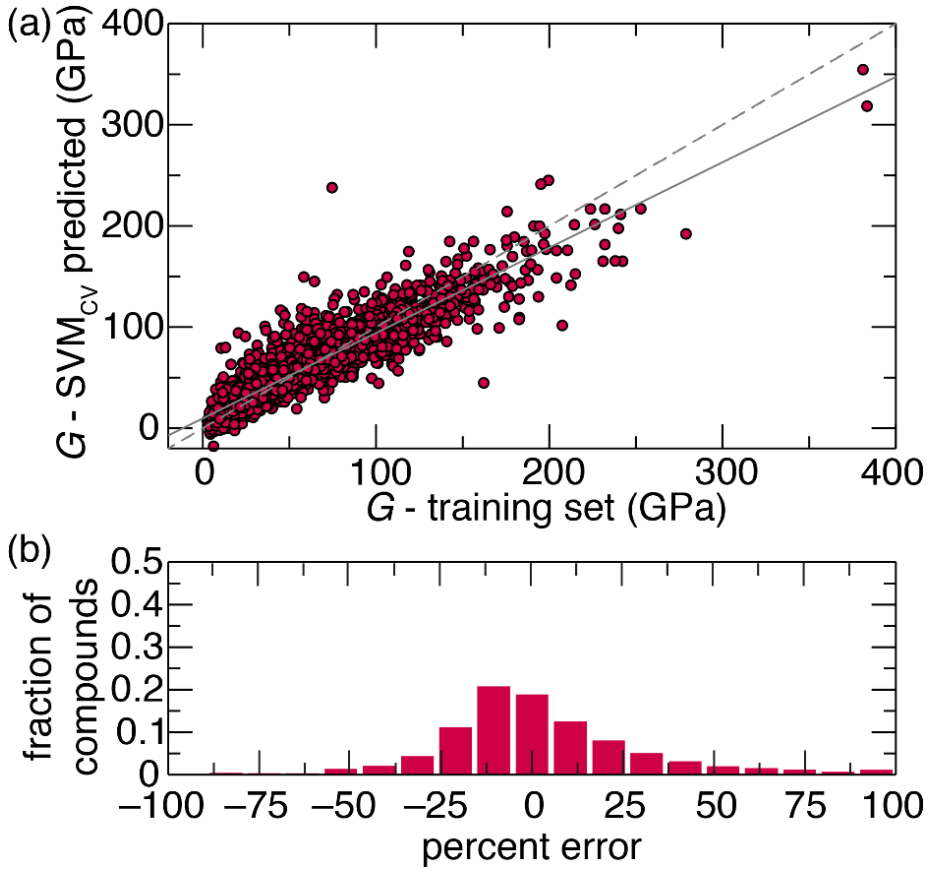
Mansouri Tehrani, Oliynyk, Parry, Rizvi, Couper, Lin, Miyagi, Sparks & Brgoch, J. Am. Chem. Soc. 2018 https://doi.org/10.1021/jacs.8b02717

Both models do surprisingly well!

Bulk modulus B: $r^2 = 0.94$, RMSE 17.2 GPa



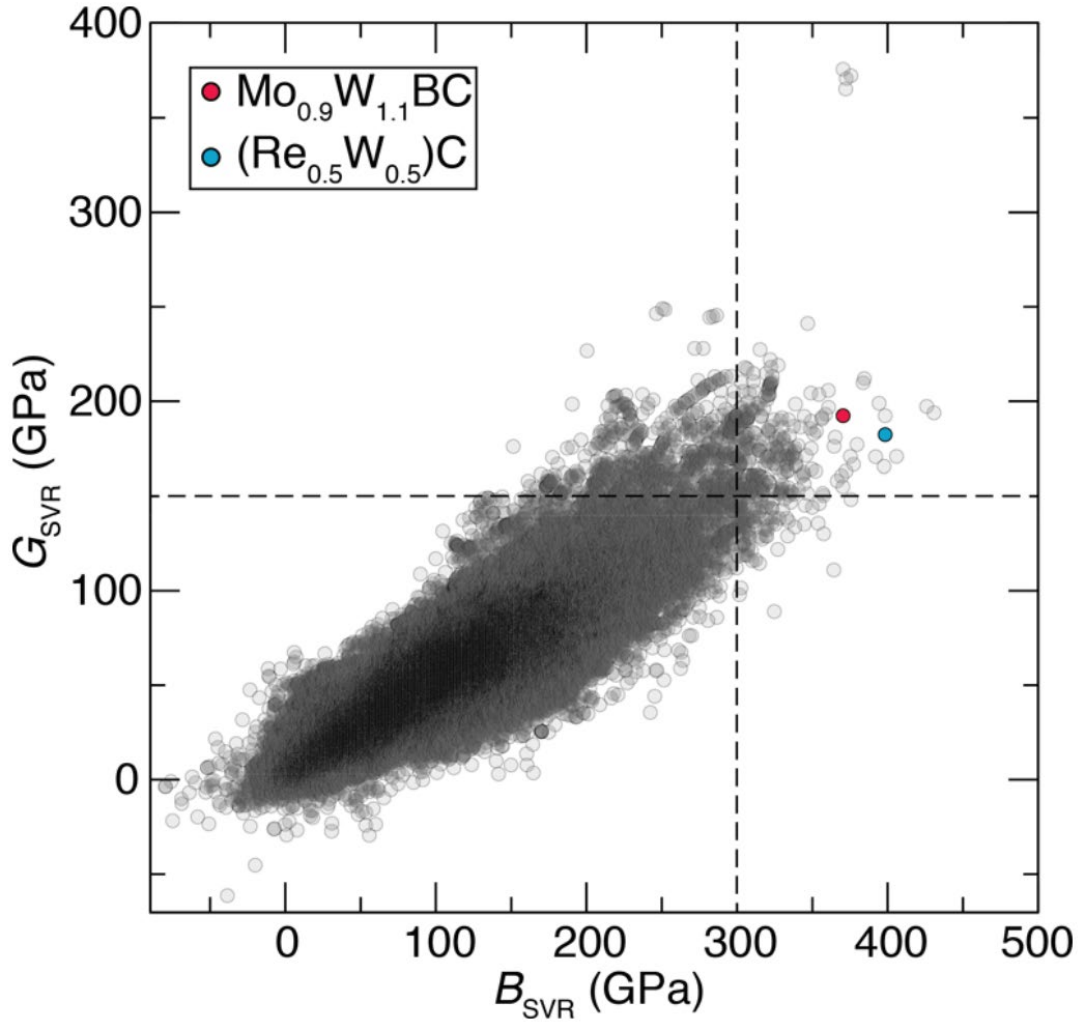
Shear modulus G: $r^2 = 0.84$, RMSE 16.5 GPa



Mansouri Tehrani, Oliynyk, Parry, Rizvi, Couper, Lin, Miyagi, Sparks & Brgoch, J. Am. Chem. Soc. 2018 https://doi.org/10.1021/jacs.8b02717

The model screened 118,287 compounds and left 258 candidates

Every binary, ternary and quaternary in Pearson's Crystal Database, scored in one pass



118,287

compounds scored

359

with B ≥ 300 GPa

258

also with G ≥ 150 GPa

Mansouri Tehrani, Oliynyk, Parry, Rizvi, Couper, Lin, Miyagi, Sparks & Brgoch, J. Am. Chem. Soc. 2018 https://doi.org/10.1021/jacs.8b02717

The model rediscovered known superhard compounds first

Known compound	Predicted B (GPa)	Predicted G (GPa)
WC	384	212
WB_4	315	227
TaC	323	210
ReB_2	330	191

New pick: $Re_{0.5}W_{0.5}C$

Top ternary (5th overall), predicted B = 398 GPa

New pick: $Mo_{0.9}W_{1.1}BC$

Top quaternary (19th overall), predicted B = 370 GPa

Why these two?

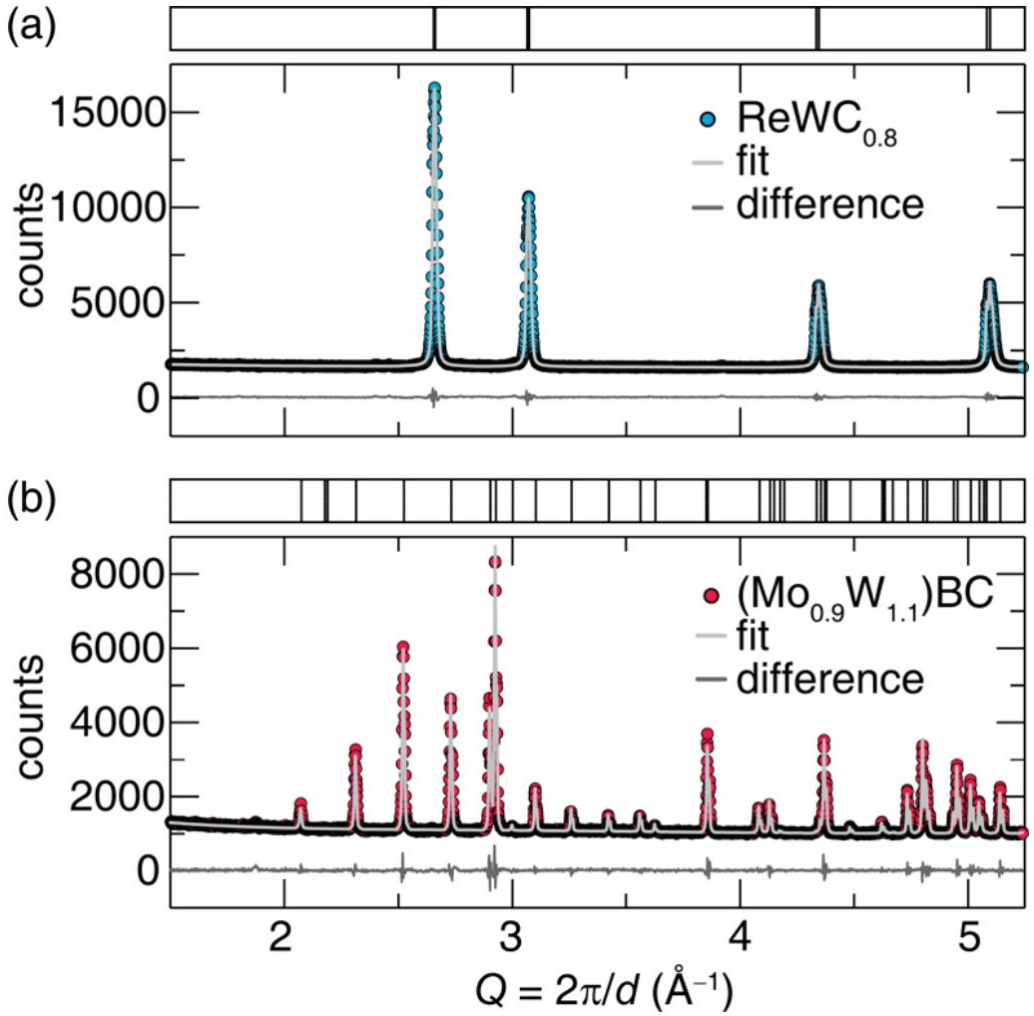
Only ever studied as superconductors (Tc 3.8 K and 4.2 K), and both arc-melt at ambient pressure.

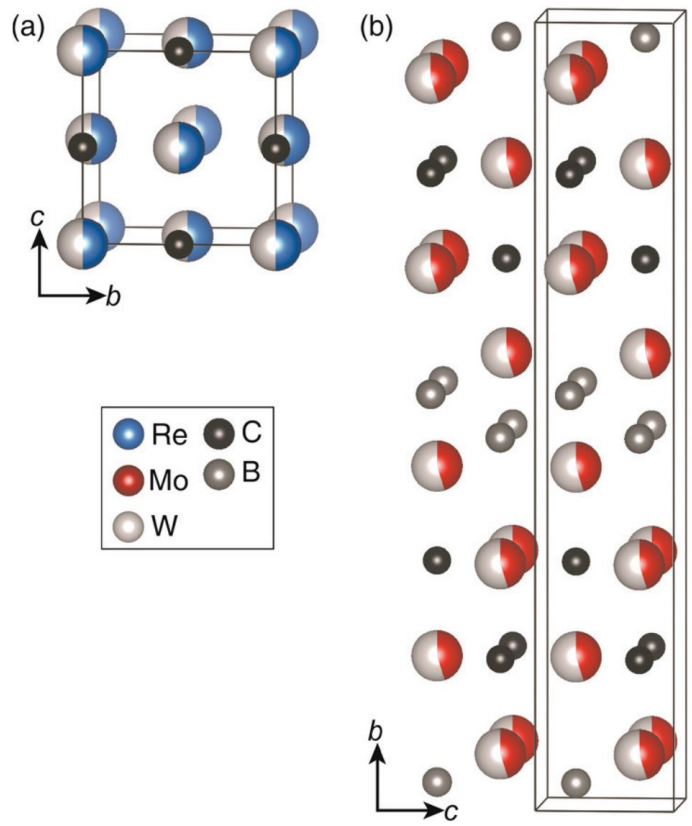
The shortlist was filtered by synthesizability and cost, not just the prediction.

Mansouri Tehrani, Oliynyk, Parry, Rizvi, Couper, Lin, Miyagi, Sparks & Brgoch, J. Am. Chem. Soc. 2018 https://doi.org/10.1021/jacs.8b02717

We were able to synthesize these both pure phase within ~1 month

Stoichiometric $ReWC_2$ left unreacted graphite; a pure phase needed carbon-deficient $ReWC_{0.8}$





$ReWC_{0.8}$

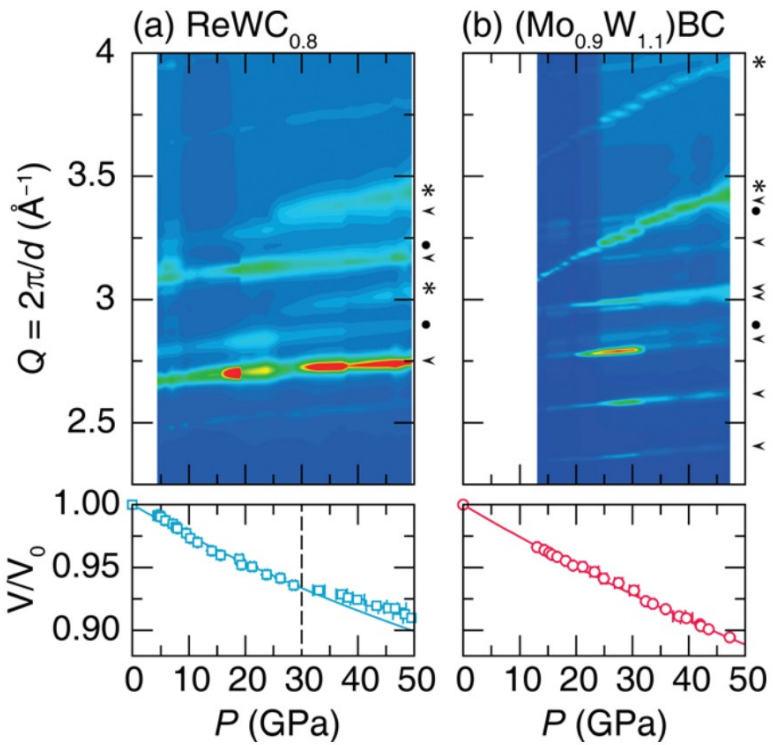
Rock-salt (NaCl) type, Re and W share the metal site, carbon site deficient

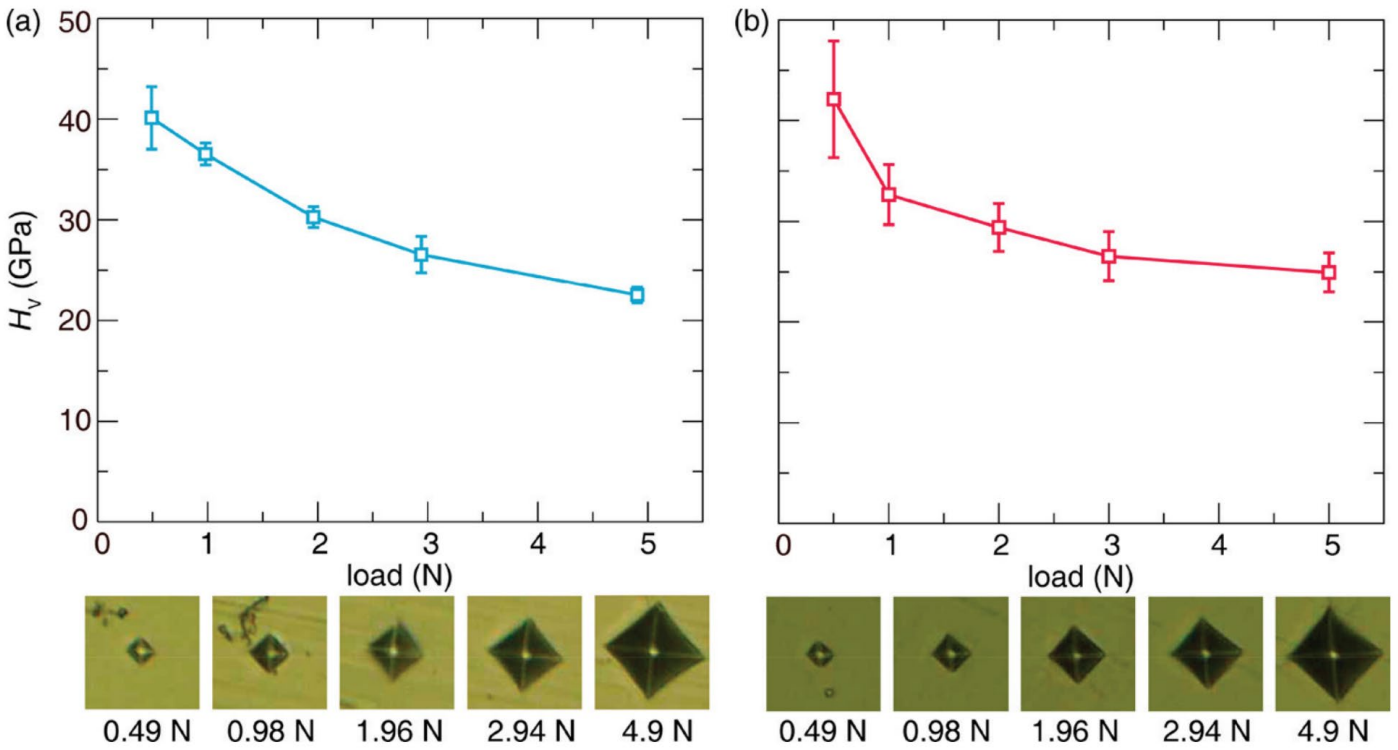
$Mo_{0.9}W_{1.1}BC$

Single phase straight from the arc melter; boron zigzag chains

Mansouri Tehrani, Oliynyk, Parry, Rizvi, Couper, Lin, Miyagi, Sparks & Brgoch, J. Am. Chem. Soc. 2018 https://doi.org/10.1021/jacs.8b02717

The predictions held: moduli within 5%, hardness above 40 GPa





380 vs 398 GPa

$ReWC_{0.8}$ B, measured vs ML

373 vs 370 GPa

$Mo_{0.9}W_{1.1}BC$ B, measured vs ML

>40 GPa

Vickers hardness at 0.49 N load

Mansouri Tehrani, Oliynyk, Parry, Rizvi, Couper, Lin, Miyagi, Sparks & Brgoch, J. Am. Chem. Soc. 2018 https://doi.org/10.1021/jacs.8b02717

Honest caveats, and where this line of work went next

Caveats

Superhard only at low load
Contested; report the full load dependence

Intrinsic moduli only
No grain-boundary or microstructure hardening

Hardness by proxy
Never predicted directly, only via B and G

What came next

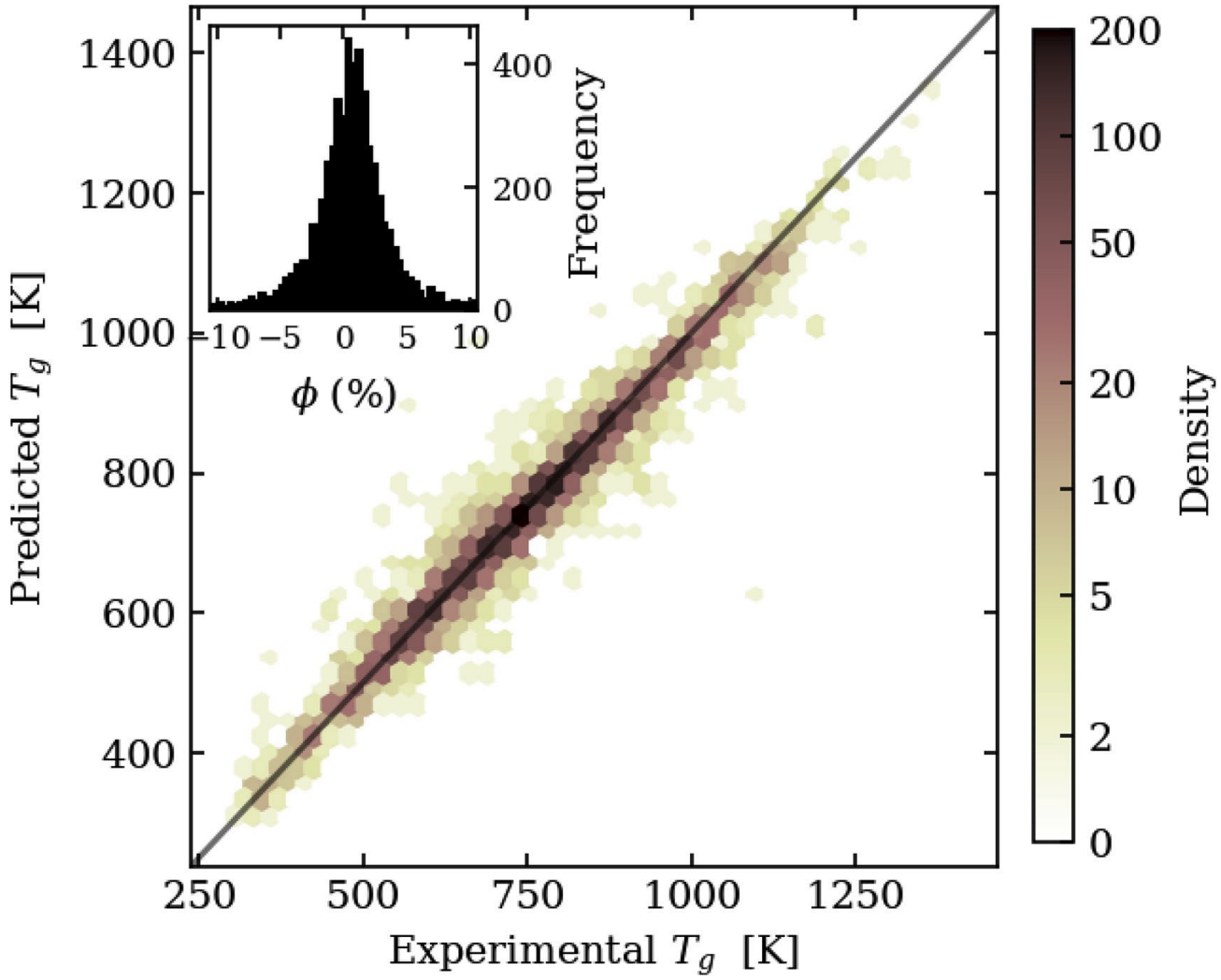
- **2019** — Atomic substitution tunes the borocarbide's hardness, ductility and cost
- **2019** — ML-informed evolutionary search finds superhard phases beyond databases
- **2021** — Ensemble model predicts load-dependent hardness directly
- **2023** — Physics-based atomic-stiffness screening for incompressibility

Chem. Mater. 2019 doi:10.1021/acs.chemmater.9b02596; npj Comput. Mater. 2019 doi:10.1038/s41524-019-0226-8; Adv. Mater. 2021 doi:10.1002/adma.202005112; Nat. Commun. 2023 doi:10.1038/s41467-023-39826-2

Let’s consider a few more examples of successful regression in ceramics and glass literature

A neural network learned Tg from 55,000 oxide glasses

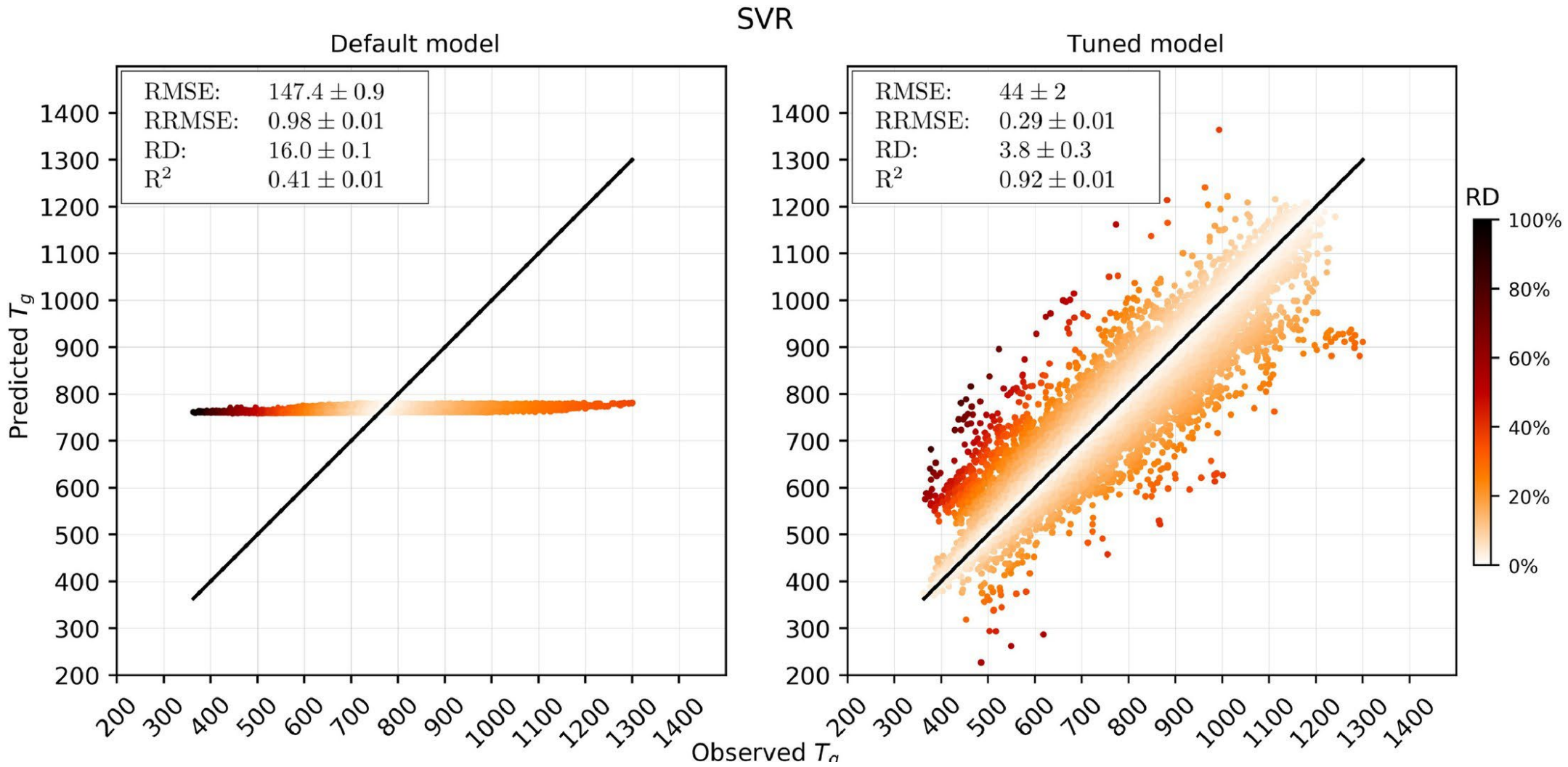
95% of predictions fall within +/-9% of the published Tg (abstract).



Cassar, de Carvalho & Zanotto, Acta Mater. 2018 https://doi.org/10.1016/j.actamat.2018.08.022

Hyperparameter tuning turned a useless model into a good one

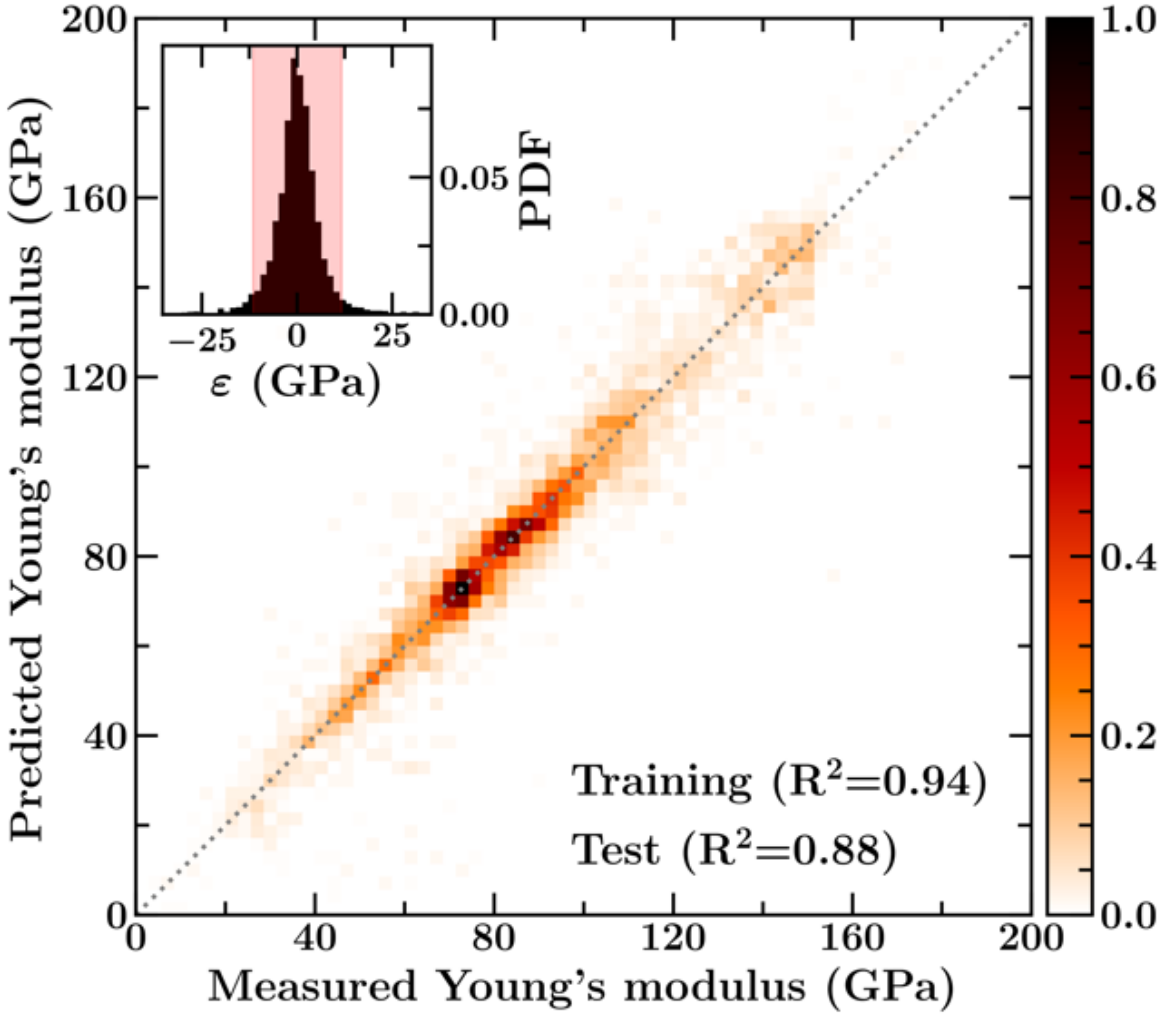
Tuned SVR: RMSE fell from 147 K to 44 K; random forest was best overall (R2 = 0.96).

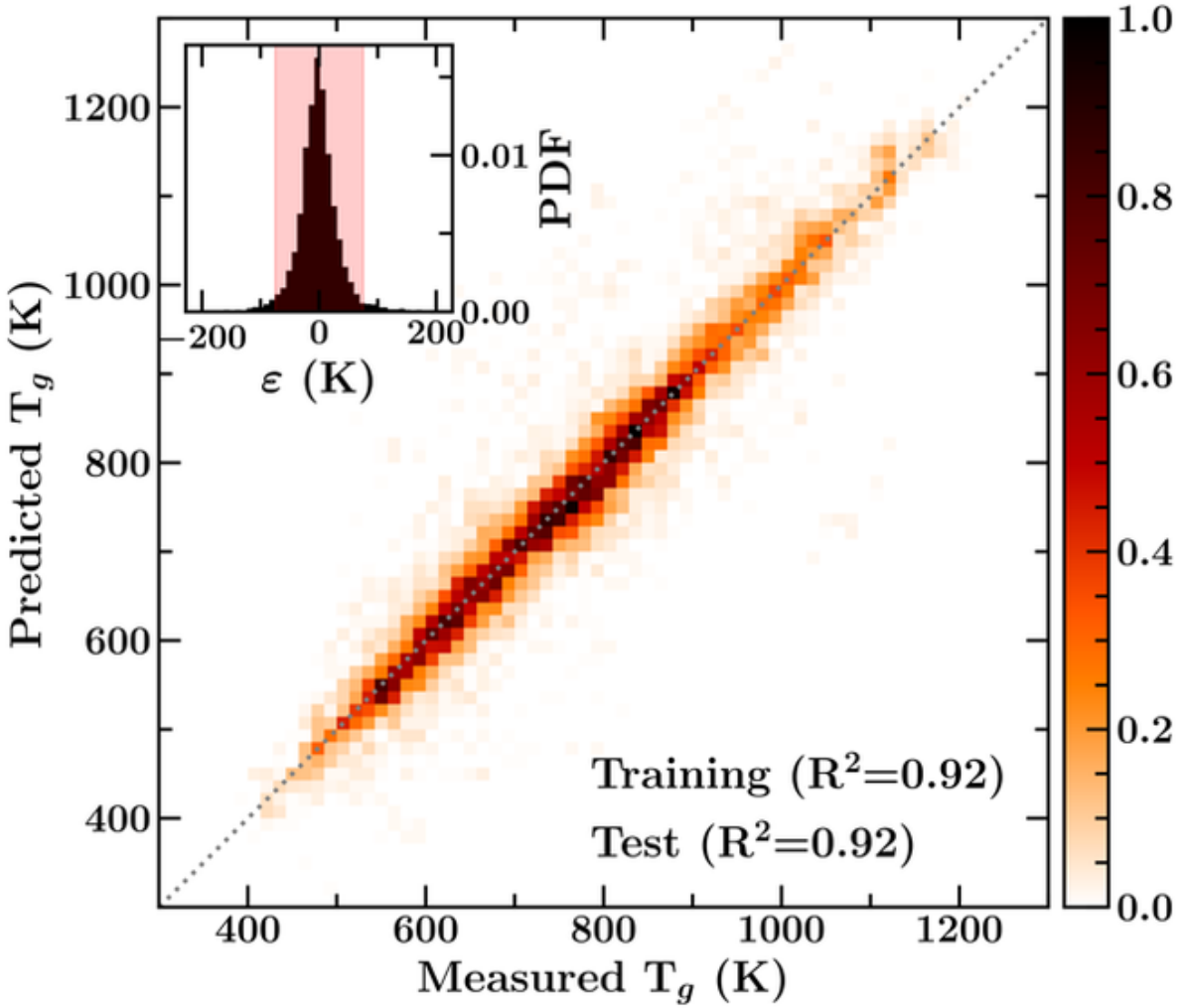


Alcobaca et al., Acta Mater. 2020 https://doi.org/10.1016/j.actamat.2020.01.047

Deep nets predict stiffness and Tg across 100,000 oxide glasses

Test R2 = 0.88 for Young's modulus and 0.92 for Tg.

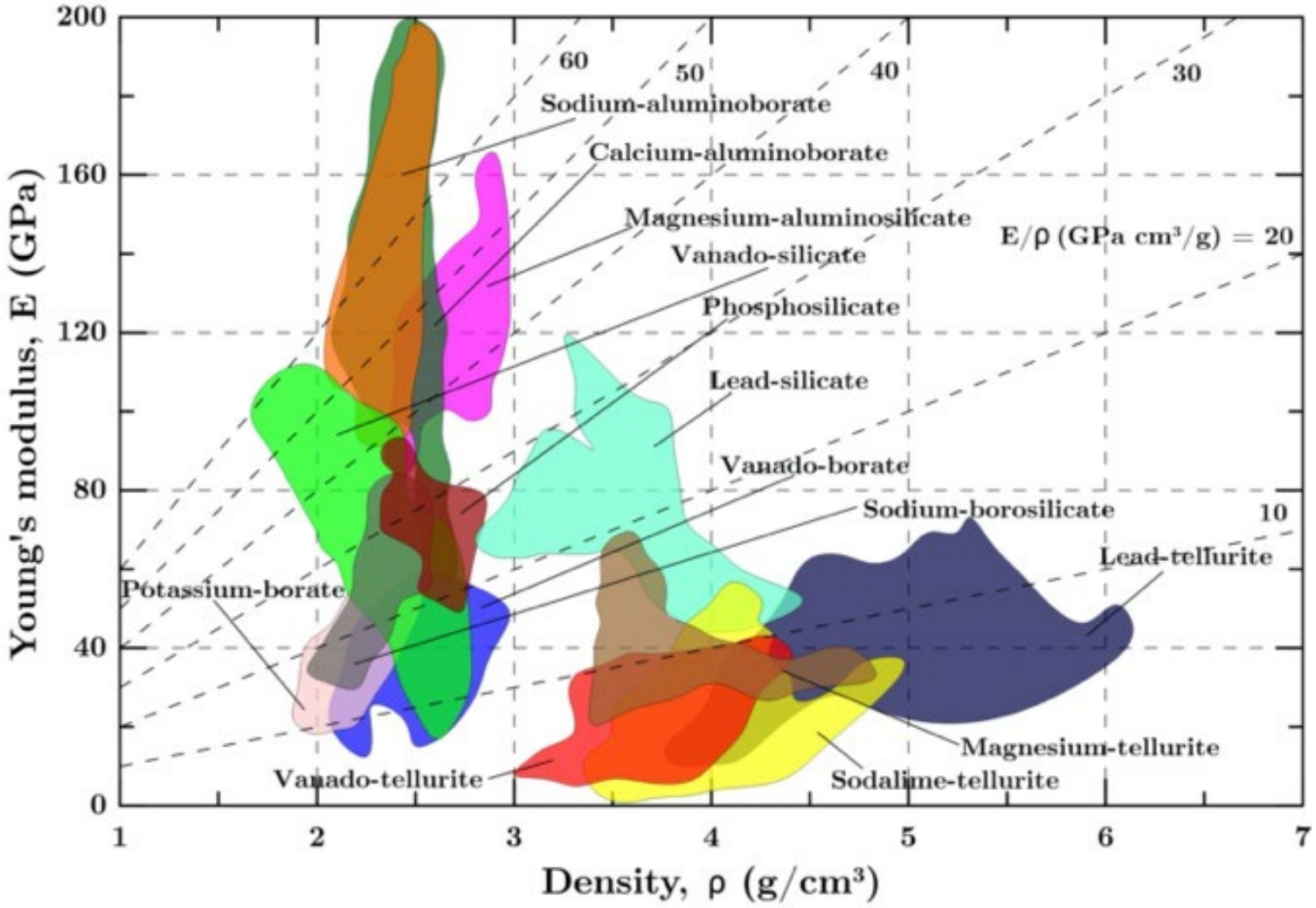




Ravinder et al., Mater. Horiz. 2020 https://doi.org/10.1039/D0MH00162G (arXiv:1912.11582)

ML models can draw Ashby charts for glasses never made

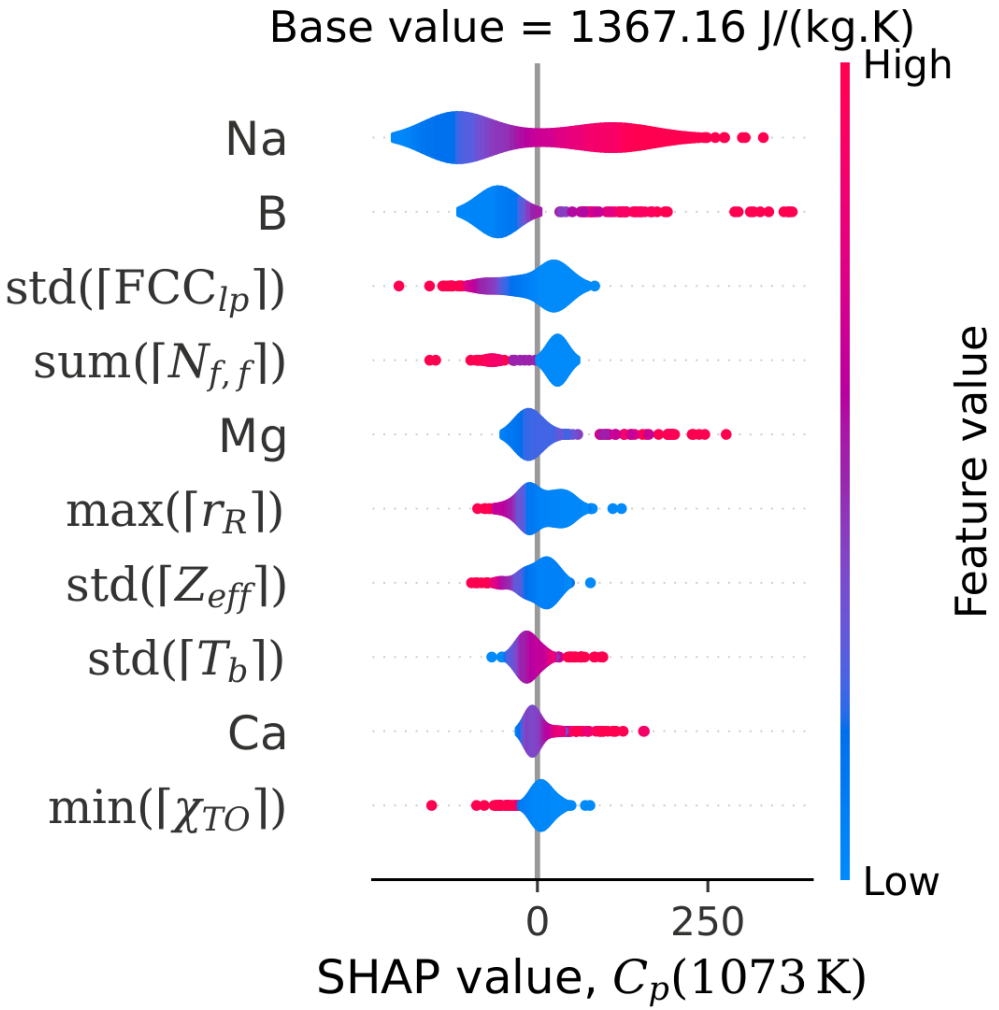
Dashed lines mark constant specific stiffness E/rho.



Ravinder et al., Mater. Horiz. 2020 https://doi.org/10.1039/D0MH00162G (arXiv:1912.11582)

One network predicts 85 glass properties from 218k glasses

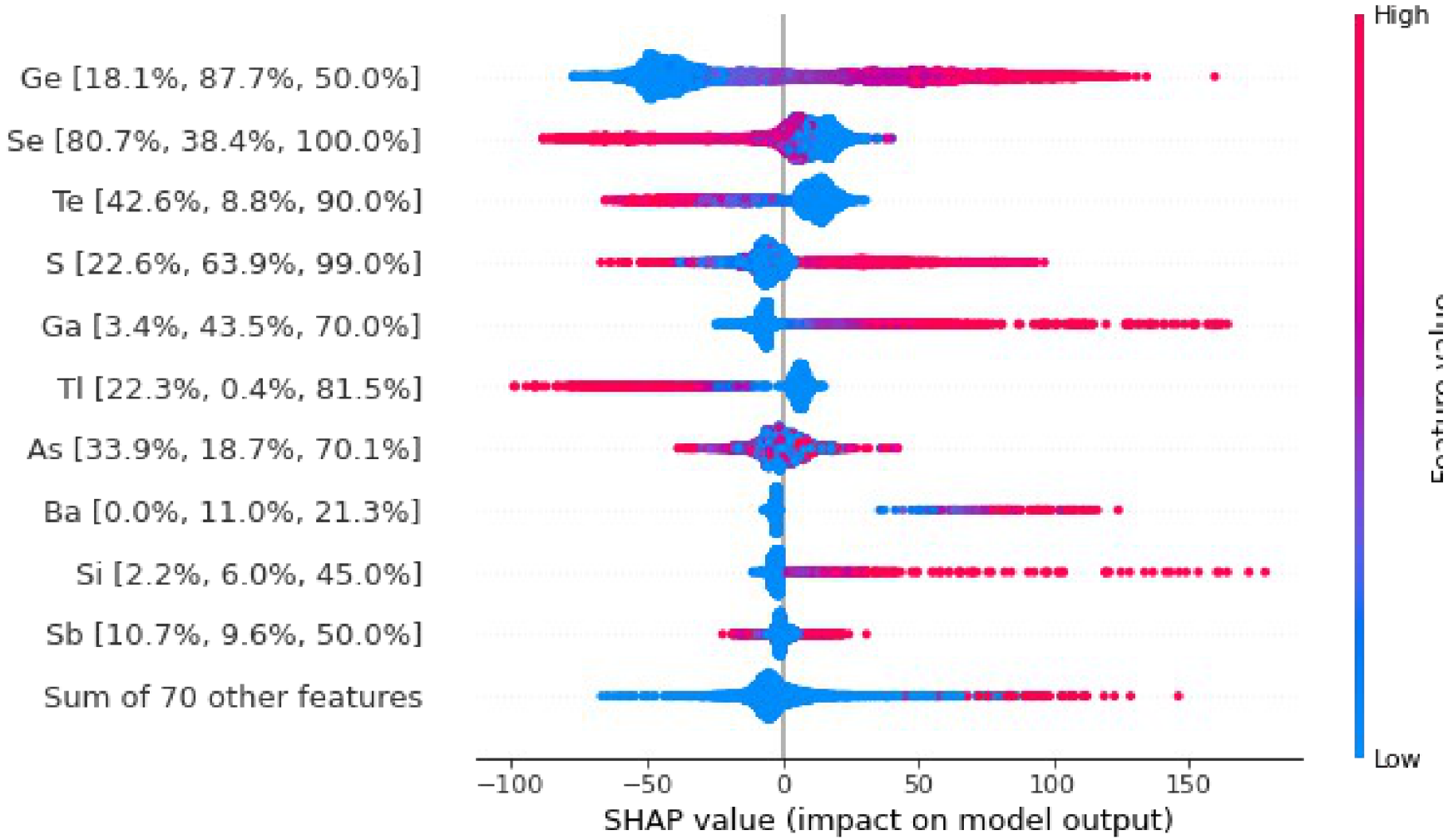
Model and data are free in the GlassPy Python package: you can run it yourself



Cassar, Ceram. Int. 2023 https://doi.org/10.1016/j.ceramint.2023.08.281 (arXiv:2303.15538)

SHAP shows which elements raise Tg in chalcogenide glasses

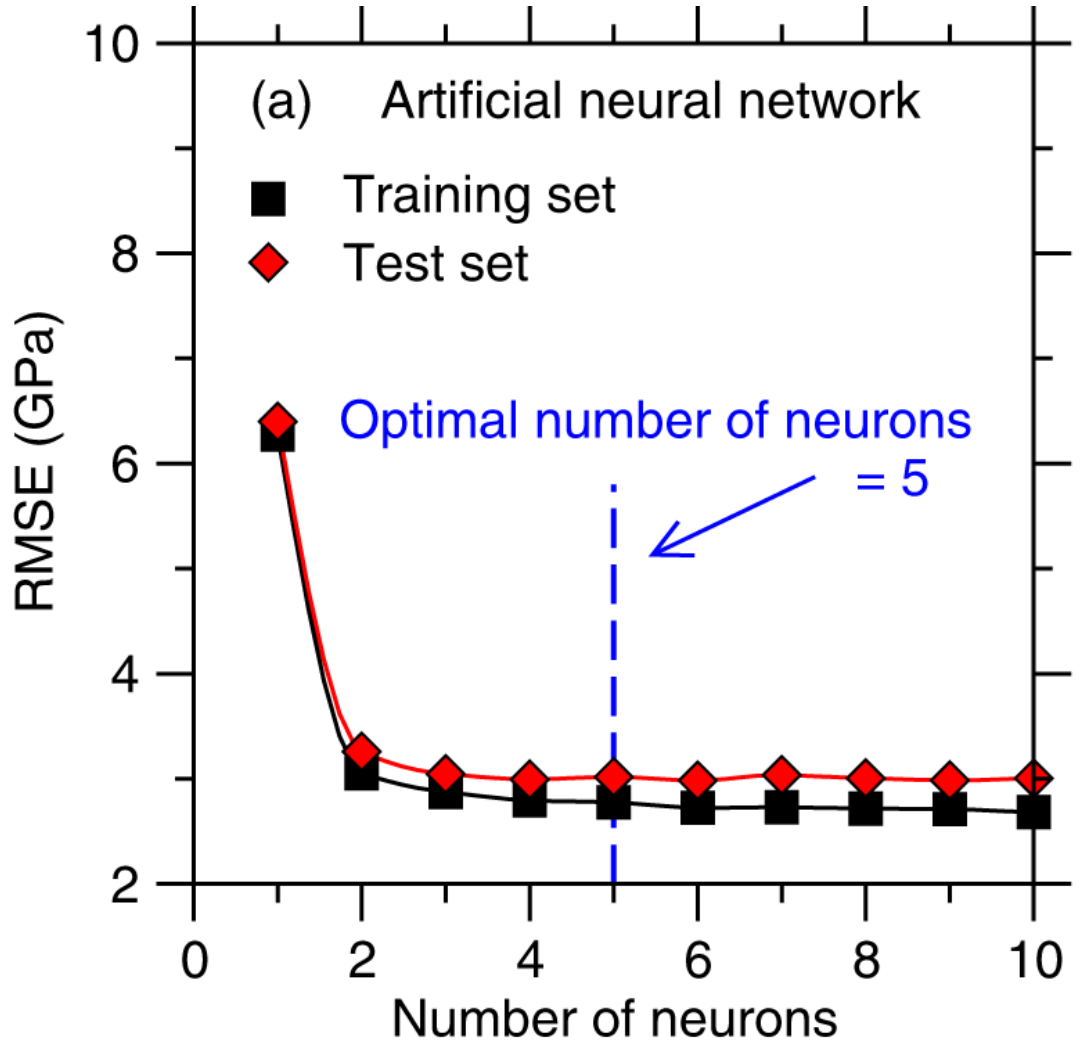
Ge and Ga raise Tg, Se lowers it; Tg model $R^2 = 0.93$ on 6,747 glasses.

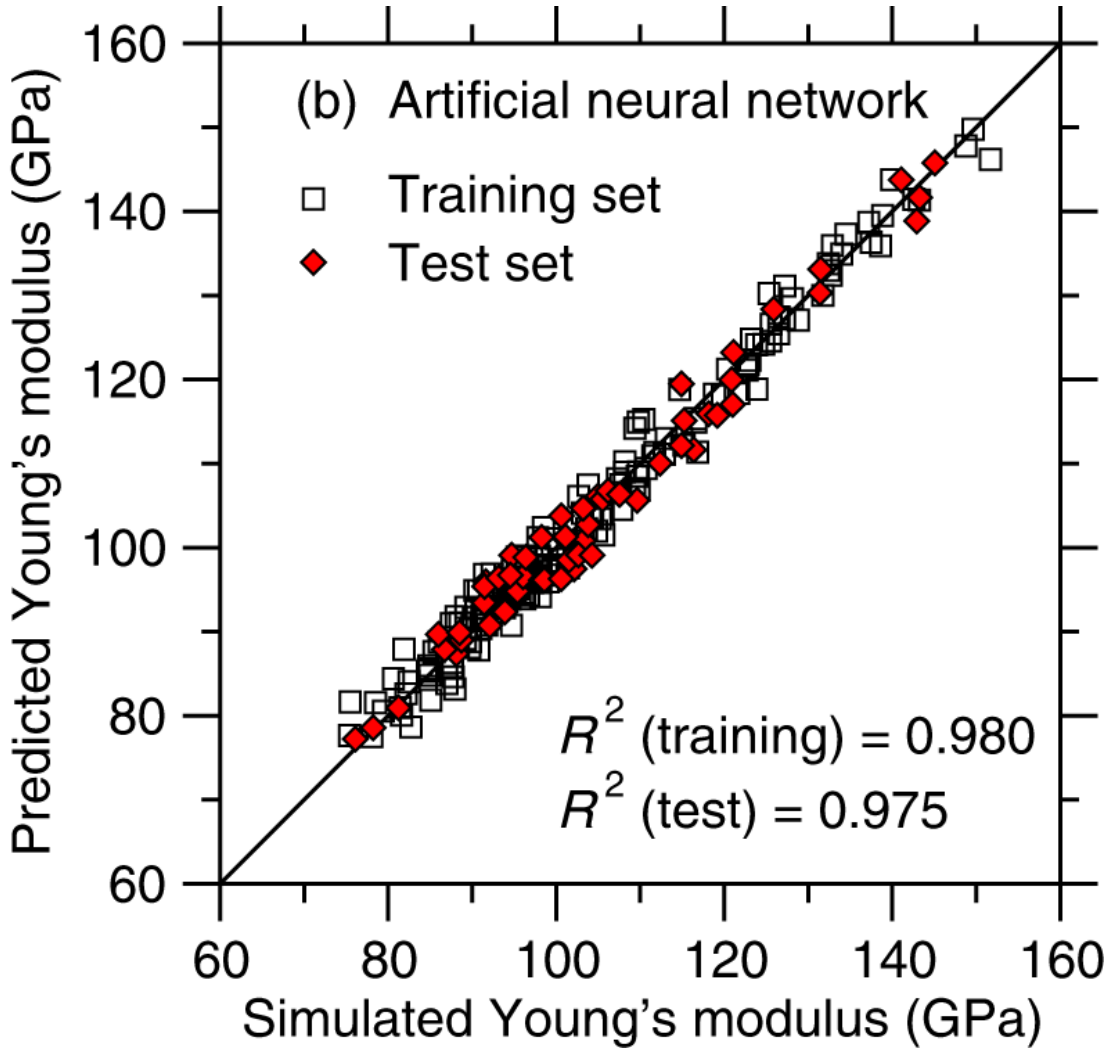


Mastelini et al., Acta Mater. 2022 https://doi.org/10.1016/j.actamat.2022.118302 (arXiv:2106.07749)

Simulated data can train ML when experiments are scarce

A 5-neuron network reaches test $R^2 = 0.975$ on 231 MD-simulated glasses.

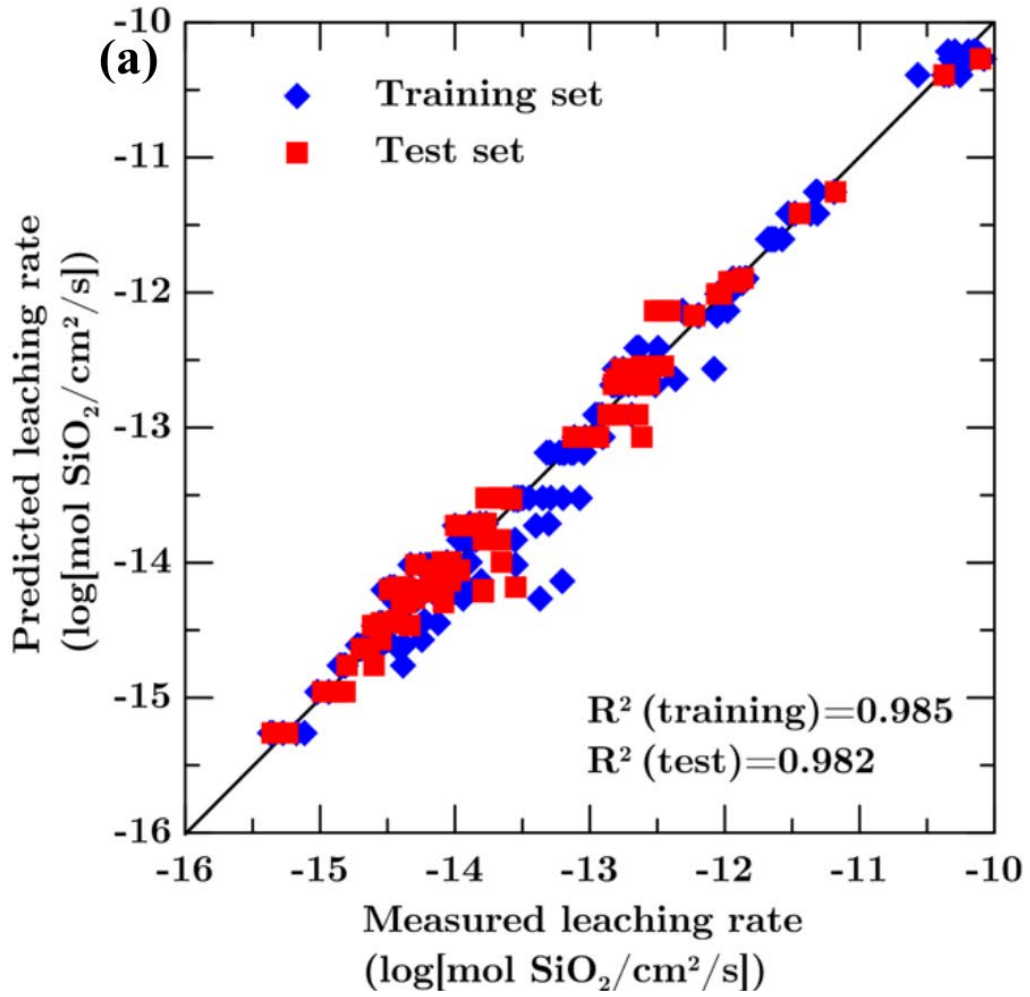


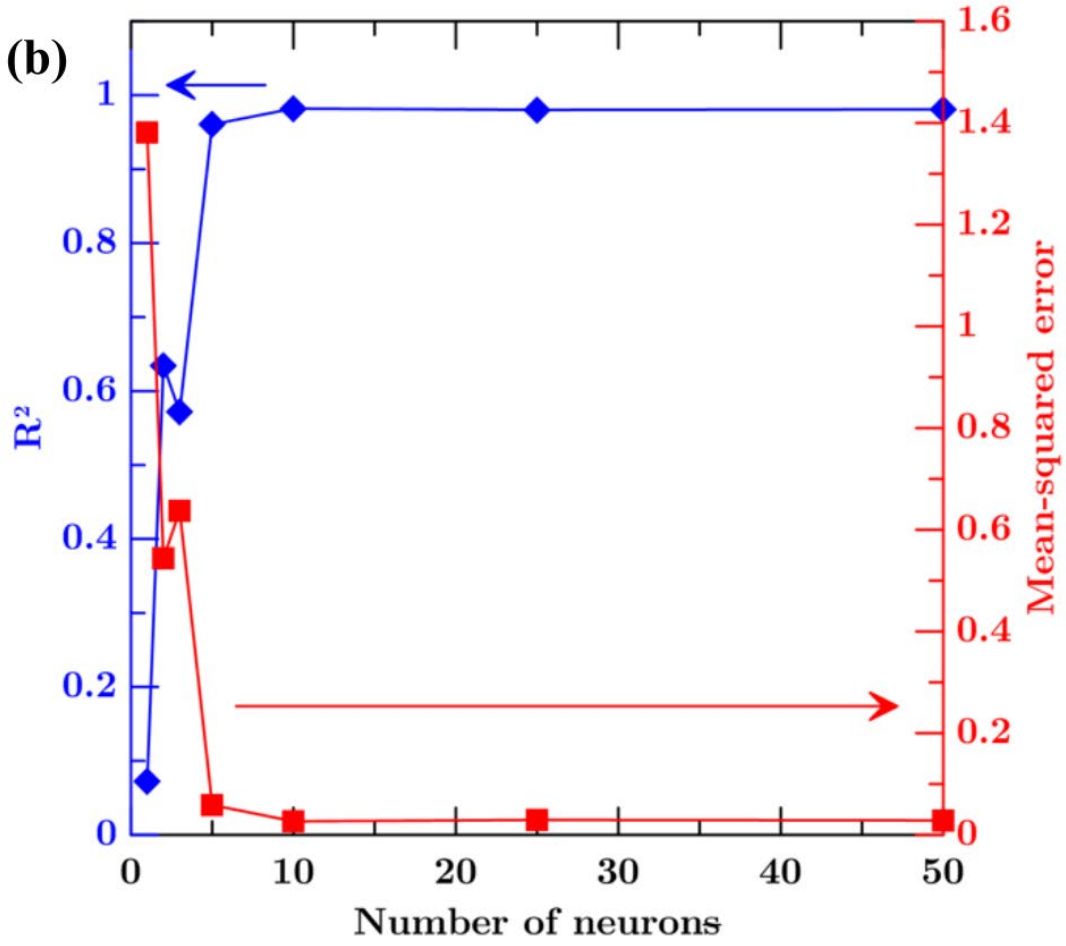


Yang et al., Sci. Rep. 2019 https://doi.org/10.1038/s41598-019-45344-3

A neural network predicts glass dissolution where linear models fail

299 leaching rates of aluminosilicate glasses, acidic to caustic pH: test $R^2 = 0.982$

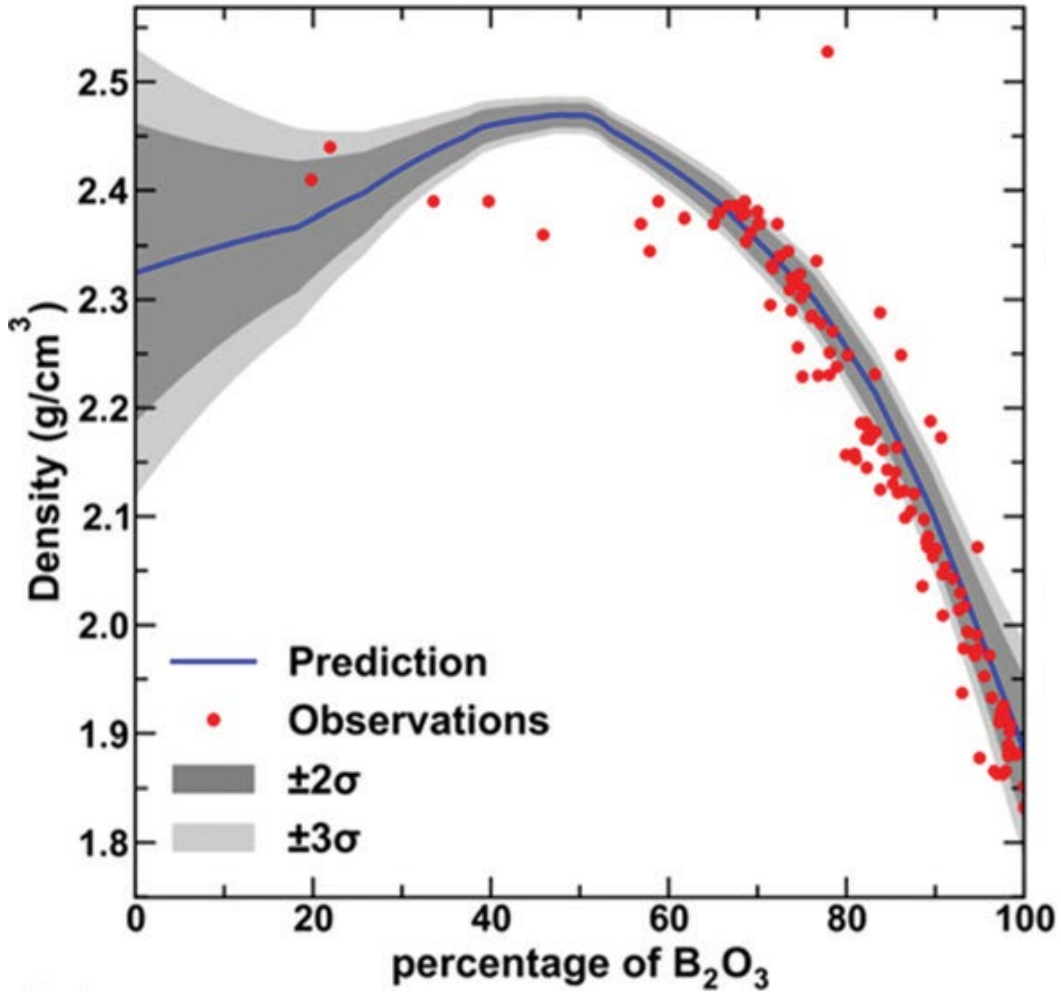


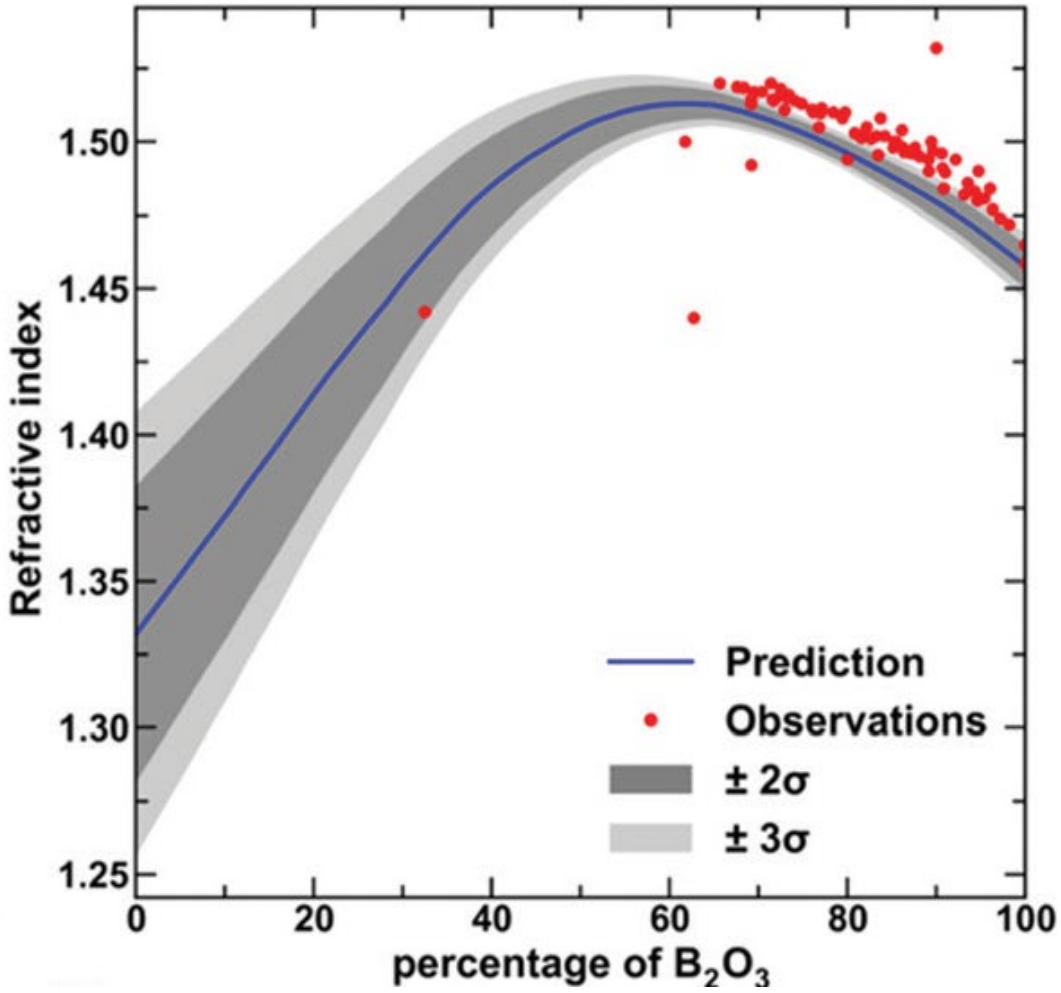


Anoop Krishnan, Mangalathu, Smedskjaer, Tandia, Burton & Bauchy, J. Non-Cryst. Solids 2018 https://arxiv.org/abs/1712.06018

Gaussian processes tell you where they are unsure

Uncertainty bands widen at low B_2O_3 content, where little data exists.

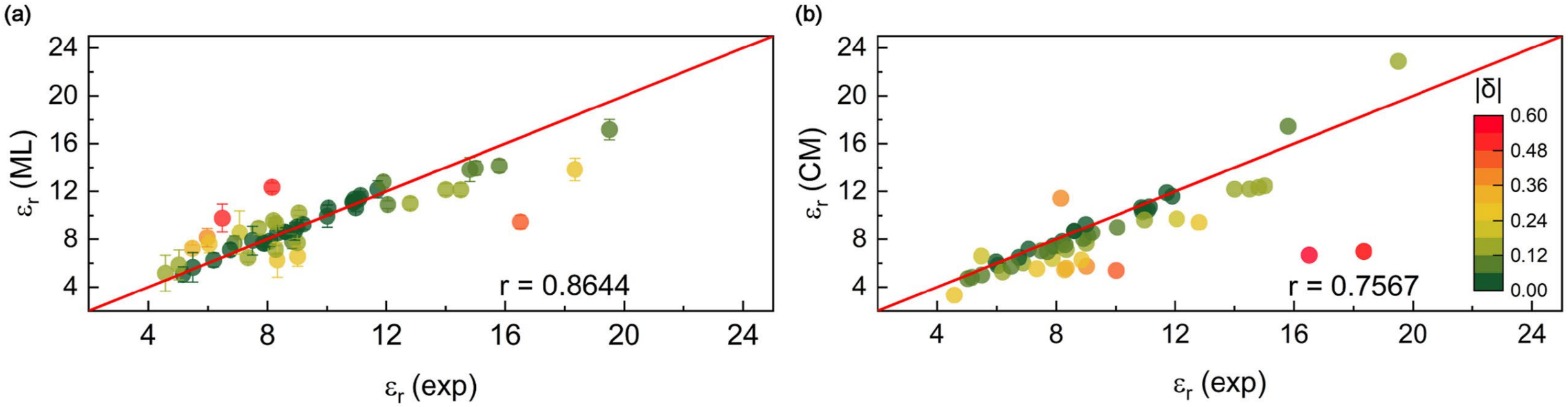




Bishnoi et al., Mater. Adv. 2021 https://doi.org/10.1039/d0ma00764a

ML beat the Clausius-Mossotti equation for microwave ceramics

Data set trained on 55,000 glasses and then tested on 254 single phase ceramics

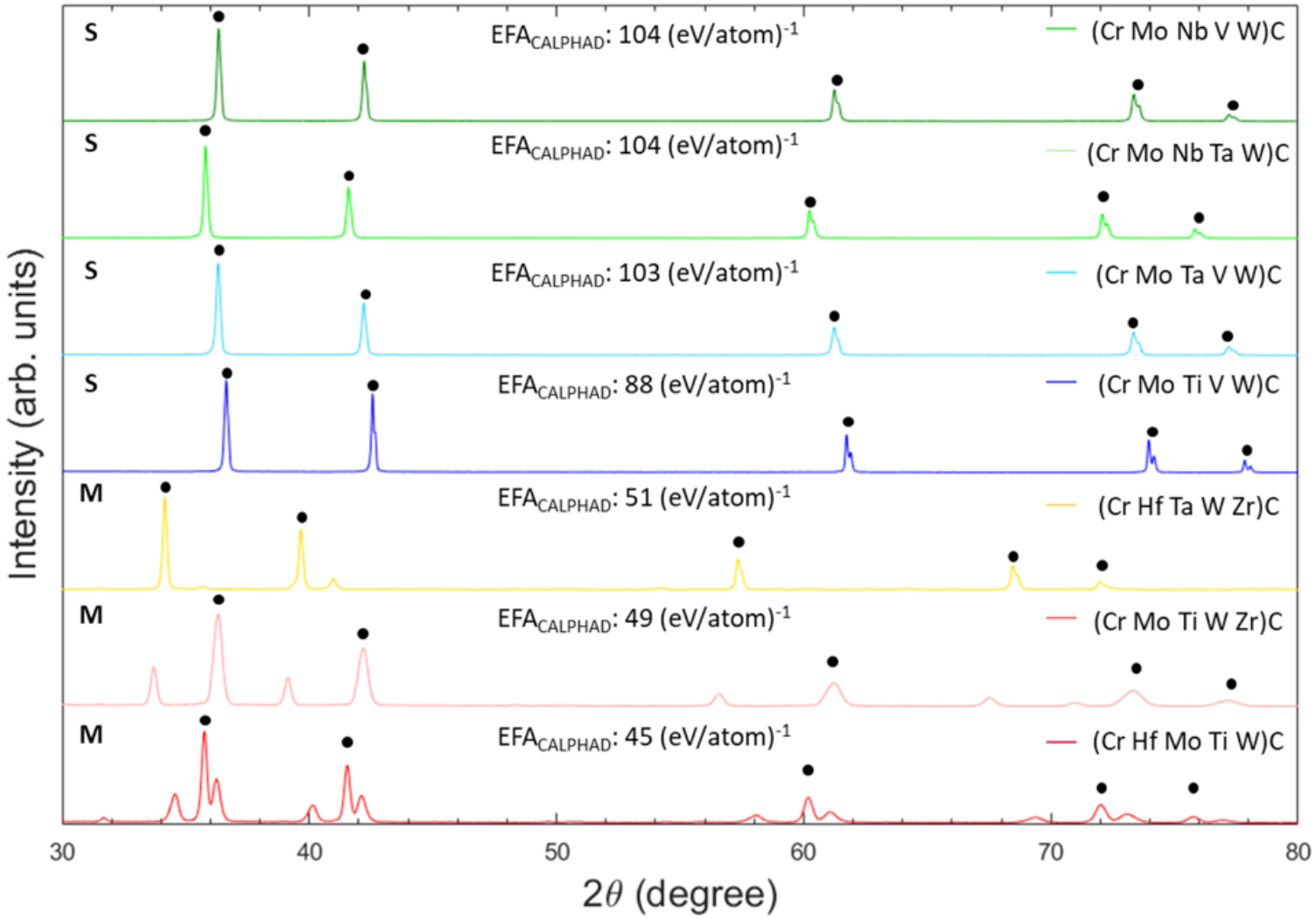


Qin et al., J. Materiomics 2021 https://doi.org/10.1016/j.jmat.2021.02.012

Classification examples:
Is it, or isn't it?

ML ranked new carbides, and the top picks formed single phases

4 of 7 synthesized compositions were single-phase: exactly the 4 with highest predicted EFA



Kaufmann et al., npj Comput. Mater. 2020 https://doi.org/10.1038/s41524-020-0317-6

A screening funnel cut 12,831 lithium solids to 21 candidates

Cheap filters first, a learned 'superionic or not' classifier next, experiments only at the end

12,831 Li-containing solids

Materials Project structures

Stability, electronic conductivity, cost

removes 92% of candidates

Logistic-regression classifier: superionic?

removes 93% of what is left

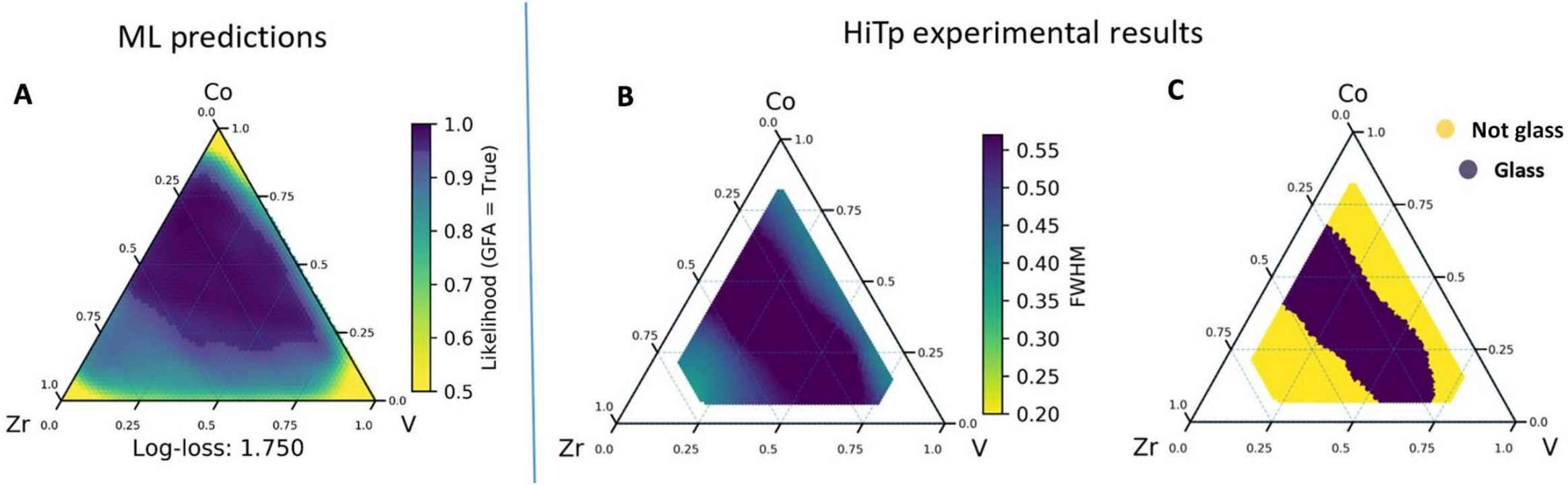
21 promising electrolytes

few had ever been tested

Sendek, Yang, Cubuk, Duerloo, Cui & Reed, Energy Environ. Sci. 2017 https://doi.org/10.1039/C6EE02697D

A glass-or-not classifier pointed experiments to a new glass system

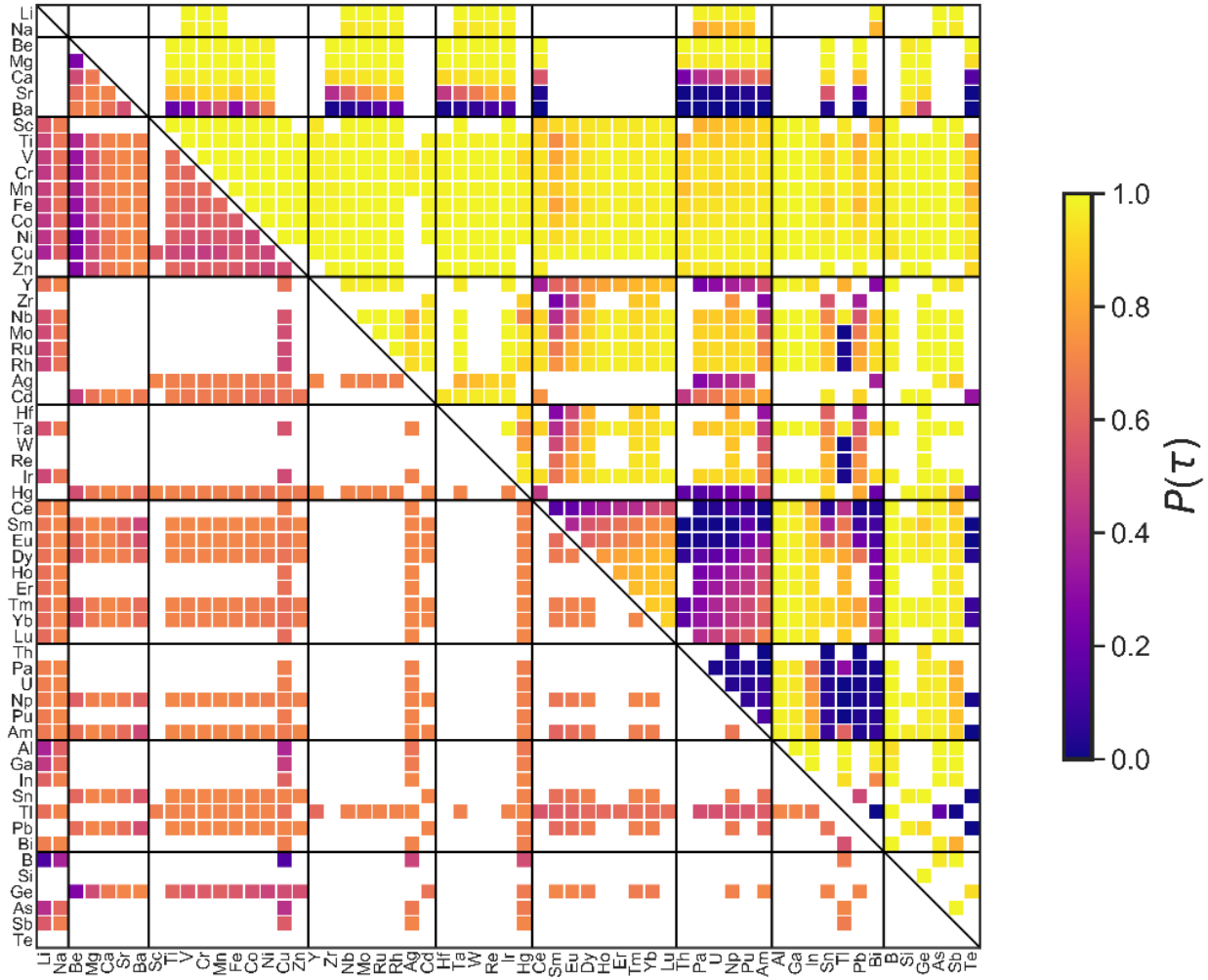
Mismatches were fed back to retrain the model, which then found two more new glass-forming ternaries



Ren et al., Sci. Adv. 2018 https://doi.org/10.1126/sciadv.aaq1566

A simple classifier can screen thousands of untested perovskites

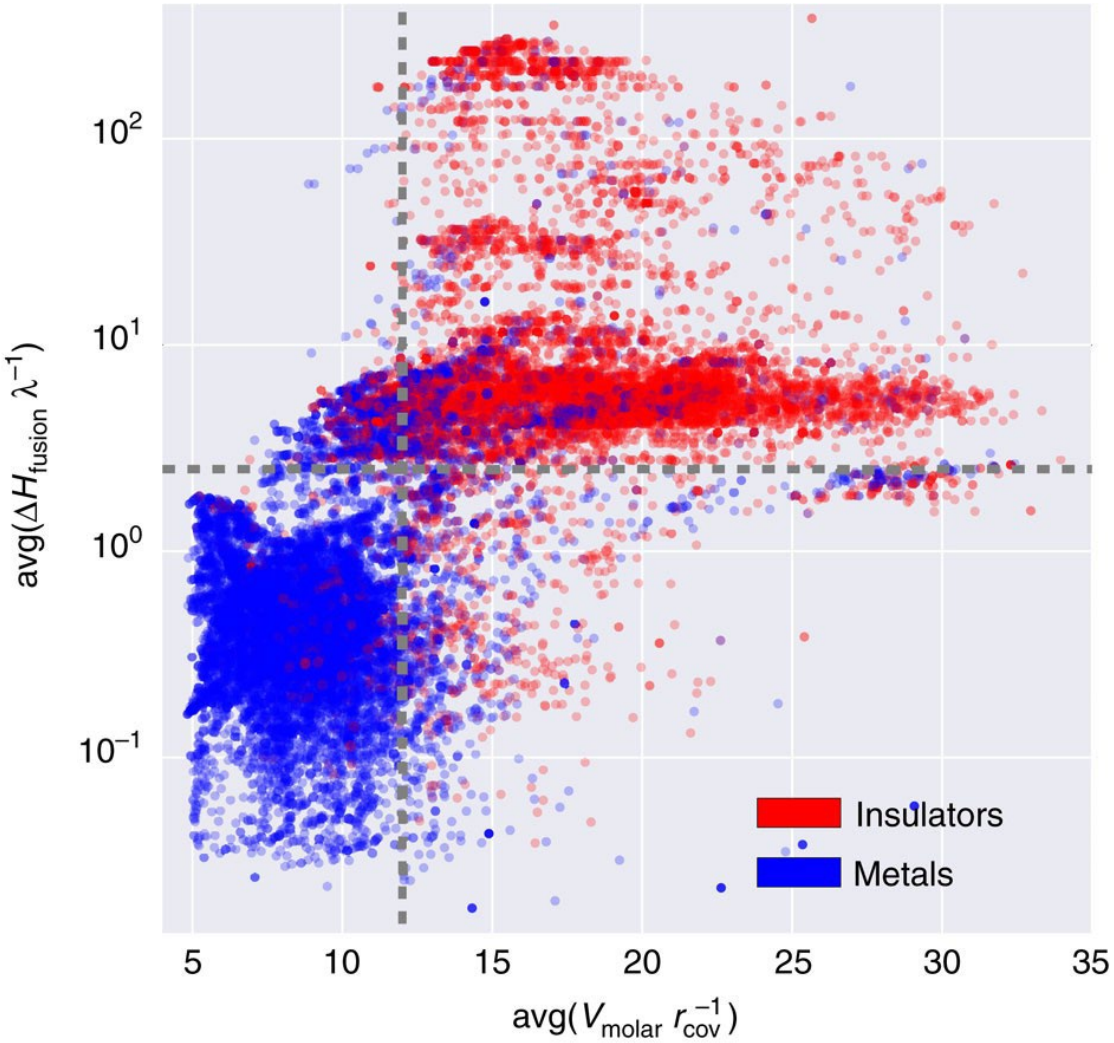
tau was used to rank 23,314 new candidate double perovskites



Bartel et al., Sci. Adv. 2019 https://doi.org/10.1126/sciadv.aav0693

Two descriptors already separate most metals from insulators

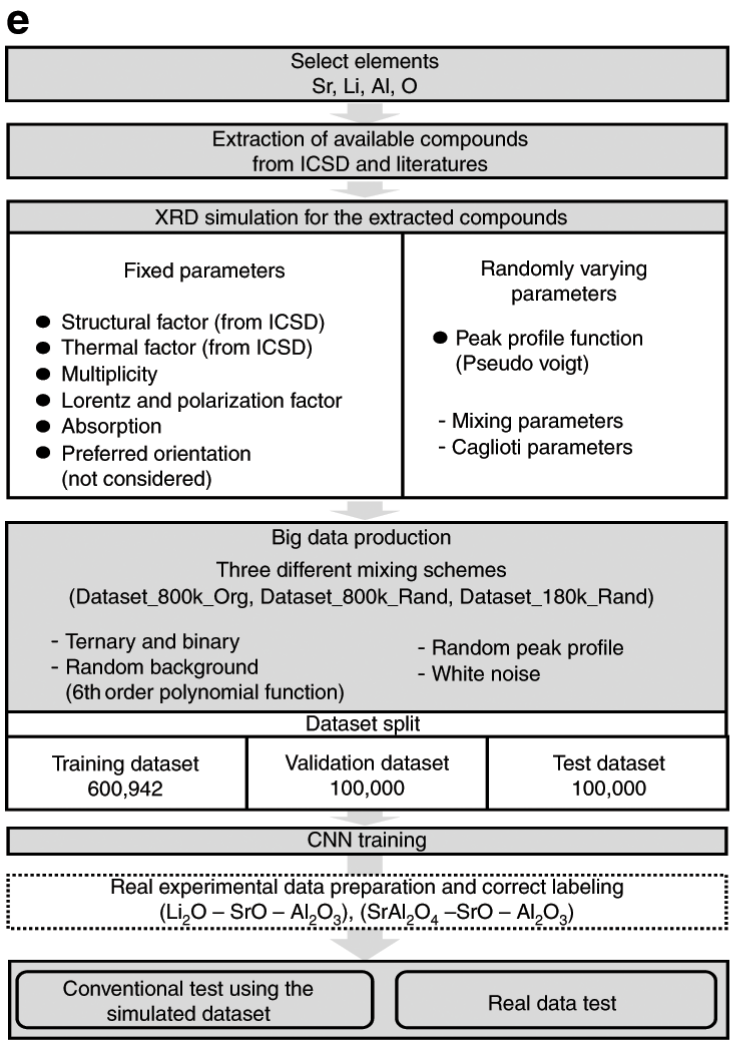
The full classifier reached AUC 0.98 and 93% overall accuracy

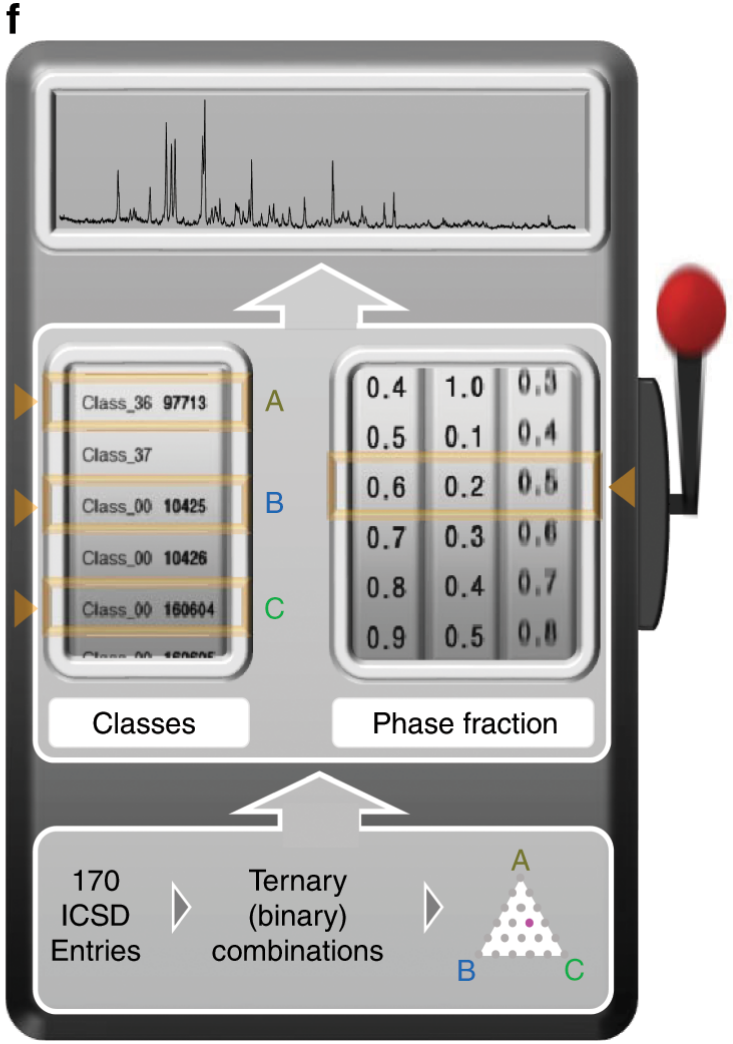


Isayev et al., Nat. Commun. 2017 https://doi.org/10.1038/ncomms15679

Simulated XRD patterns can train a CNN to identify real phases

1,785,405 synthetic patterns in total; near-100% phase ID on real experimental mixtures

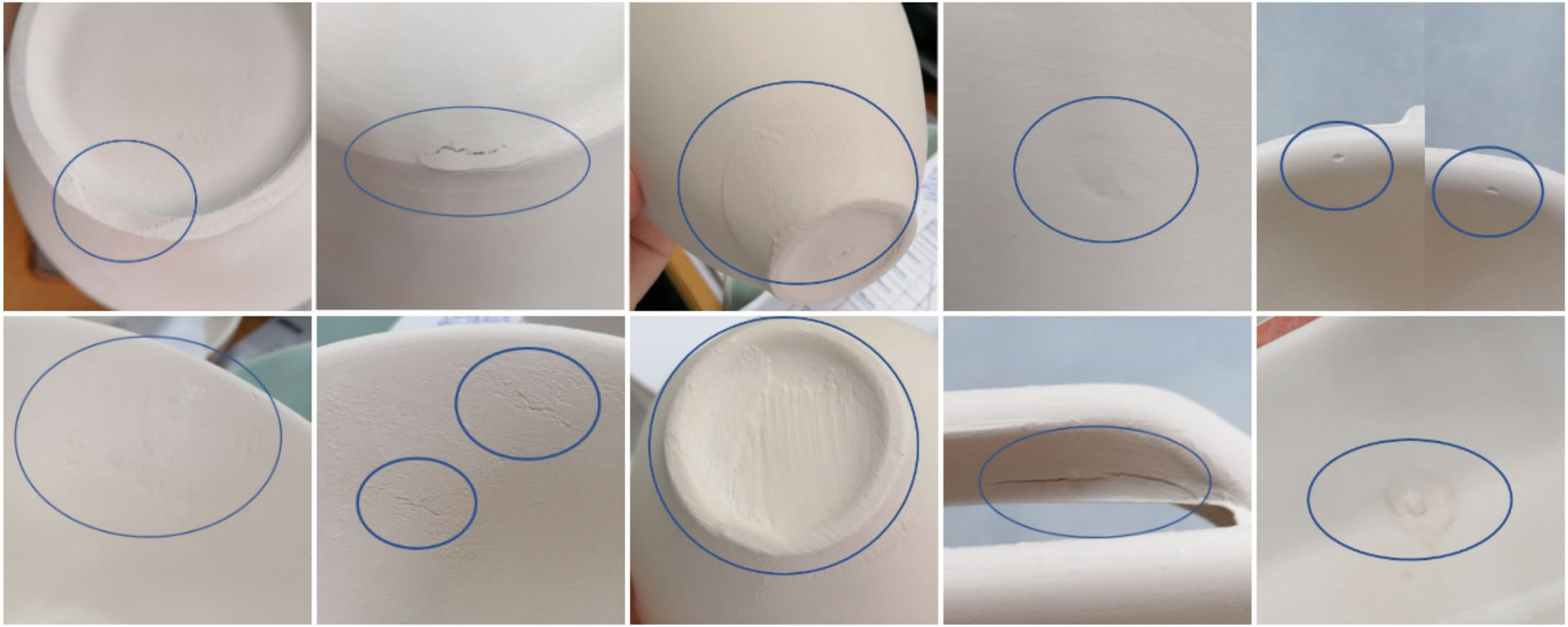




Lee et al., Nat. Commun. 2020 https://doi.org/10.1038/s41467-019-13749-3

CNNs can flag defective ceramic pieces on the production line

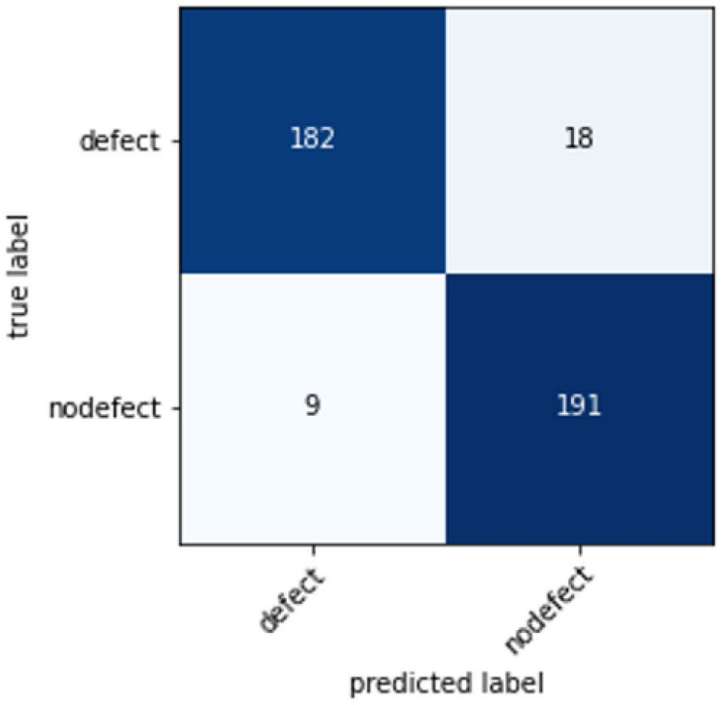
Reported accuracy 98.00% and F1-score 97.29% for the deployed system



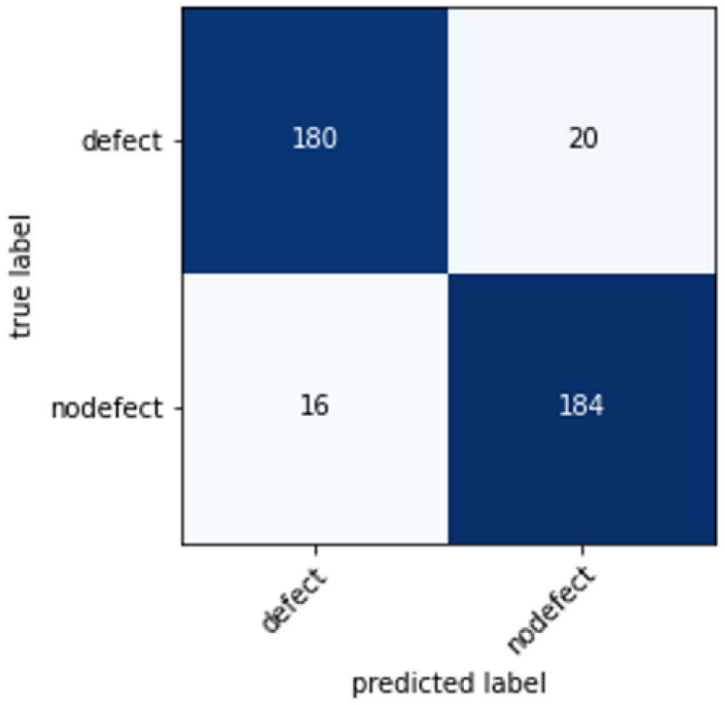
Cumbajin et al., Sensors 2024 https://doi.org/10.3390/s24010232

A confusion matrix shows which mistakes a defect classifier makes

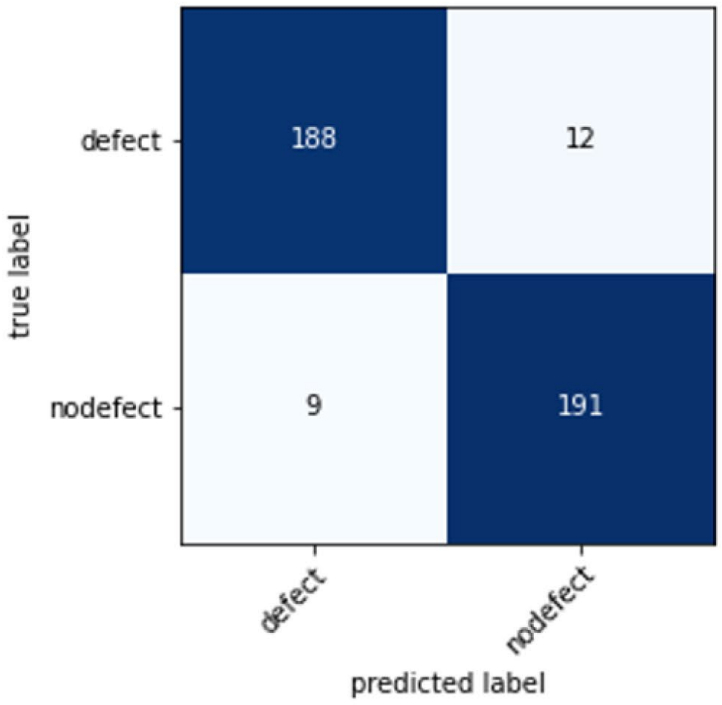
Fine-tuned model: 12 missed defects and 9 false alarms out of 400 images



(a) Train from scratch



(b) Transfer learning



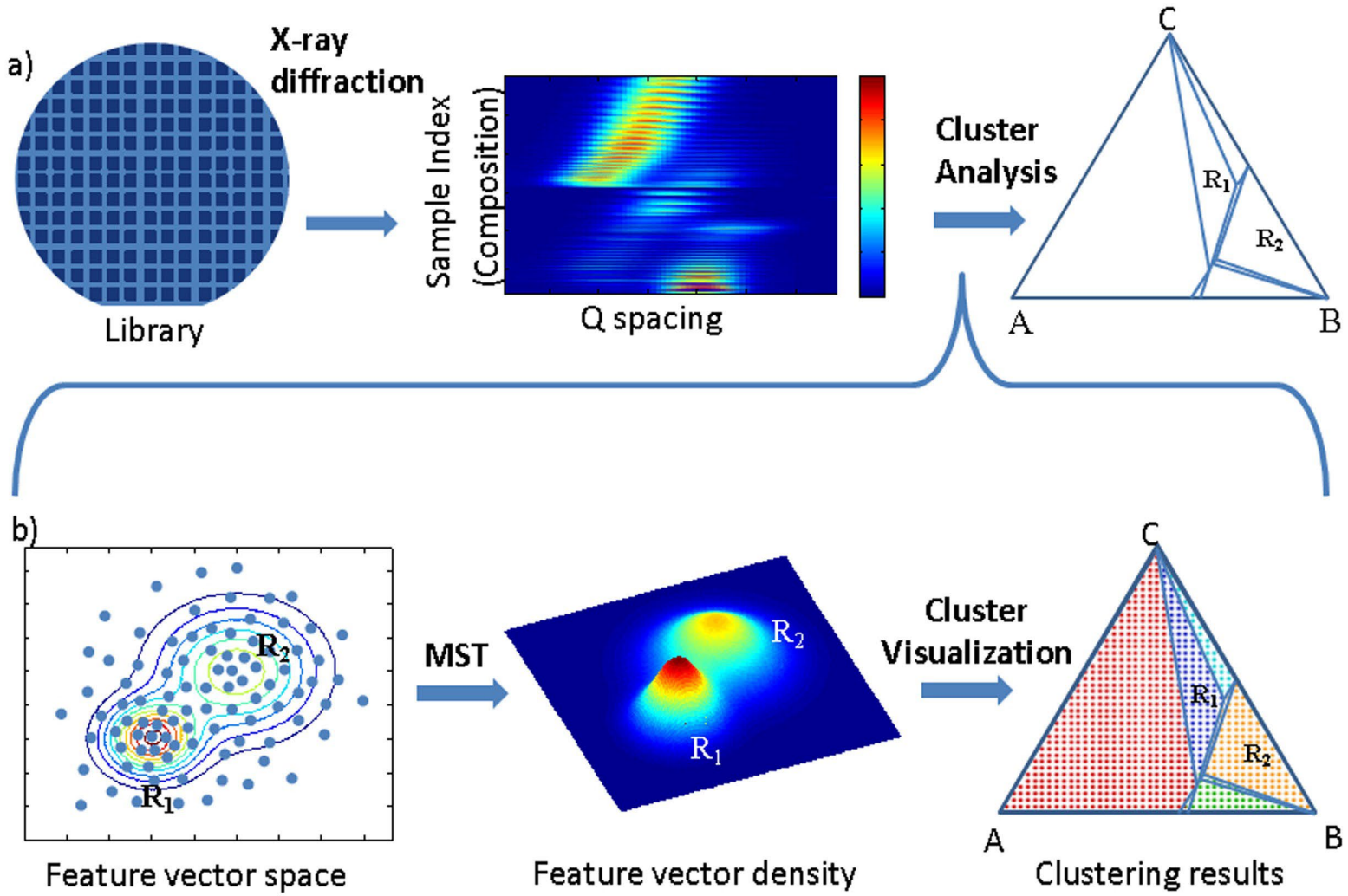
(c) Transfer learning with FT

Cumbajin et al., Sensors 2024 https://doi.org/10.3390/s24010232

Unsupervised learning:
What can we learn without labels?

Clustering diffraction patterns sketches a phase diagram automatically

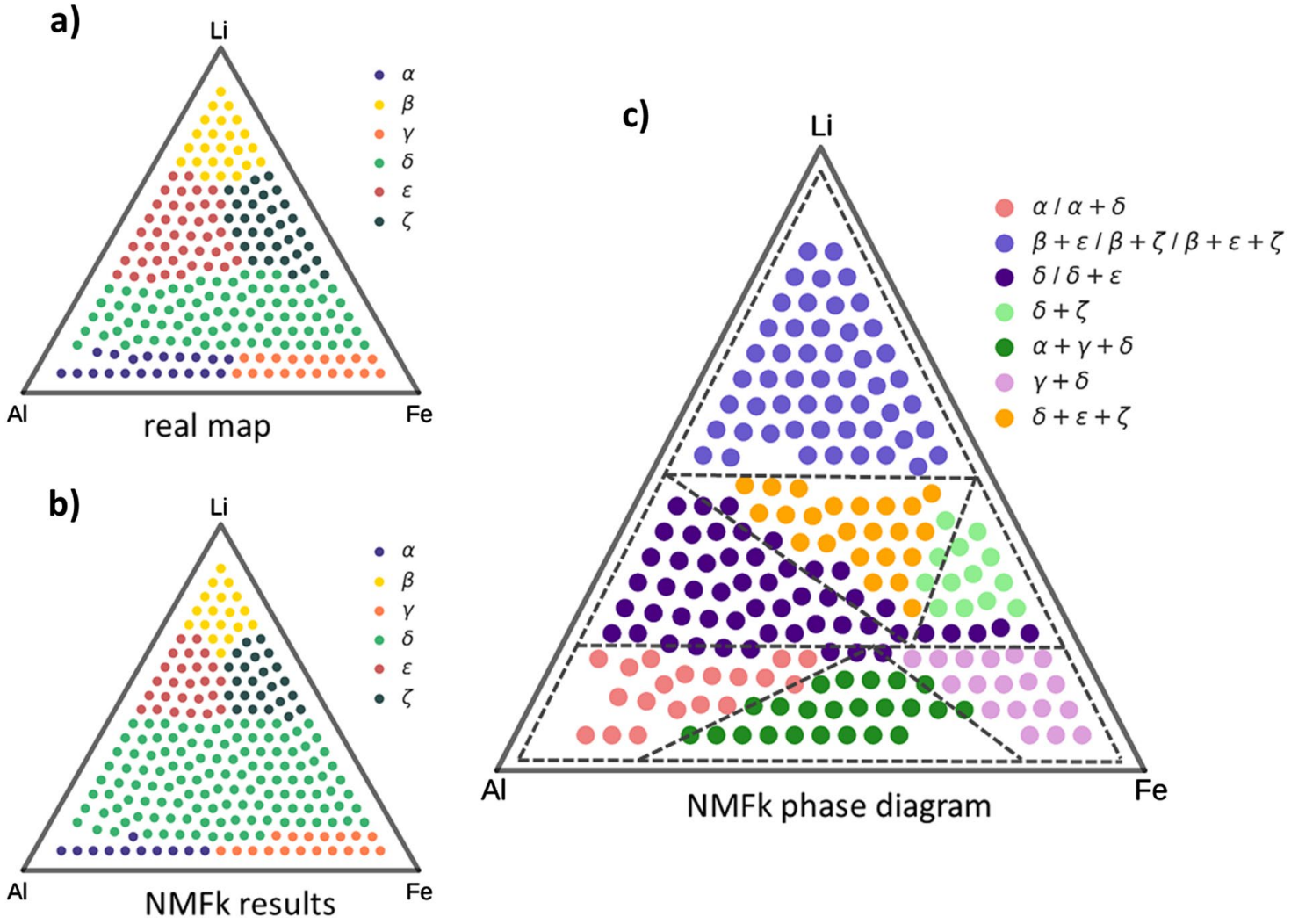
Ran on the fly at a synchrotron beamline to guide a magnet search



Kusne et al., Sci. Rep. 2014 https://doi.org/10.1038/srep06367

Unsupervised NMF recovers an oxide phase diagram without labels

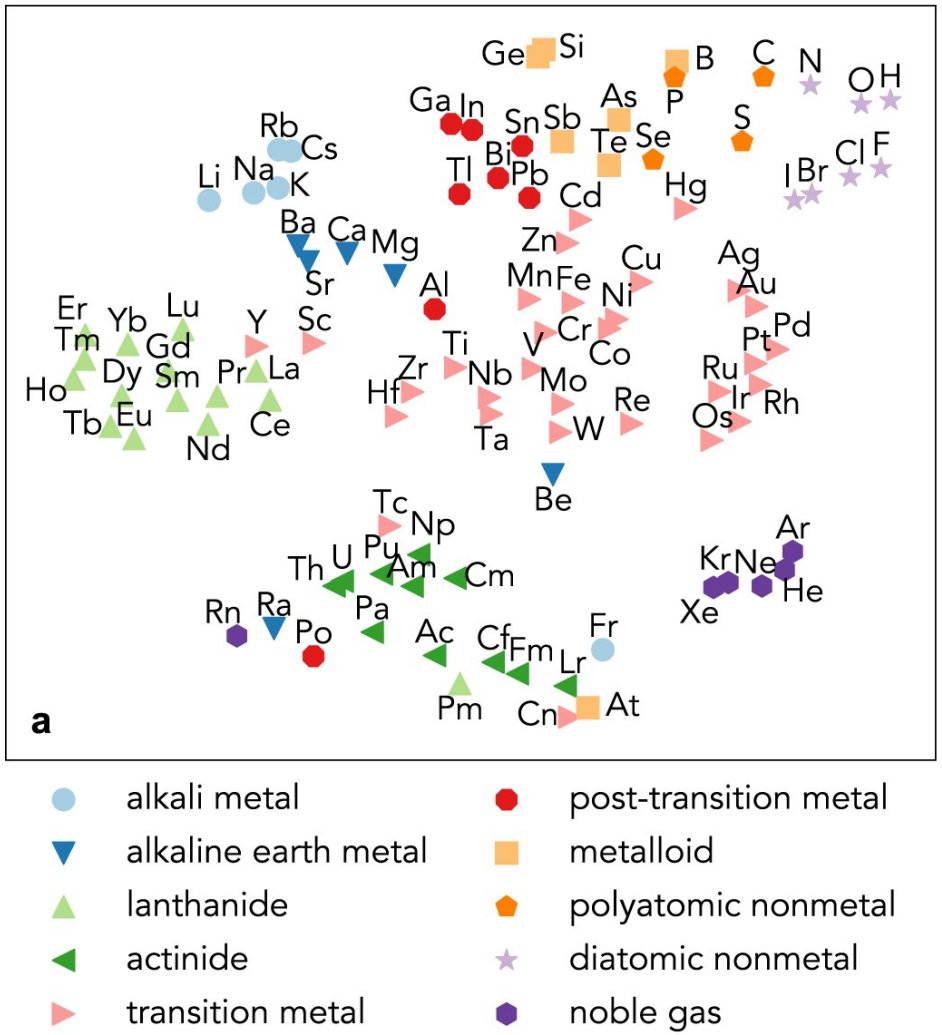
NMFk chooses the number of phases itself, using silhouette scores



Stanev et al., npj Comput. Mater. 2018 https://doi.org/10.1038/s41524-018-0099-2

Reading only abstracts, an algorithm rediscovered the periodic table

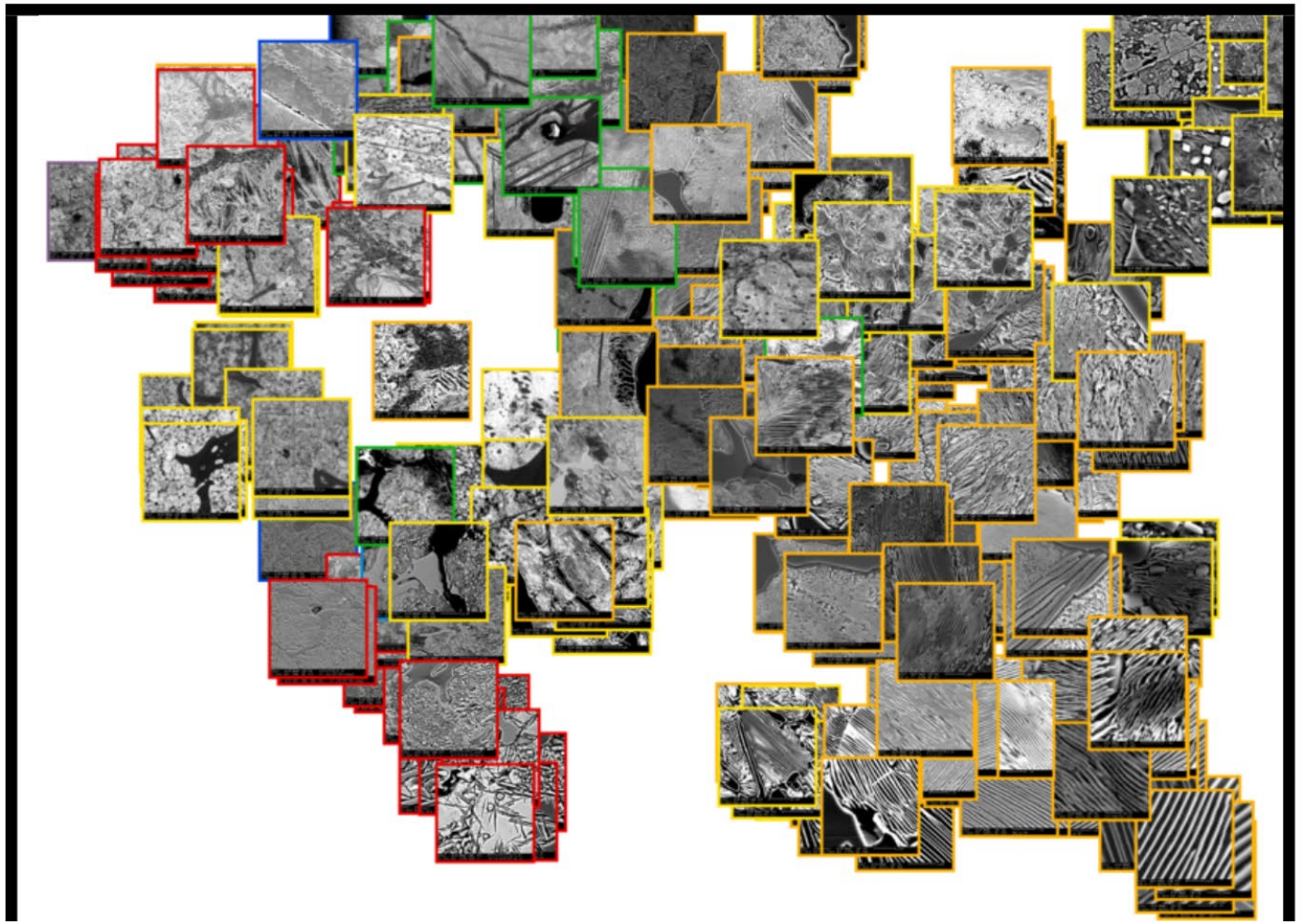
No chemical knowledge was given to the model, only text

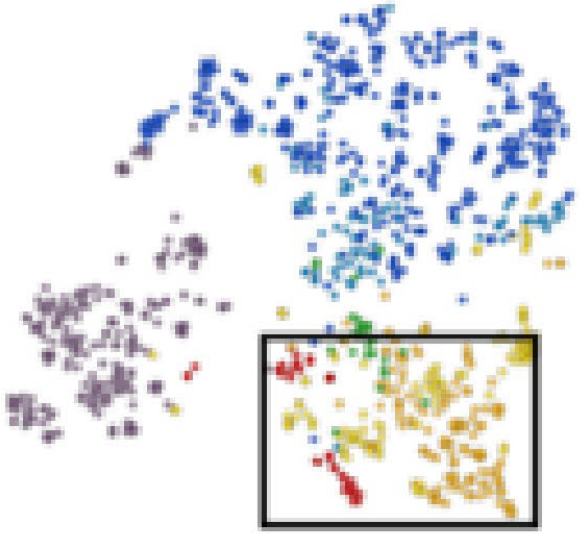


Tshitoyan et al., Nature 2019 https://doi.org/10.1038/s41586-019-1335-8

t-SNE groups similar micrographs together without any labels

Shown here for steels; the same approach works for SEM images of ceramic microstructures

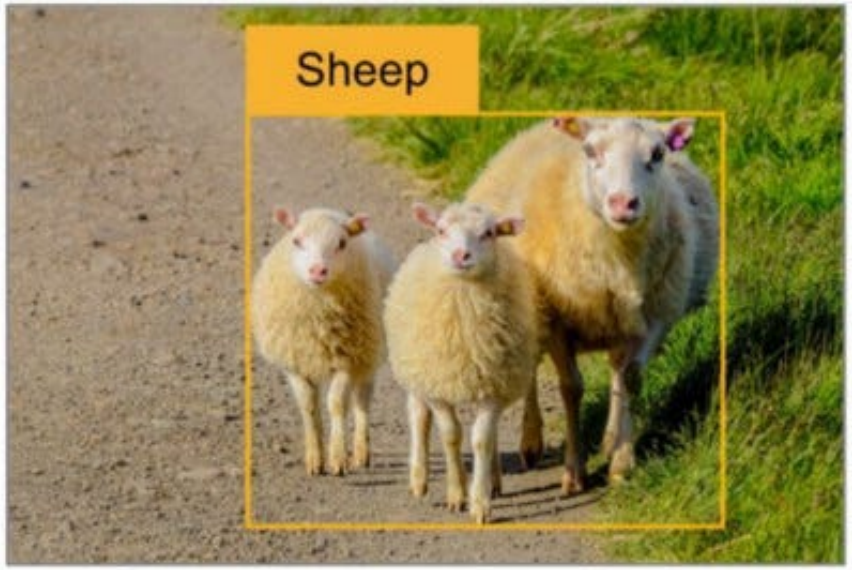




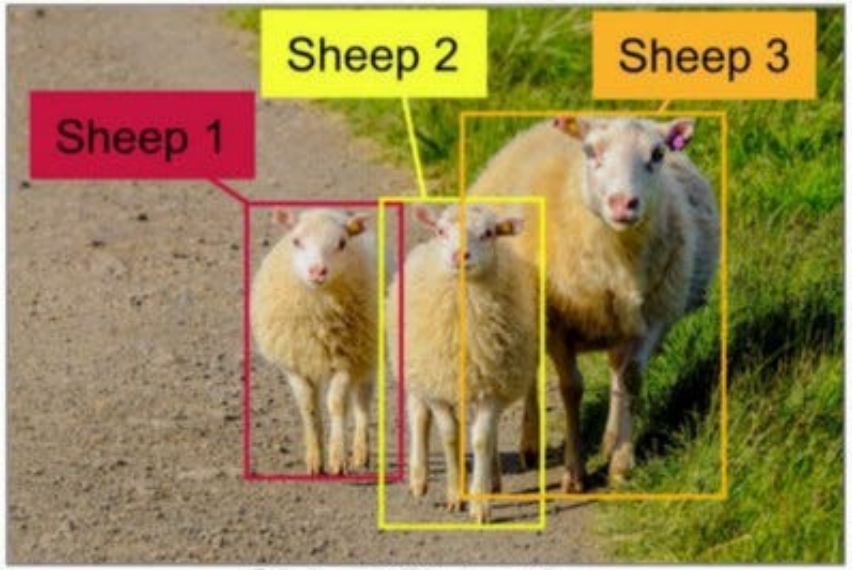
DeCost et al., Acta Mater. 2017 https://doi.org/10.1016/j.actamat.2017.05.014

Computer vision:
Quantifying micrographs

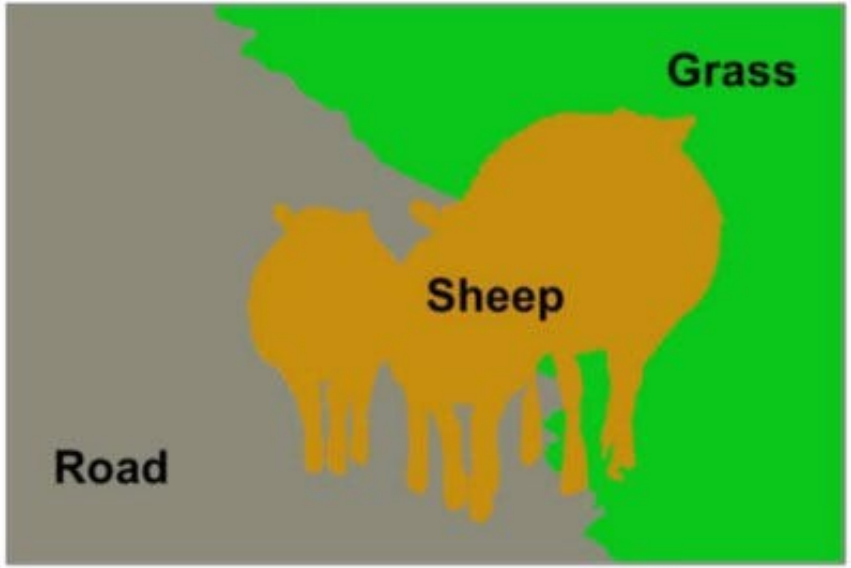
Recognition vs semantic segmentation vs object detection vs instance segmentation



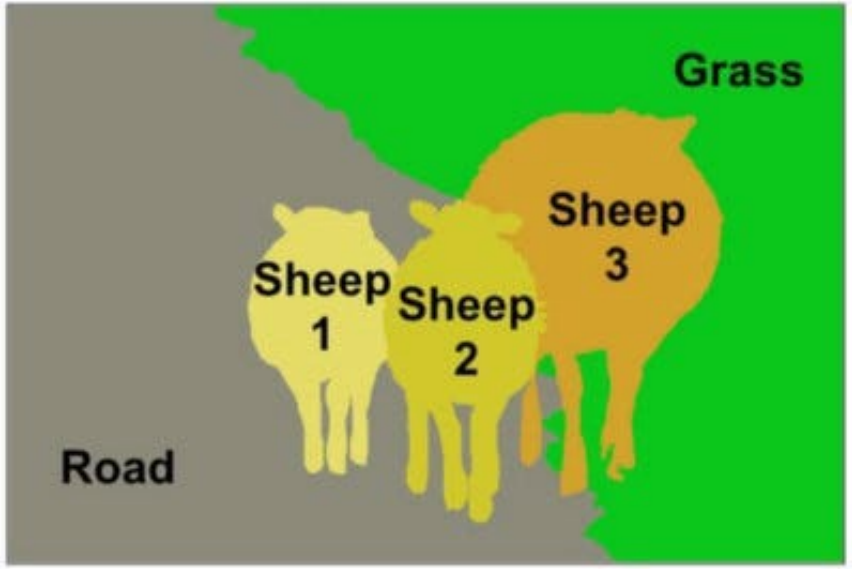
Classification + Localization



Object Detection



Semantic Segmentation



Instance Segmentation

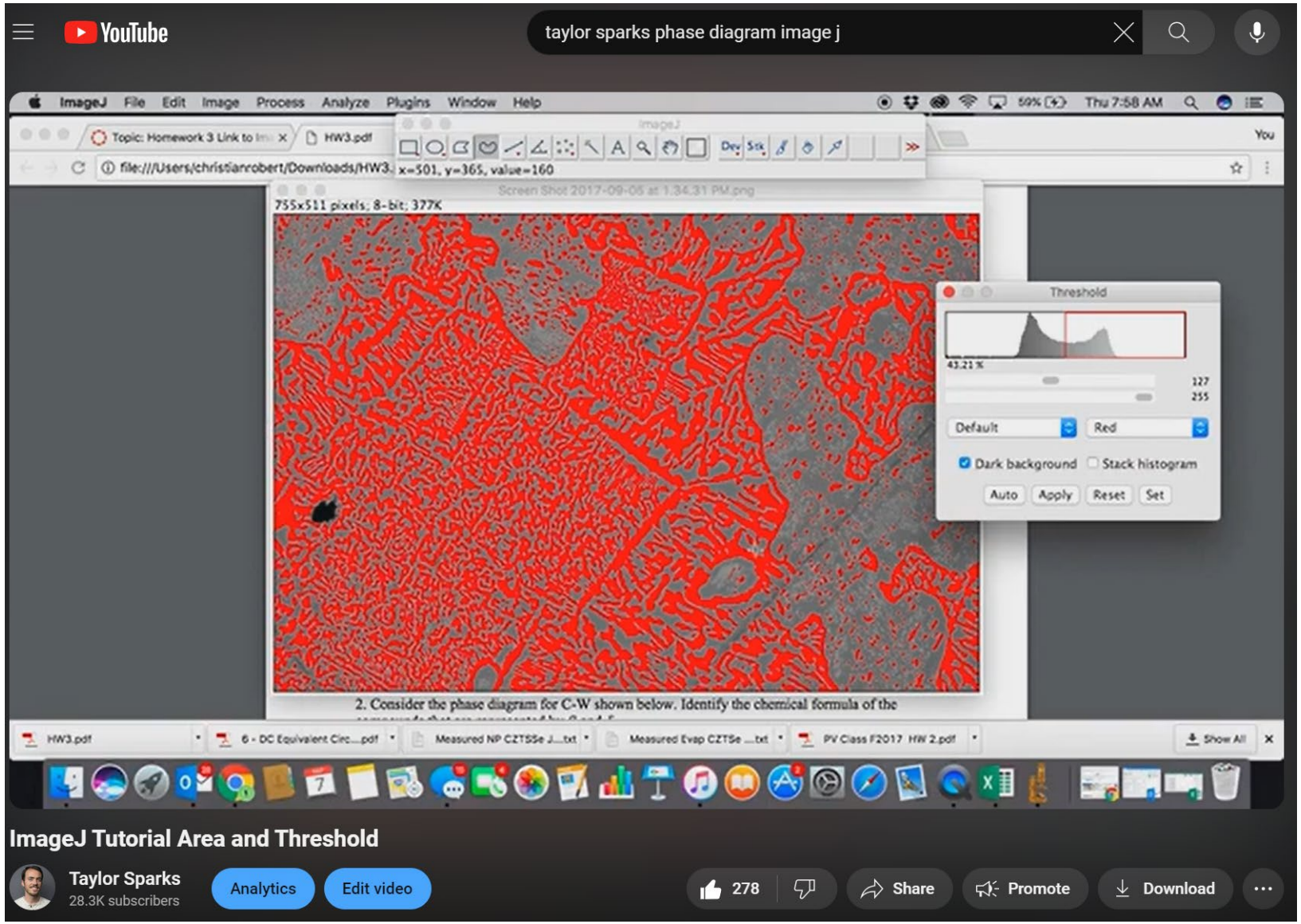
Thresholding is something you've probably come across before

Thresholding: The simplest form of segmentation, based on pixel intensity. The basic idea is to select a threshold value, and then all pixels above this value are considered part of one segment, while those below are considered another.

Math is simple for thresholding

$$I(x,y) \rightarrow S(x,y)$$

$$S(x,y) = \begin{cases} 1 \; if \; I(x,y) > T \\ 0 \quad otherwise \end{cases}$$

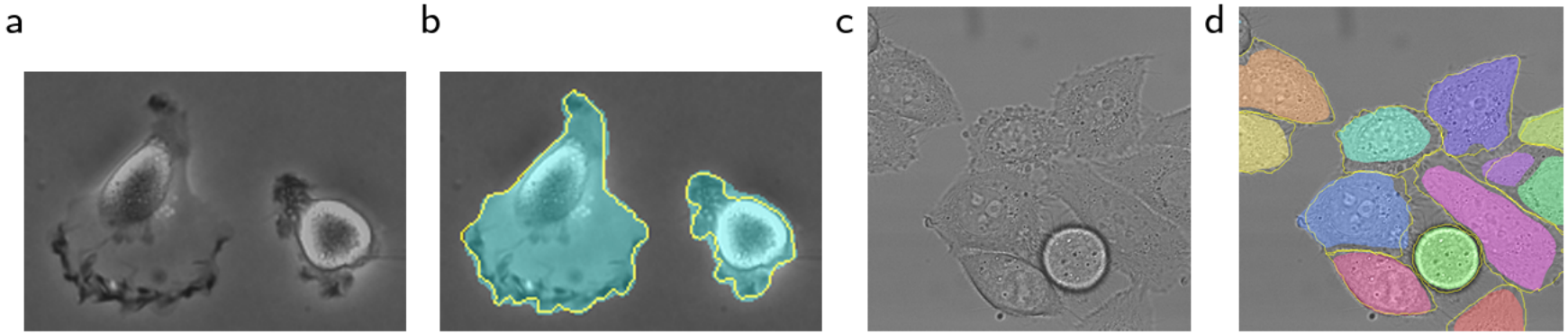


U-Net has become a very popular tool for image segmentation

U-Net: Convolutional Networks for Biomedical Image Segmentation

Olaf Ronneberger, Philipp Fischer, and Thomas Brox

Computer Science Department and BIOSS Centre for Biological Signalling Studies,
University of Freiburg, Germany
ronneber@informatik.uni-freiburg.de,
WWW home page: http://lmb.informatik.uni-freiburg.de/



U-net: Convolutional networks for biomedical image segmentation

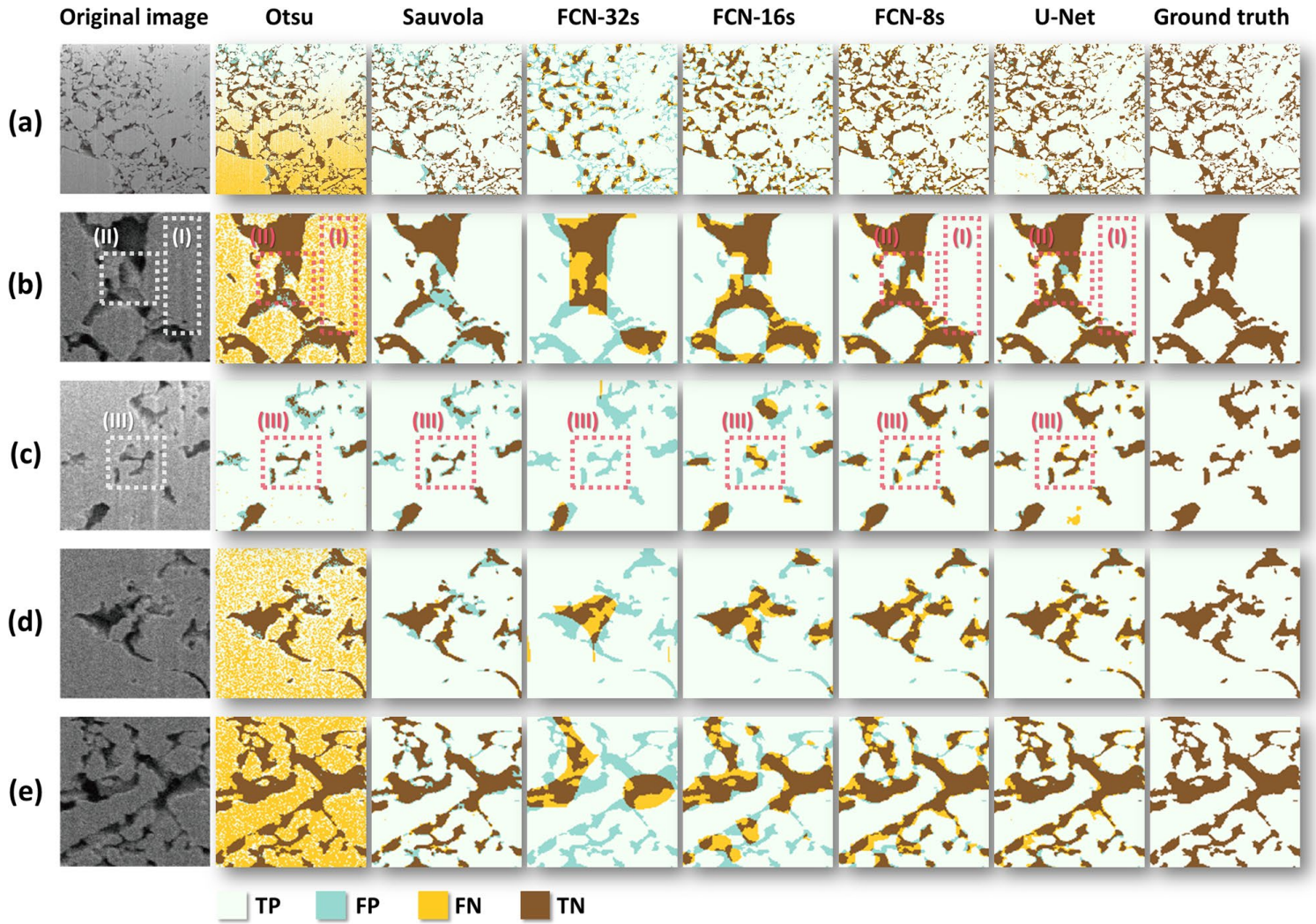
O **Ronneberger**, P Fischer, T Brox - … , Munich, Germany, October 5-9, 2015 …, 2015 - Springer

There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies …

☆ Save Cite Cited by 106165 Related articles All 30 versions

Thresholding breaks on real SEM; a U-Net nearly matches the expert

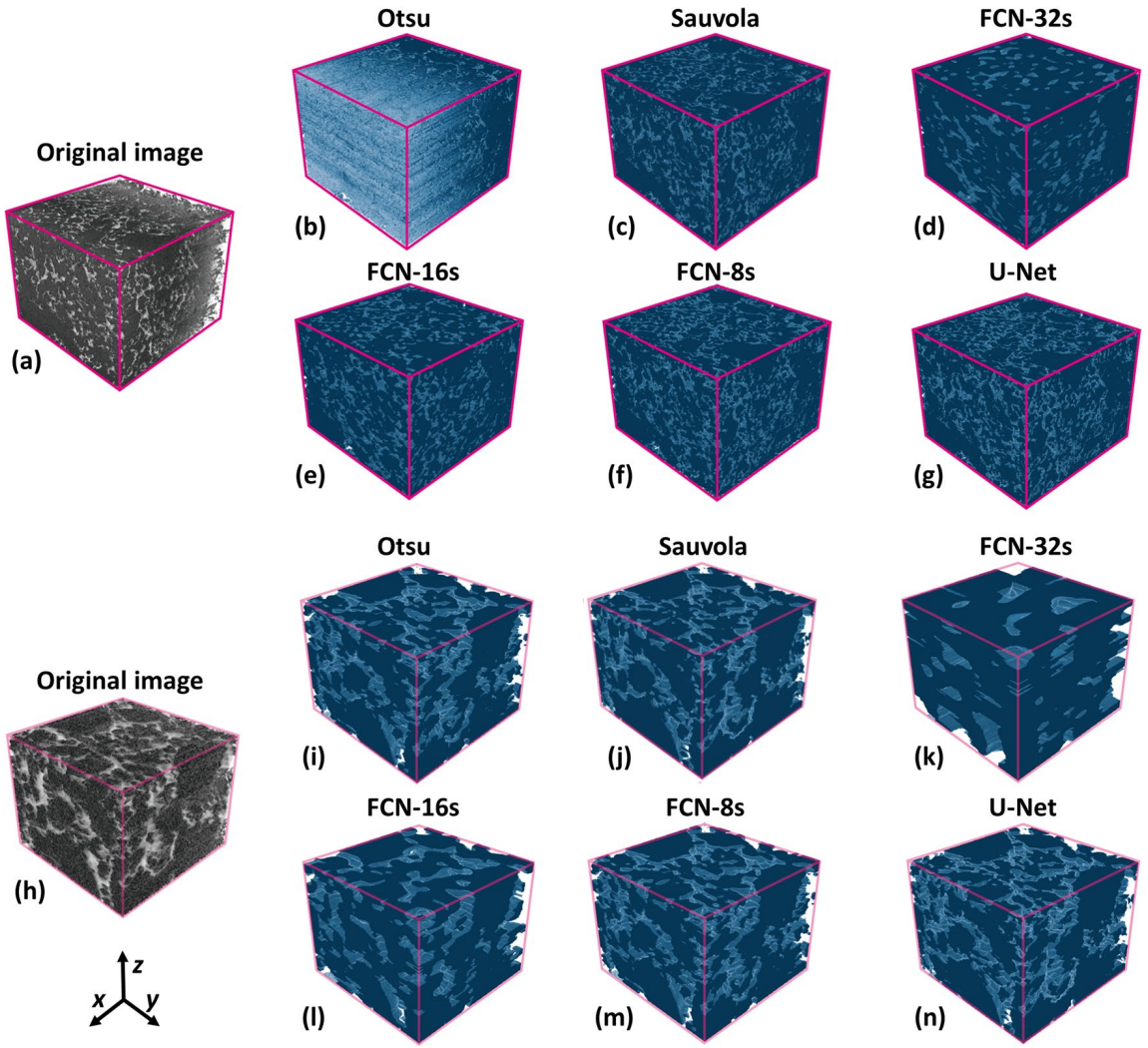
U-Net reached 94.6% intersection-over-union against hand-segmented images.



Hirabayashi et al., npj Comput. Mater. 10, 46 (2024) https://doi.org/10.1038/s41524-024-01226-5

A trained network turns a FIB-SEM stack into a 3D microstructure

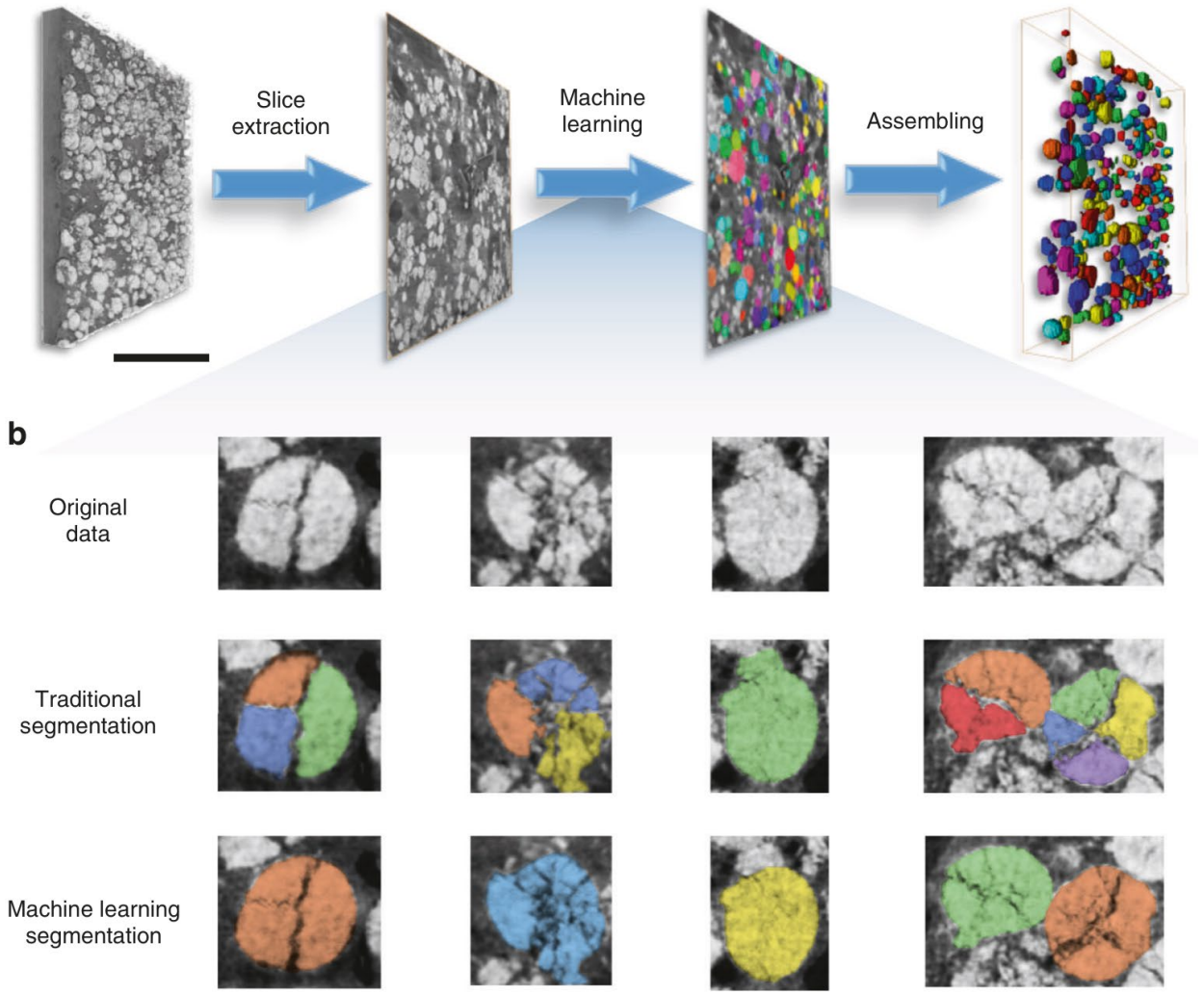
Giga-voxel volumes at 20 nm voxel size were segmented in a few minutes.



Hirabayashi et al., npj Comput. Mater. 10, 46 (2024) https://doi.org/10.1038/s41524-024-01226-5

Mask R-CNN segmented 650+ battery-cathode particles in 3D

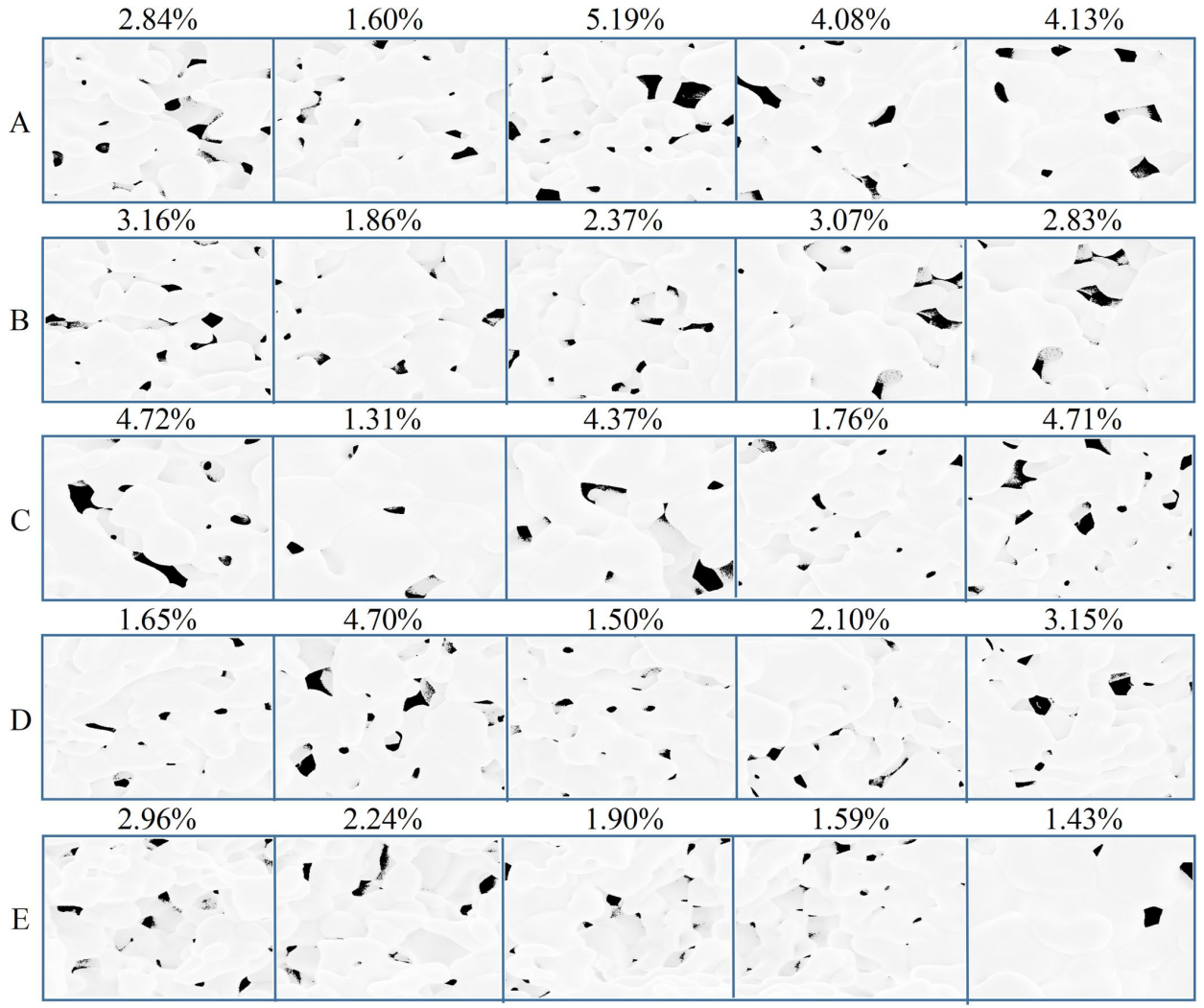
Particle-by-particle statistics: detachment from the carbon/binder grows with charging rate



Jiang et al., Nat. Commun. 2020 (CC BY 4.0) https://doi.org/10.1038/s41467-020-16233-5

Segment the pores and you get a porosity map, tile by tile

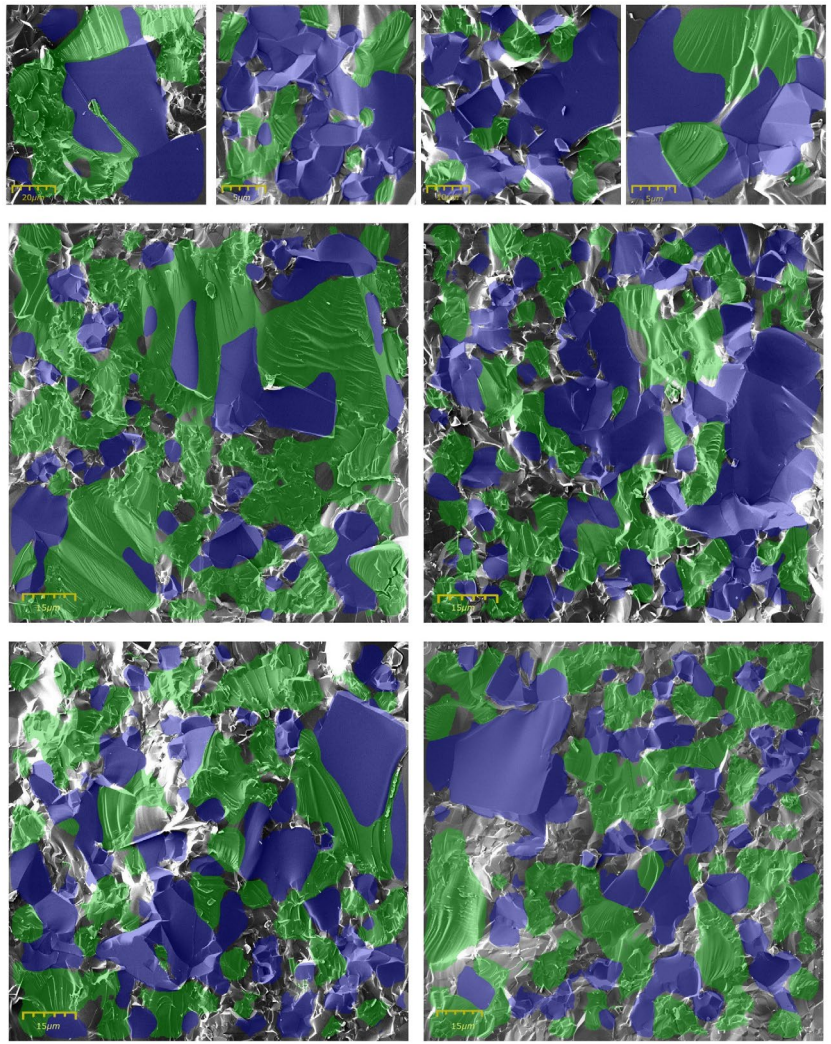
PSTNet's average relative porosity error on Al_2O_3 was 6.36%, vs 9.12% for U-Net (ImageJ as reference).



Xia et al., PLoS ONE 19, e0306385 (2024) https://doi.org/10.1371/journal.pone.0306385

A CNN maps intergranular vs transgranular fracture in ceramics

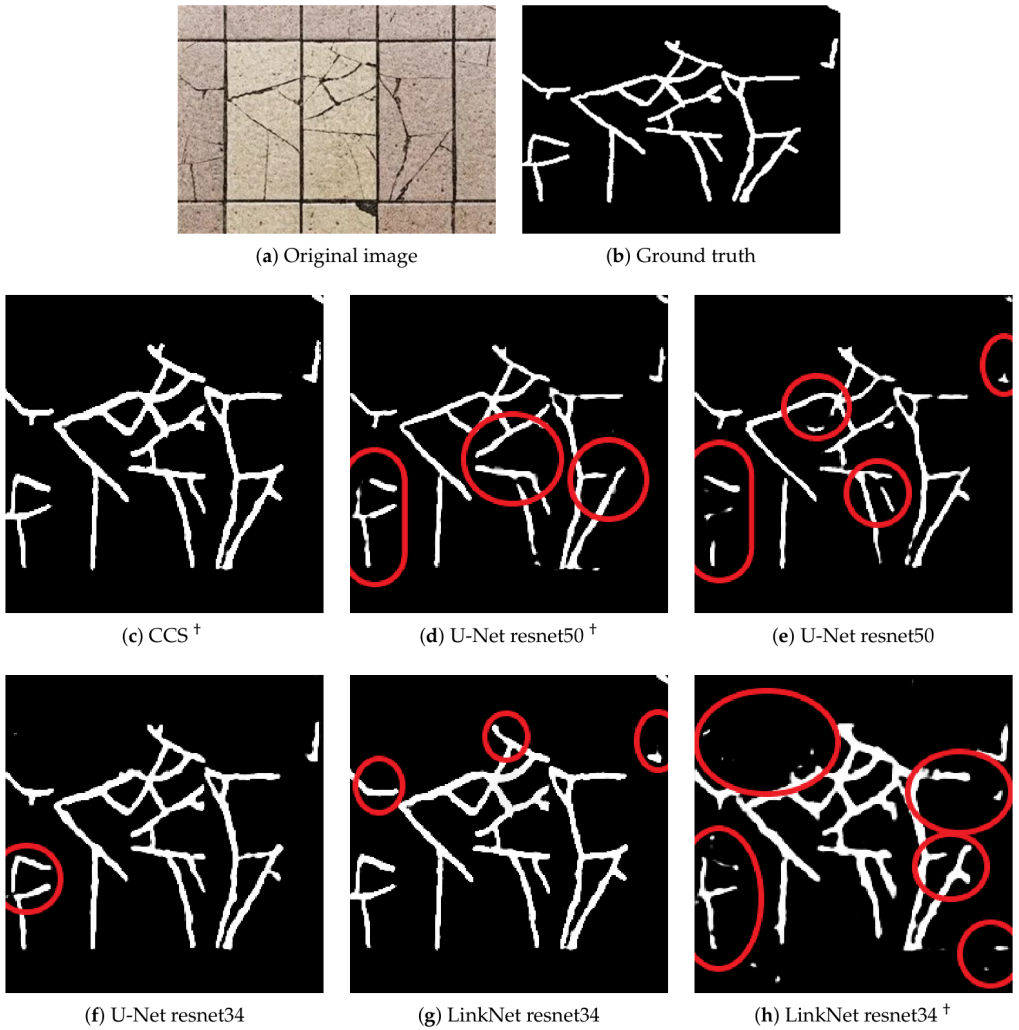
Mean IoU of 91.1% (93% intergranular, 87.4% transgranular) once background pixels were excluded.



Tsopanidis, Herrero Moreno & Osovski, Eng. Fract. Mech. 231, 106992 (2020); arXiv:1908.02242 https://doi.org/10.1016/j.engfracmech.2020.106992

Crack masks for ceramic tiles, straight from photos

The custom CCS model reached IoU 0.865 vs about 0.67-0.70 for standard U-Net/LinkNet backbones.



(a) Original image (b) Ground truth

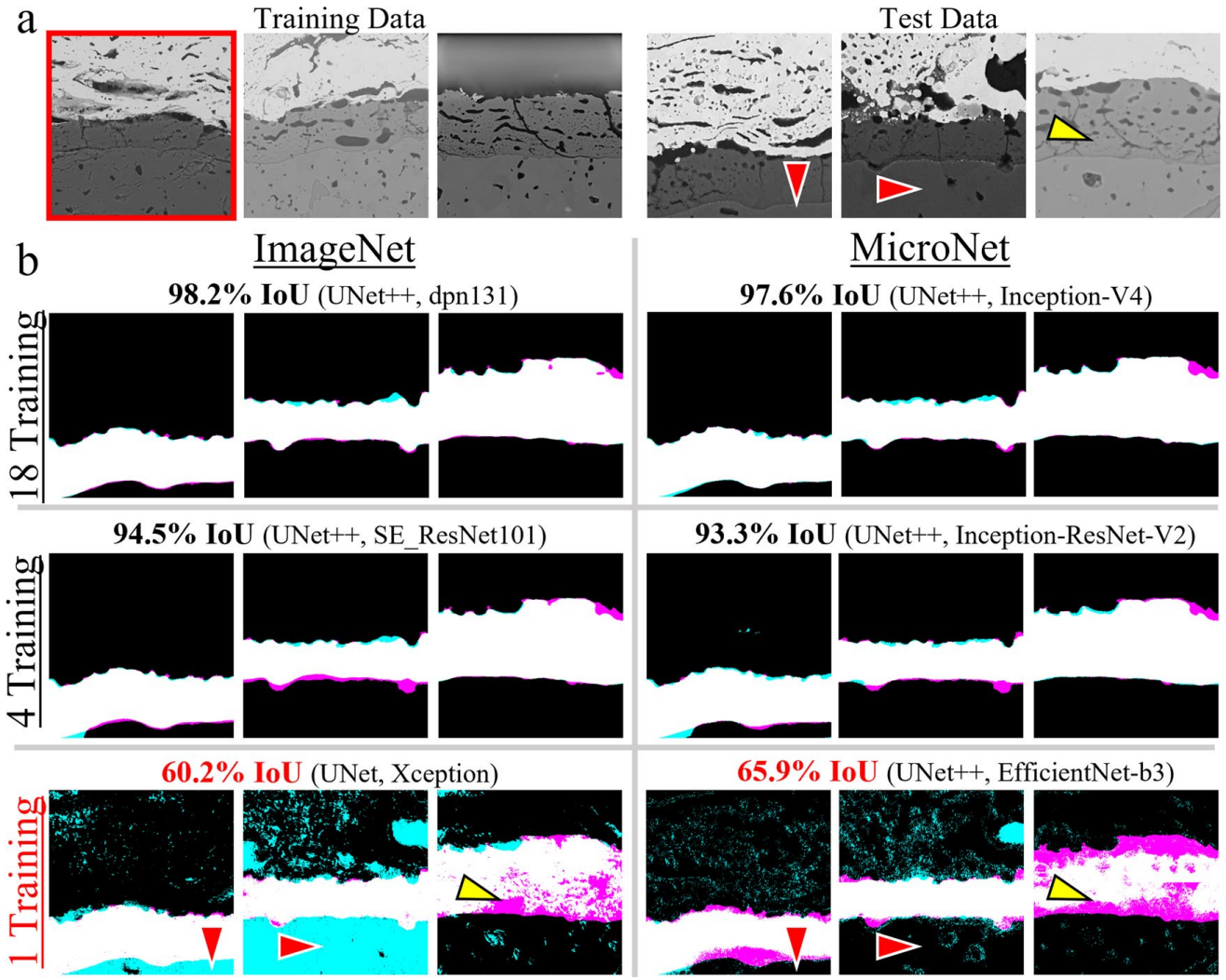
(c) CCS † (d) U-Net resnet50 † (e) U-Net resnet50

(f) U-Net resnet34 (g) LinkNet resnet34 (h) LinkNet resnet34 †

Junior et al., Appl. Sci. 11, 6017 (2021) https://doi.org/10.3390/app11136017

Pre-training on micrographs helps when you only have one label

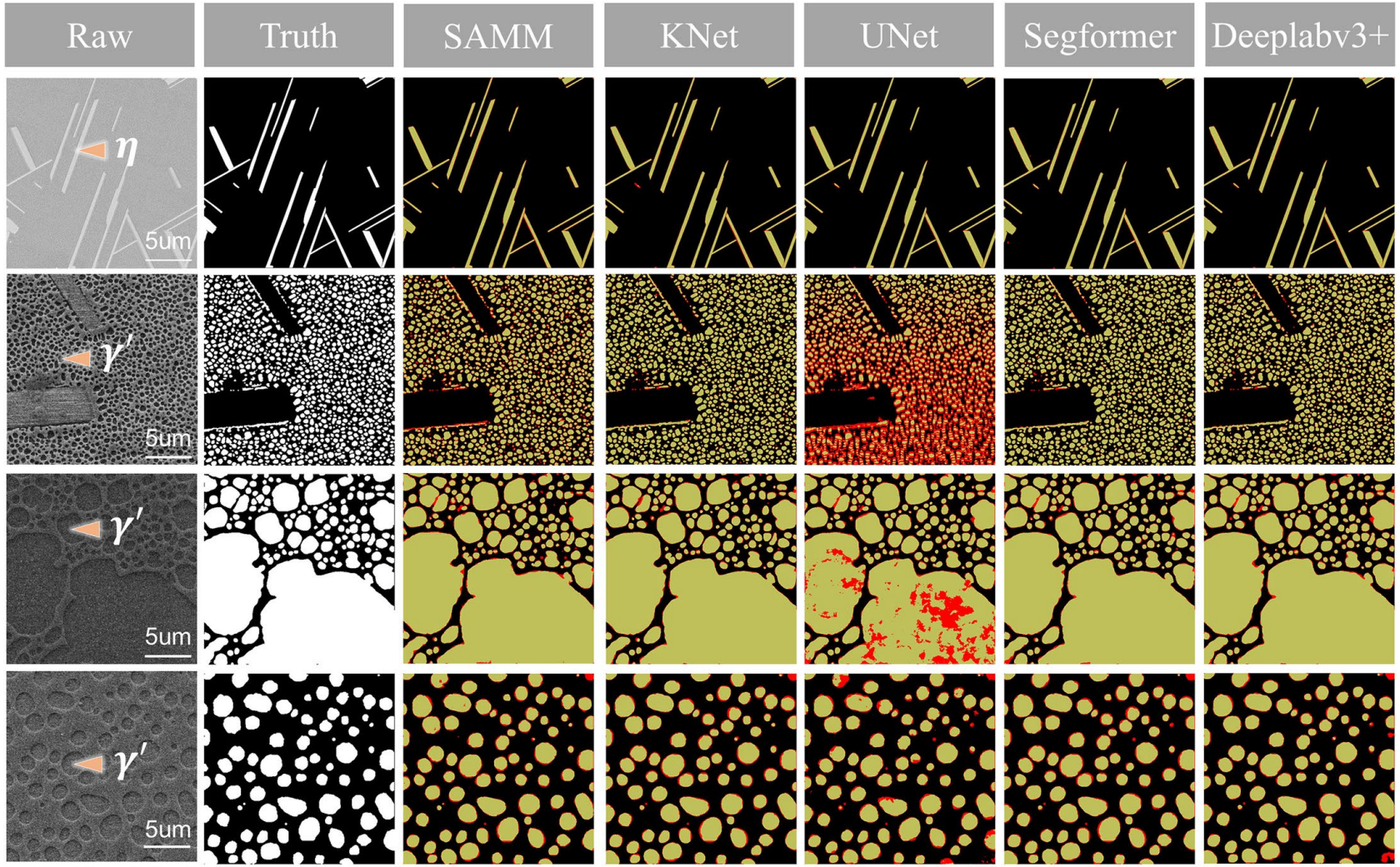
With one training image: 65.9% IoU (MicroNet) vs 60.2% (ImageNet); with 18 images both are about 98%.



Stuckner, Harder & Smith, npj Comput. Mater. 8, 200 (2022) https://doi.org/10.1038/s41524-022-00878-5

Foundation models like Segment Anything are now tuned for micrographs

SAMM reports up to 97.41% mIoU on unseen datasets (shown here on superalloy micrographs)



Tu et al., Adv. Powder Mater. 5, 100404 (2026) https://doi.org/10.1016/j.apmate.2026.100404

Generative models:
Towards inverse design

Generative models are increasing at an incredible rate!

2014

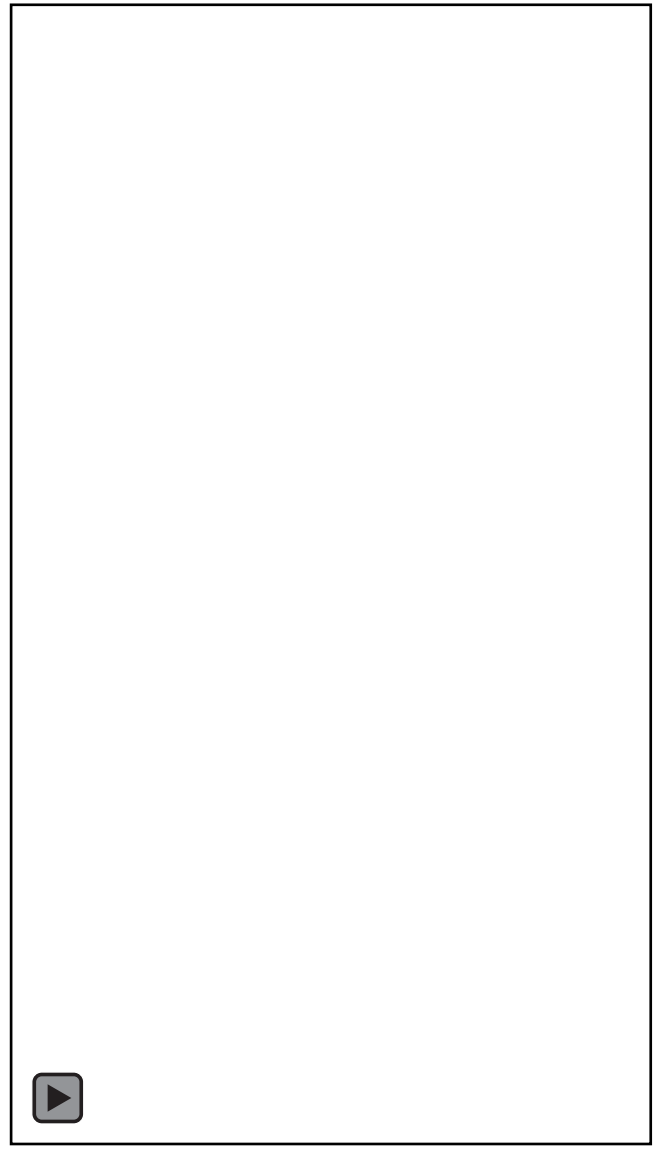
2015

2016

2017

2018

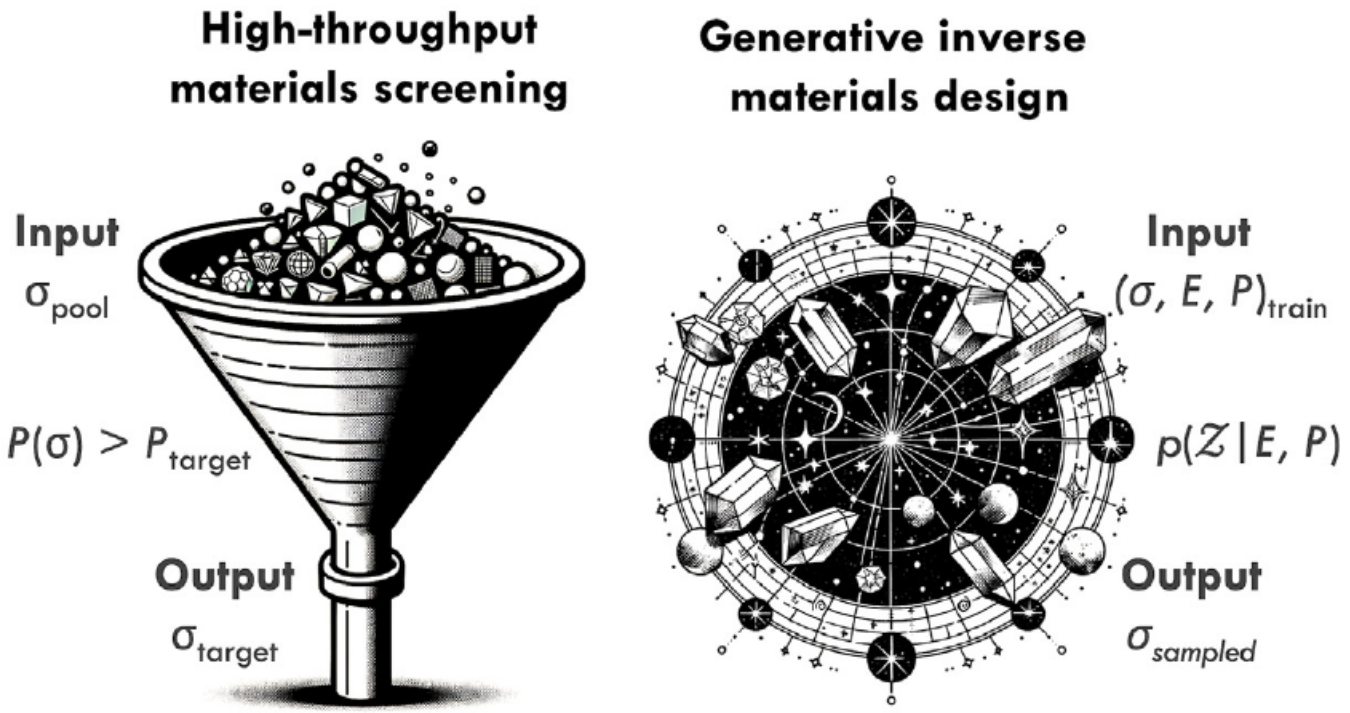
2021



2023

https://twitter.com/goodfellow_ian/status/1084973596236144640

Generative discovery is different than high throughput screening

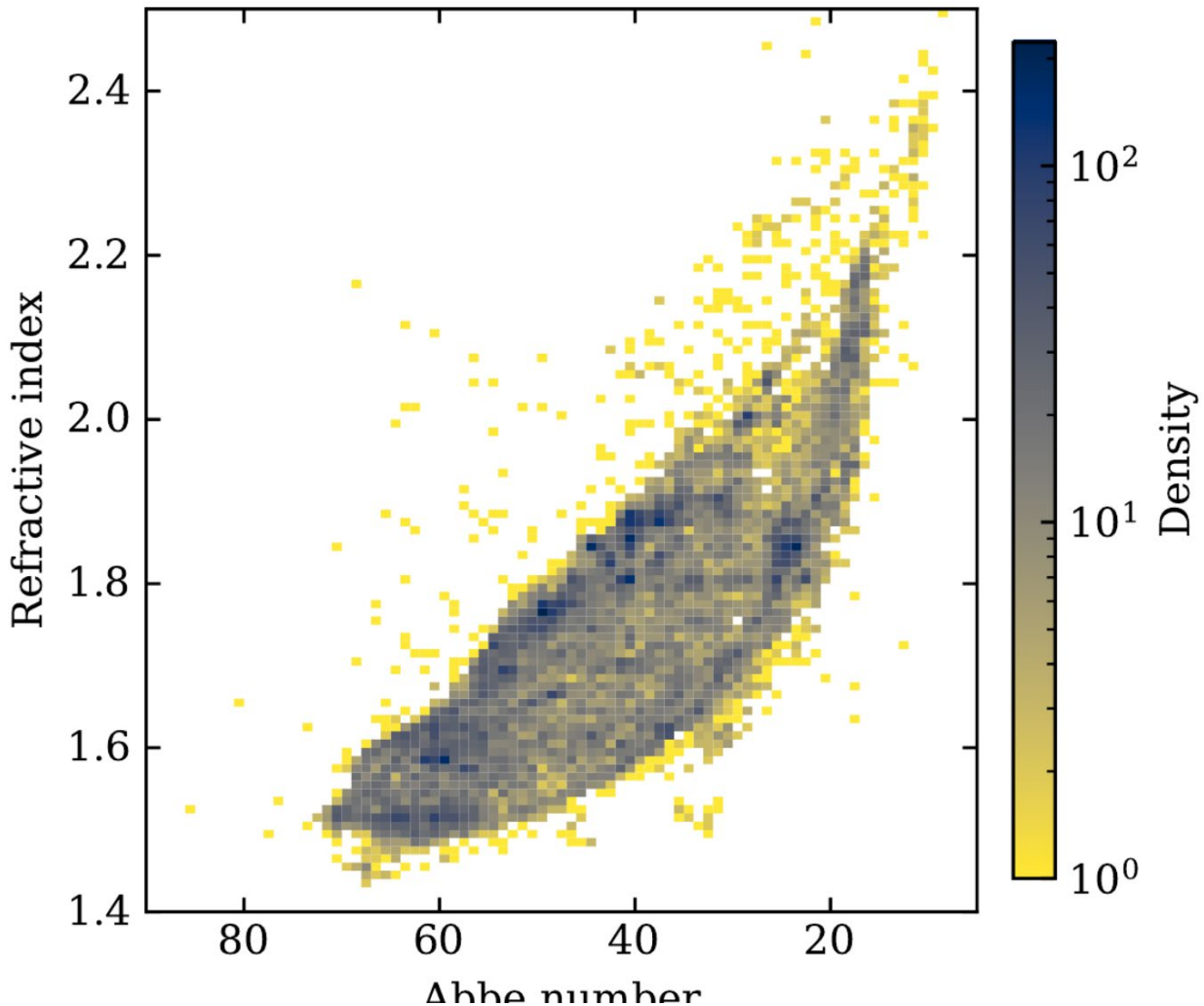


Input	Pool of candidates	Diverse training data
Model	Property calculator	Probability distributions
Selection	Hand-built filters	Probability sampling

Park, Li, Walsh “Has Generative artificial intelligence solved inverse design?” Matter 2024.

Neural nets plus a genetic algorithm designed new optical glasses

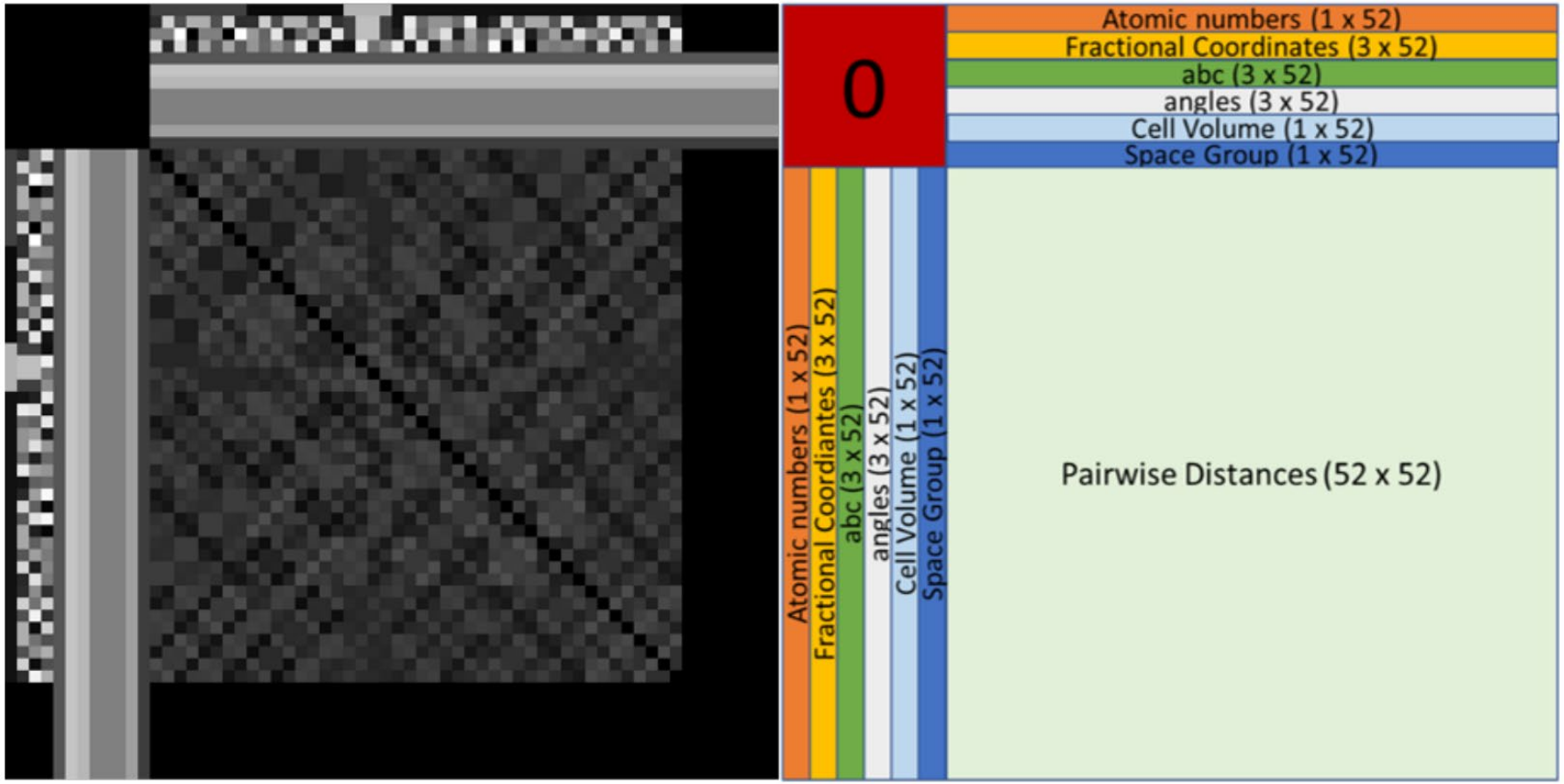
NN property models + a genetic algorithm: two lab glasses hit n = 1.713 and 1.749 at Tg = 438 and 450 °C



Cassar, Santos & Zanotto, Ceram. Int. 47, 10555-10564 (2021) https://doi.org/10.1016/j.ceramint.2020.12.167

Turn a crystal into an image and reuse image AI tools

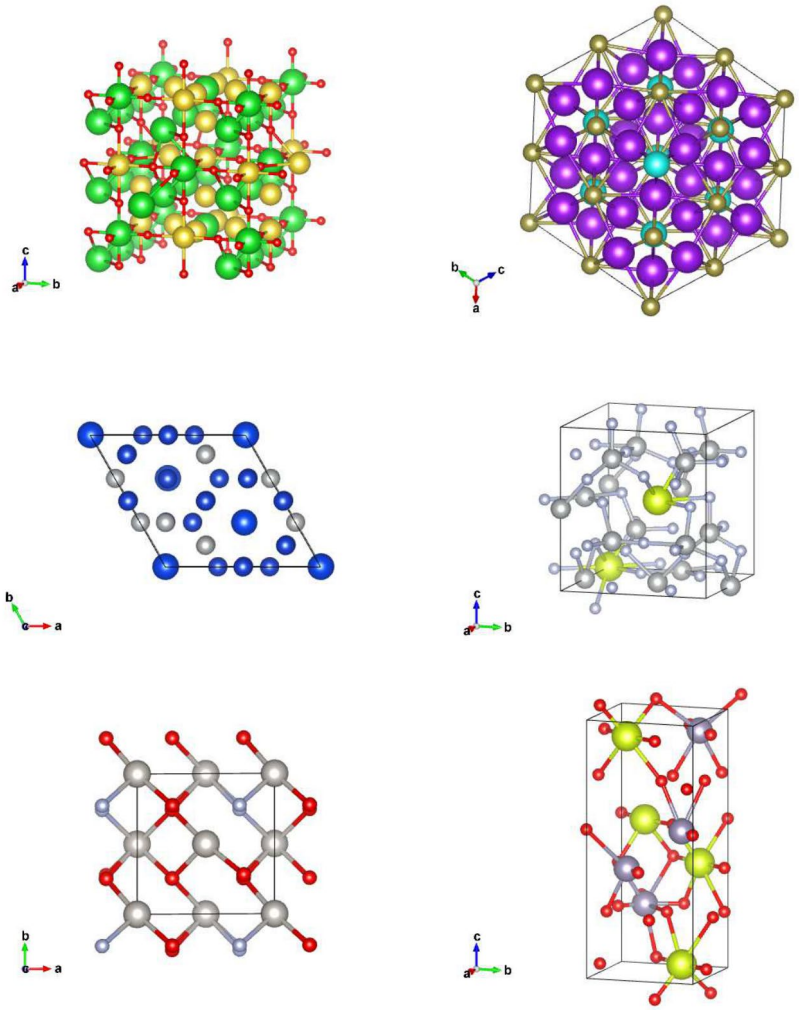
Sparks group, University of Utah

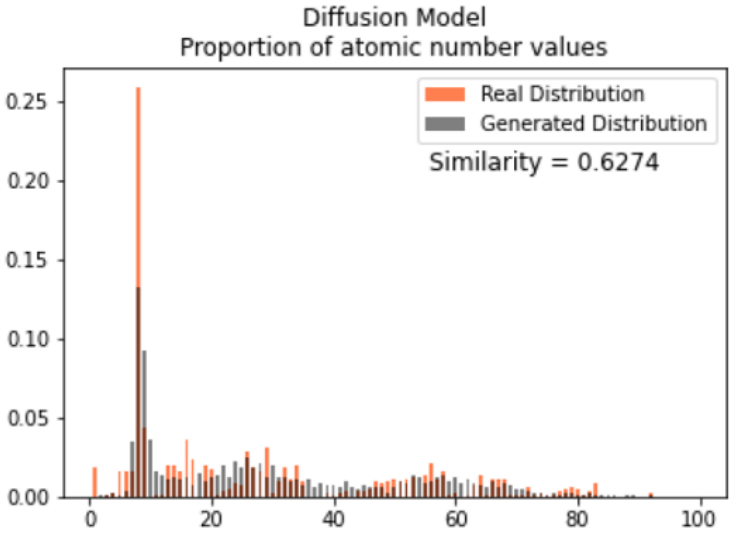


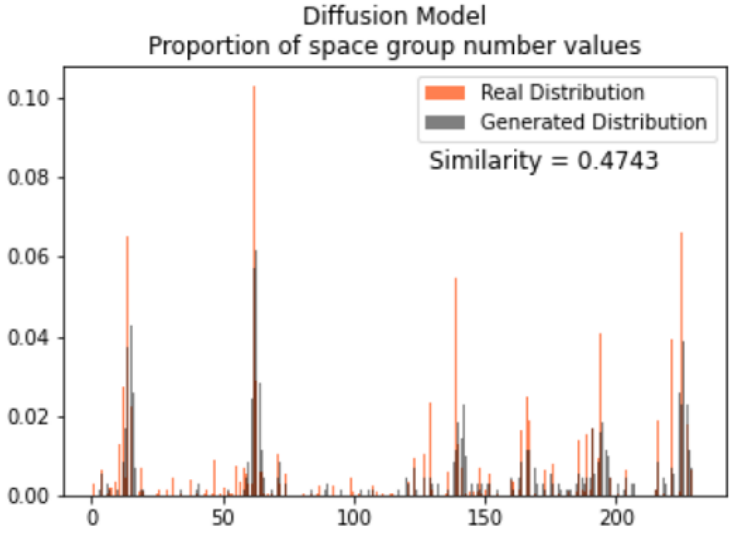
Baird et al., J. Open Source Softw. 7, 4528 (2022) https://doi.org/10.21105/joss.04528

Diffusion models made more realistic crystals than GANs did

Sparks group: diffusion vs. GANs on 50,000+ CIFs encoded as CrysTens



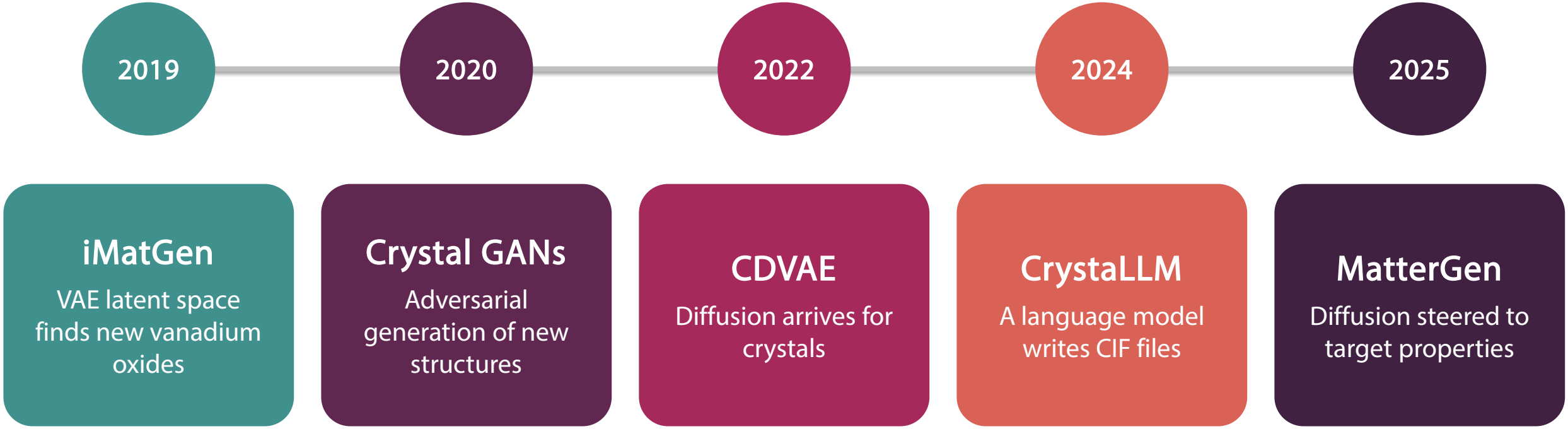




Alverson et al., Digital Discovery 3 (2024) https://doi.org/10.1039/D3DD00137G

Crystal generators: autoencoders, then diffusion, then LLMs

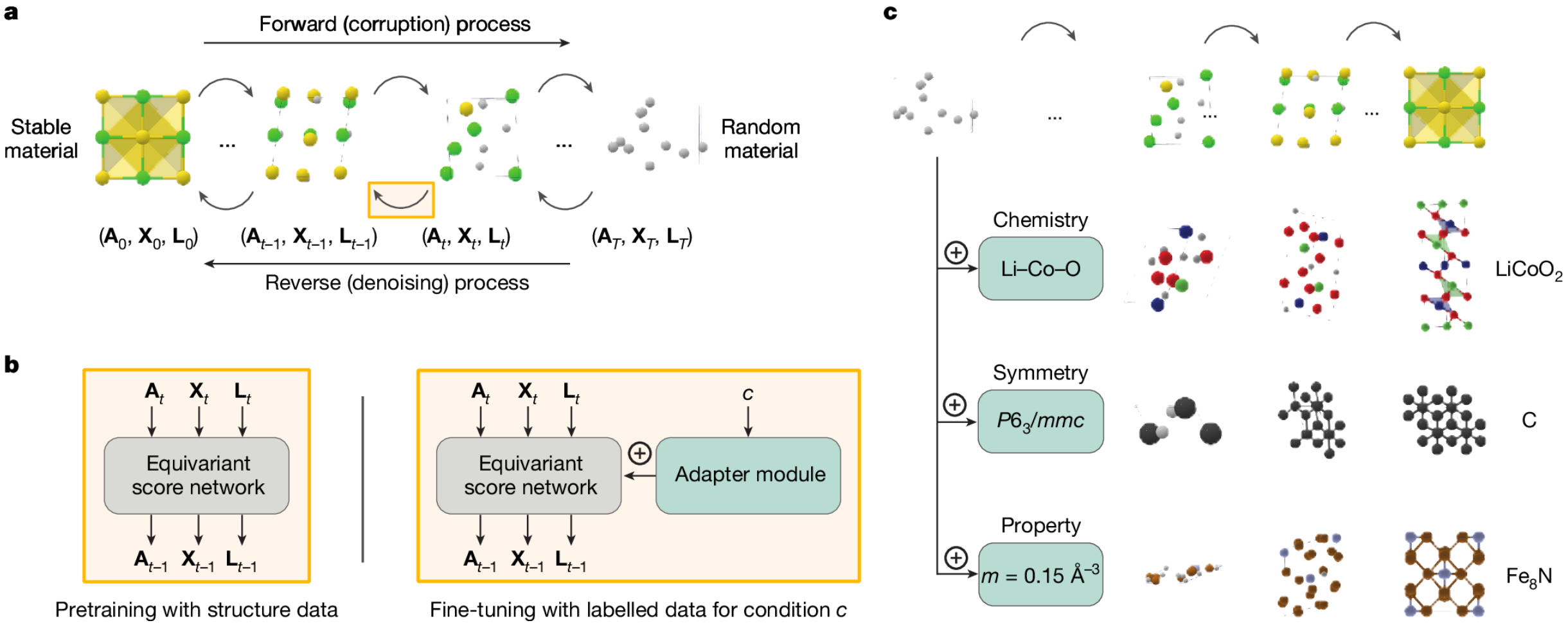
Each generation borrowed the newest idea from image and text AI



Noh et al., Matter 2019; Kim et al., ACS Cent. Sci. 2020; Xie et al., ICLR 2022; Antunes et al., Nat. Commun. 2024; Zeni et al., Nature 2025

Diffusion models build crystals by denoising random atoms

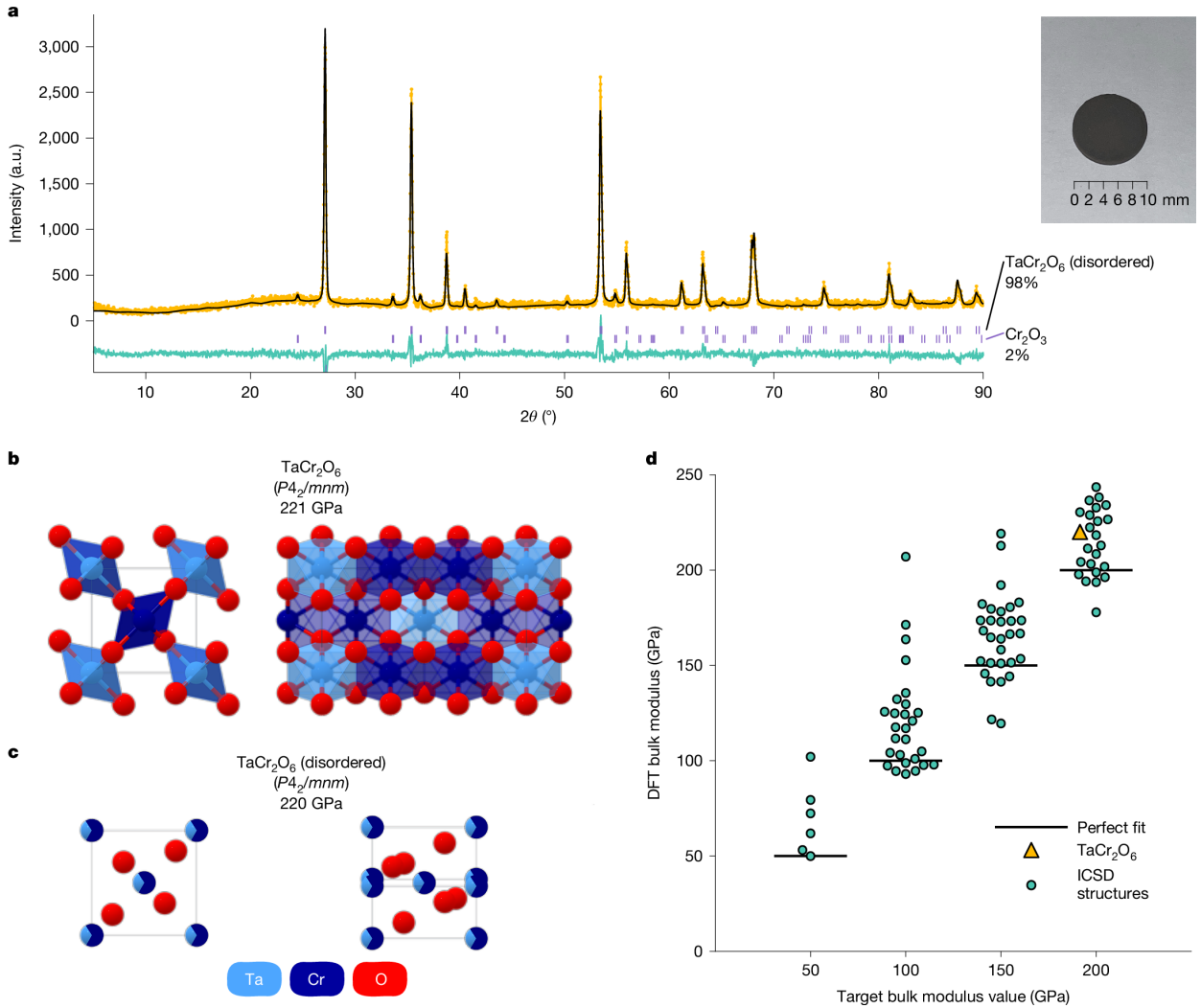
Fine-tuning steers generation toward a chosen chemistry, symmetry or property



Zeni et al., Nature 639, 624-632 (2025) https://doi.org/10.1038/s41586-025-08628-5

In some cases, AI designed crystals are actually synthesized

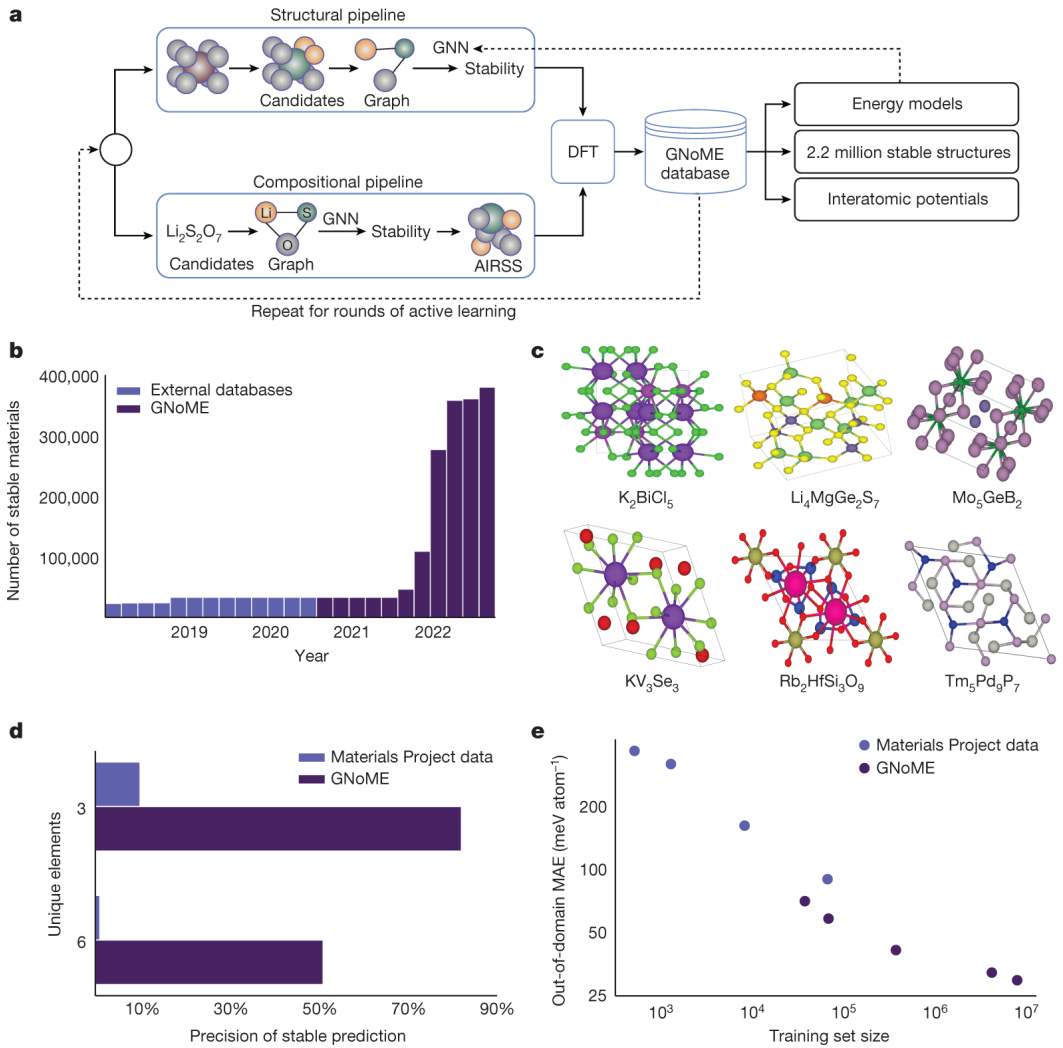
Targeted a 200 GPa bulk modulus; measured property within 20% of target



Zeni et al., Nature 639, 624-632 (2025) https://doi.org/10.1038/s41586-025-08628-5

Graph networks grew known stable crystals almost tenfold

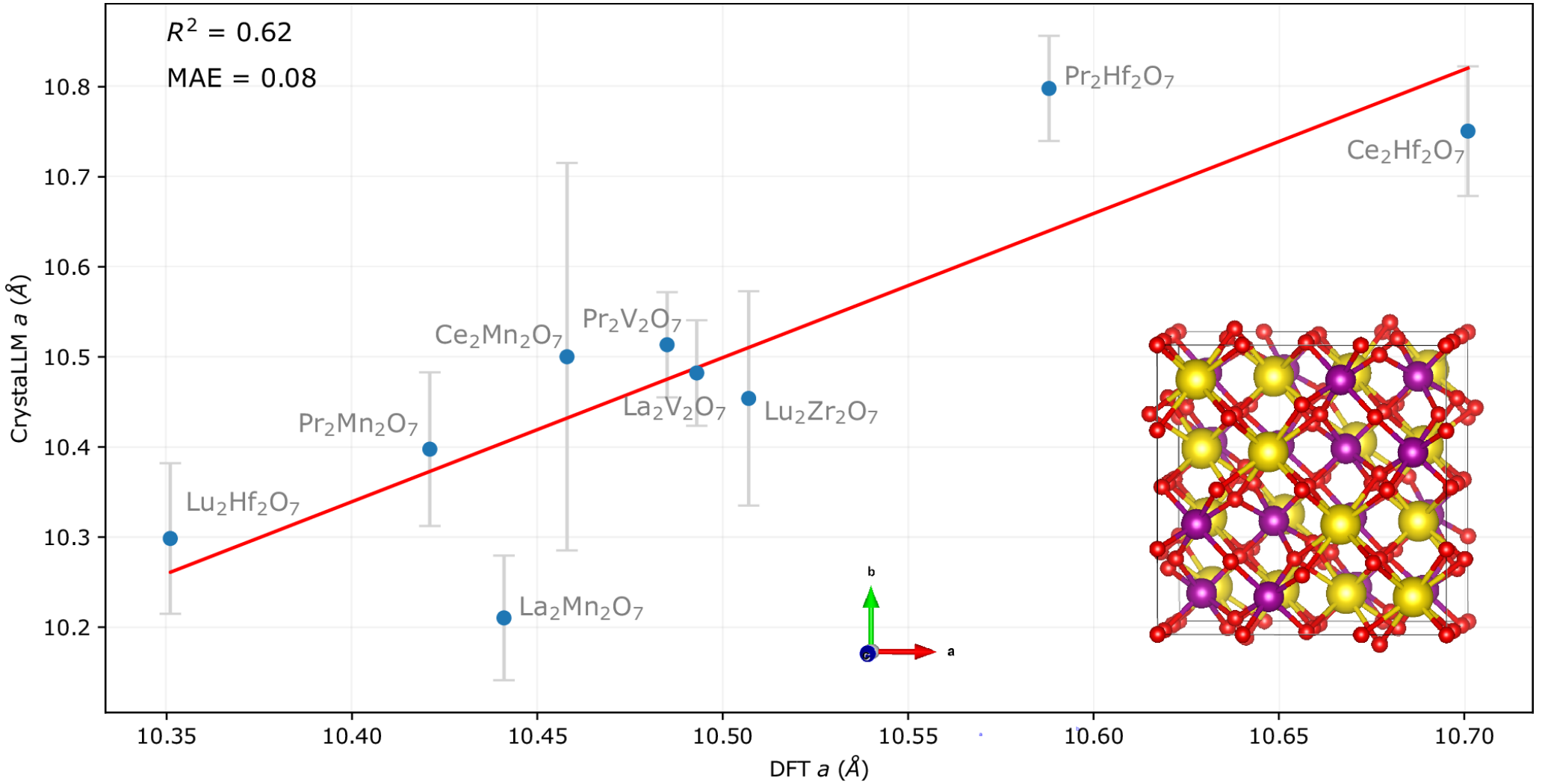
381,000 new stable materials; 736 independently verified experimentally



Merchant et al., Nature 624, 80-85 (2023) https://doi.org/10.1038/s41586-023-06735-9

LLM's were even capable of structure generation!

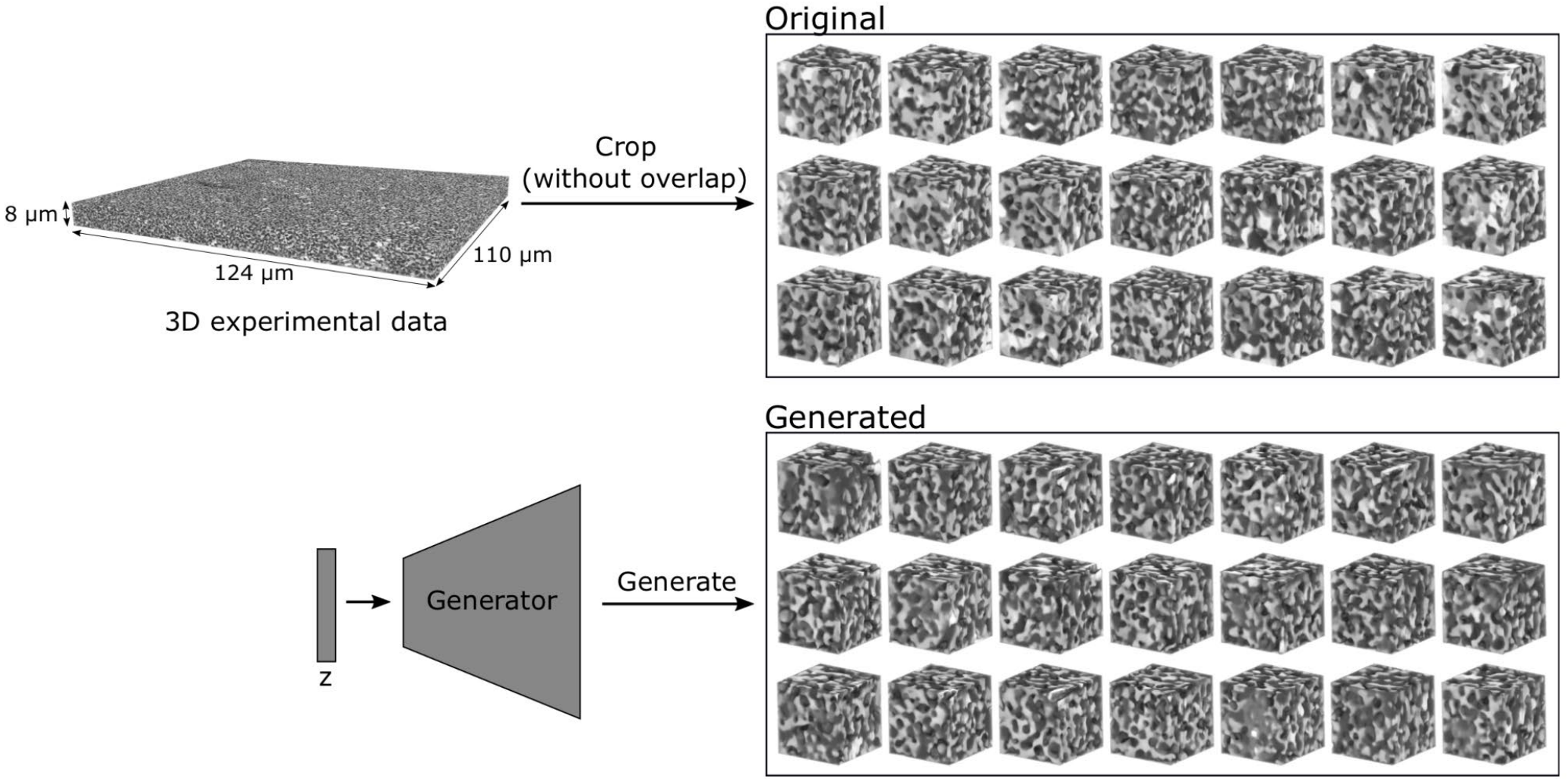
Lattice parameters match DFT with MAE = 0.08 A



Antunes, Butler & Grau-Crespo, Nat. Commun. 15 (2024) https://doi.org/10.1038/s41467-024-54639-7

GANs generate realistic 3D ceramic fuel-cell microstructures

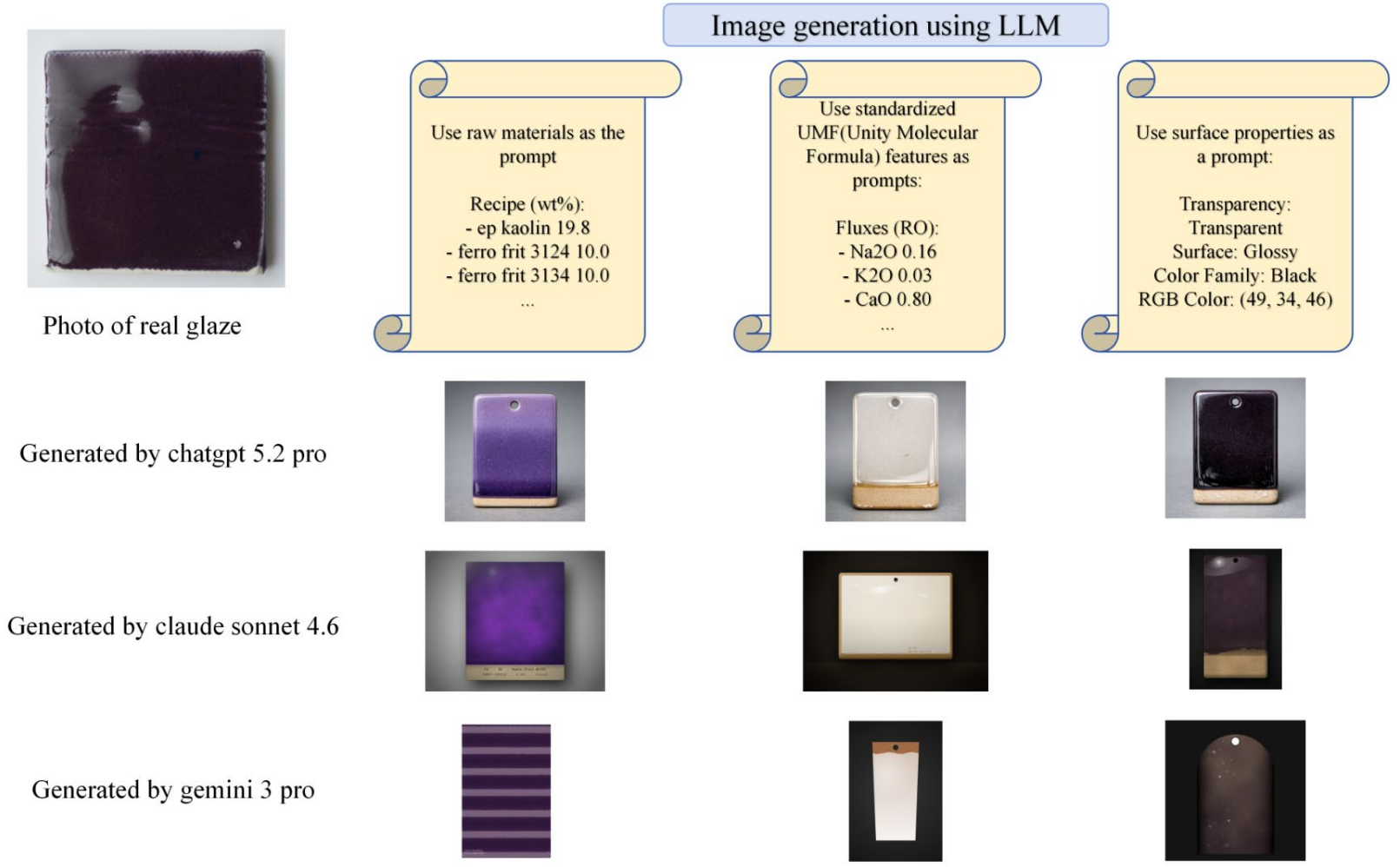
Generated volumes match real ones in tortuosity, particle size and TPB density



Hsu et al., JOM 73, 90-102 (2021) https://arxiv.org/abs/2006.13886

AI can picture a fired glaze from its properties, not yet from its recipe

GlazyBench: 23,148 real glaze recipes for predicting color and finish, and for drawing the tile



From the recipe

Generated tiles miss the color and finish

From the properties

Generated tiles look close to the real glaze

Why

Too few datasets pair recipes, firing and photos

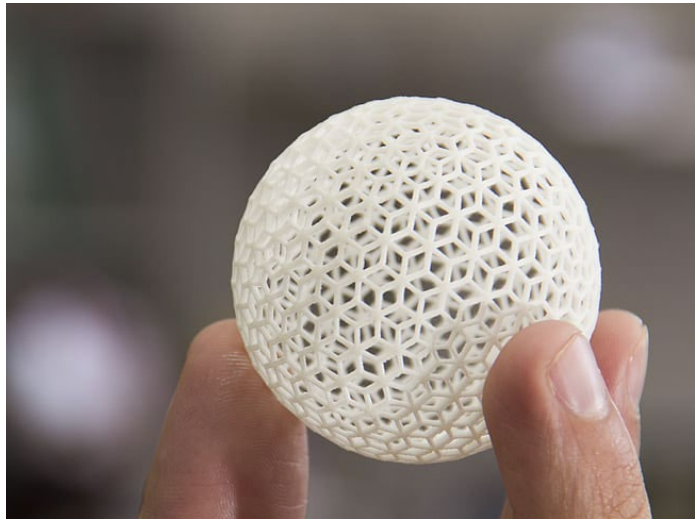
Zhai, Li, Shao & Yu, GlazyBench, arXiv 2026 https://doi.org/10.48550/arXiv.2605.06641

Self-driving labs and automation

Materials design is filled with difficult property tradeoff decisions

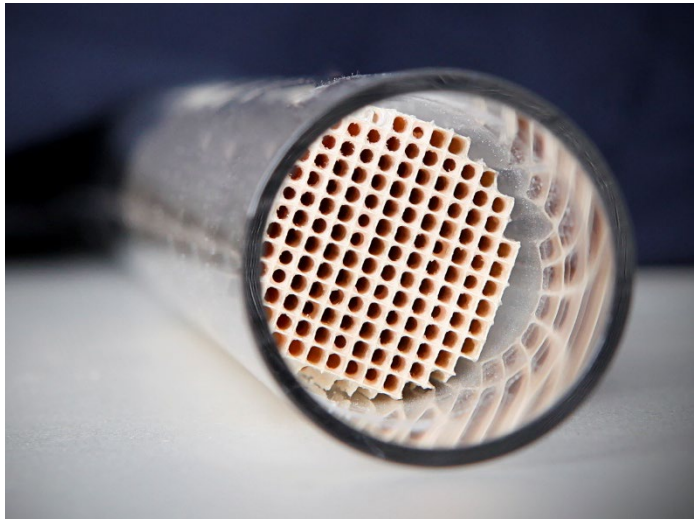
Strength vs ductility

3d print parameters

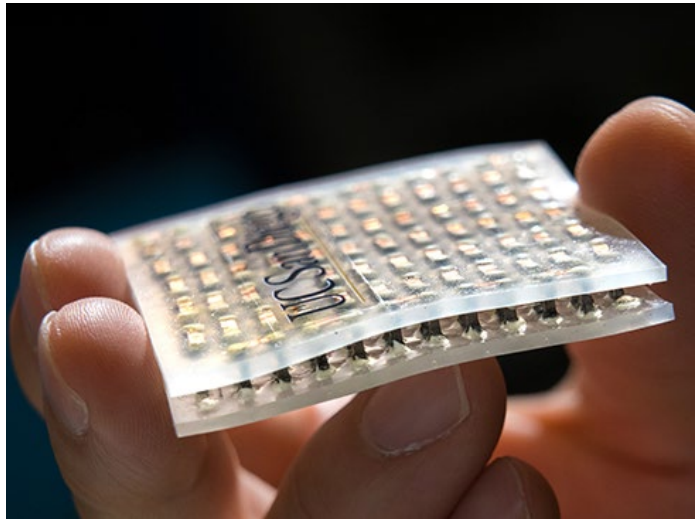


Performance vs biodegradability

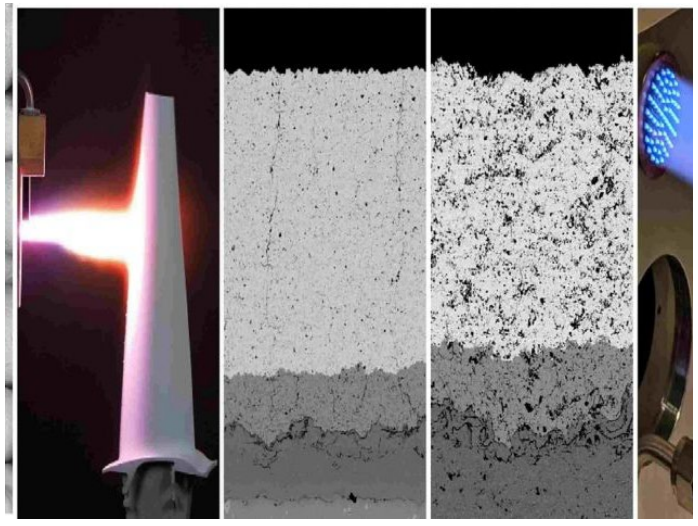
Catalytic activity vs selectivity



Thermal vs electrical

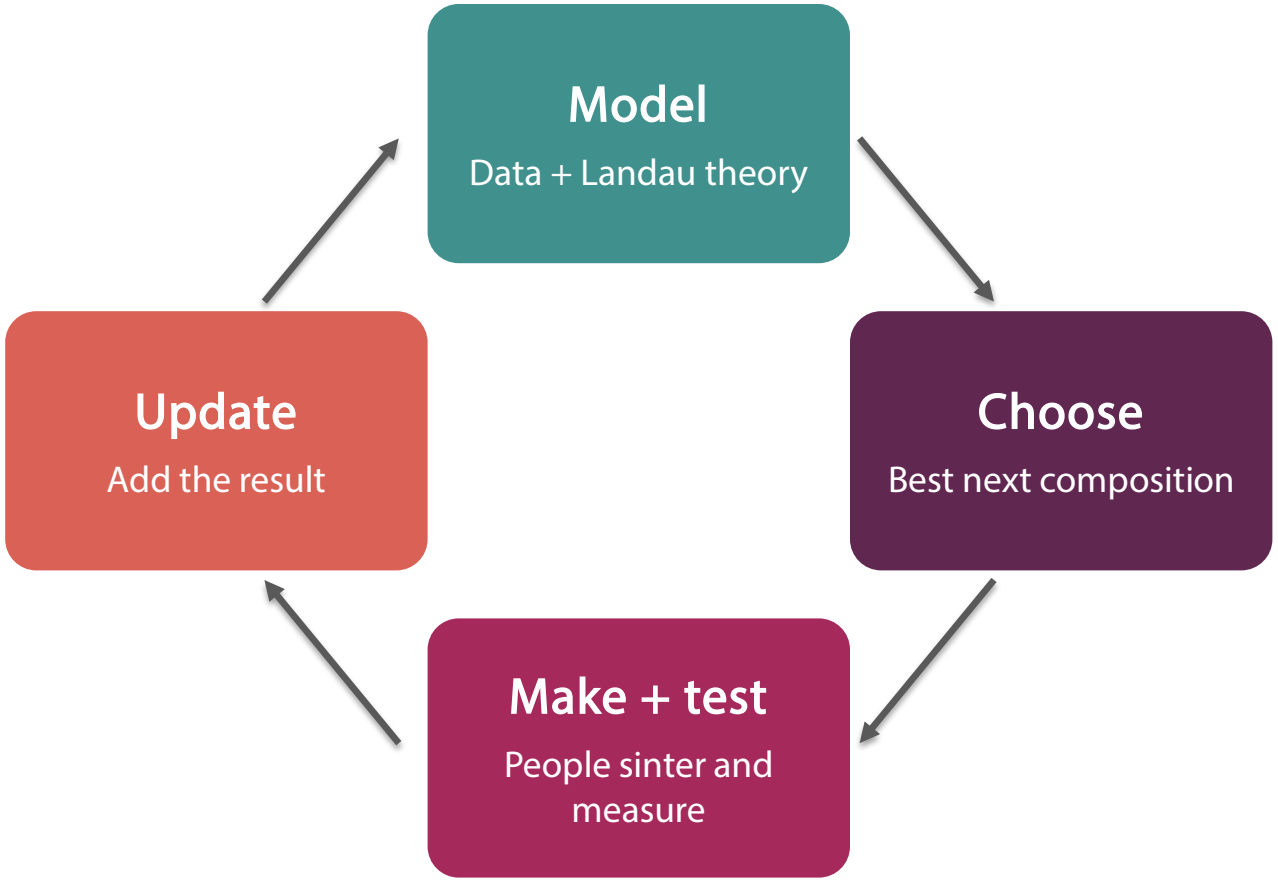


Thermal vs mechanical



Bayesian learning chose the next piezoceramic before robots joined in

The algorithm picked each composition; people still did the sintering and measuring



Result

$(Ba_{0.5}Ca_{0.5})TiO_3–Ba(Ti_{0.7}Zr_{0.3})O_3$

Better temperature stability than any $BaTiO_3$ piezoelectric in the training data

Physics (Landau theory, known phase diagrams) guided the search, not data alone.

Xue, Balachandran, Yuan, Hu, Qian, Dougherty & Lookman, PNAS 2016 https://doi.org/10.1073/pnas.1607412113

A mobile robot ran 688 experiments in 8 days in an ordinary lab

Batched Bayesian optimization steered a free-roaming robot through a ten-variable search

688
experiments

8 days
172 of 192 hours working

10
variables searched

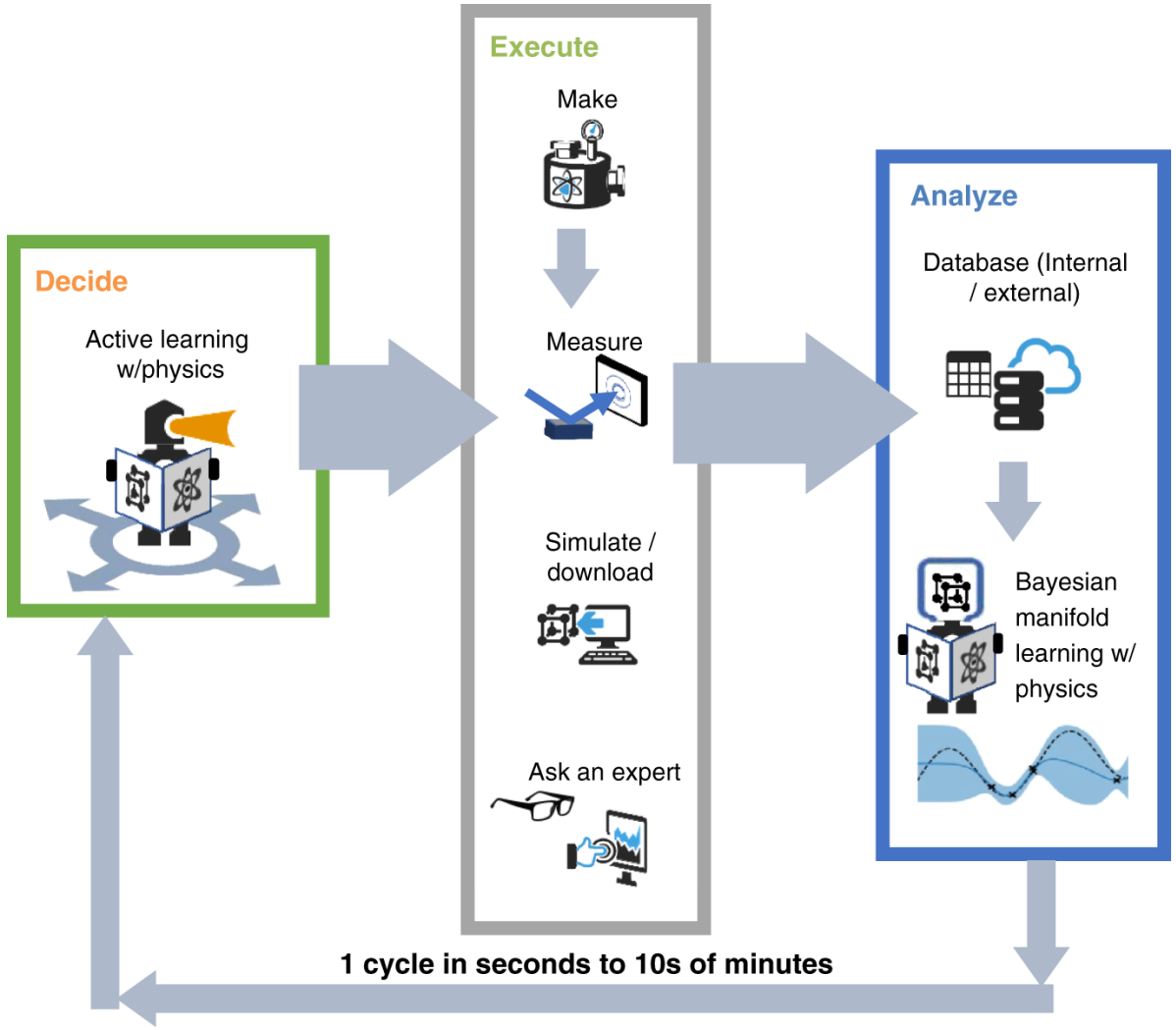
6×
more active photocatalyst

Same benches and instruments as the humans: autonomy retrofitted into an ordinary lab. The catch: about two years to set up.

Burger et al., Nature 2020 https://doi.org/10.1038/s41586-020-2442-2

Active learning lets the algorithm choose the next experiment

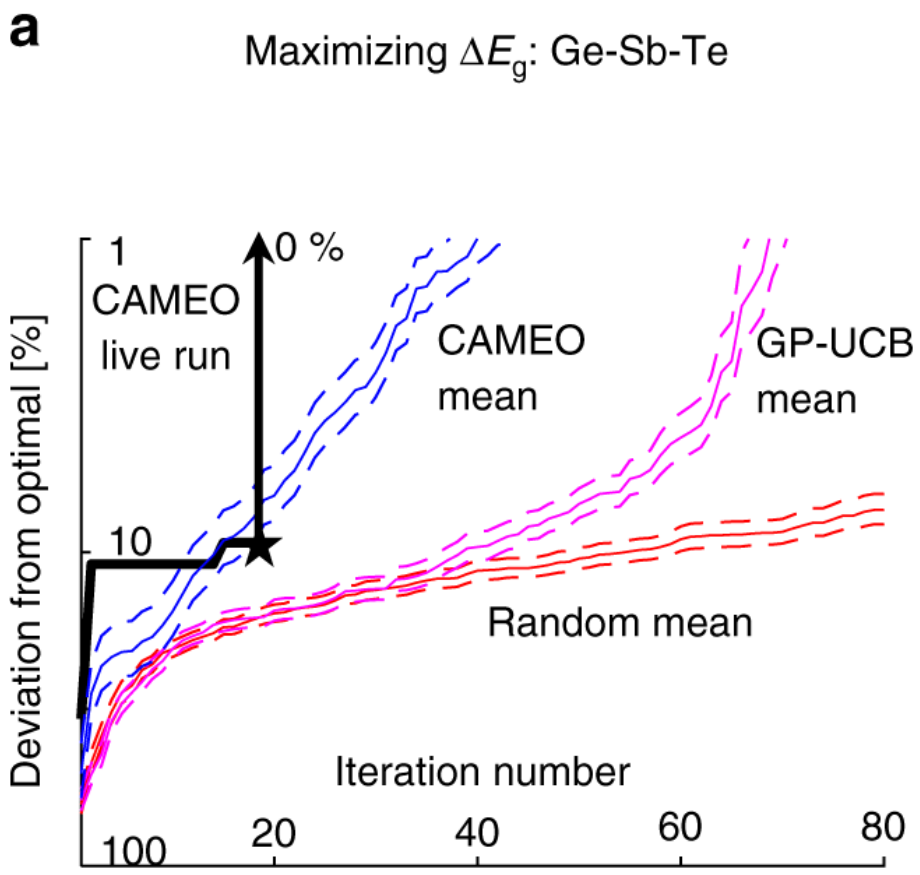
Each CAMEO cycle takes seconds to tens of minutes at the beamline.

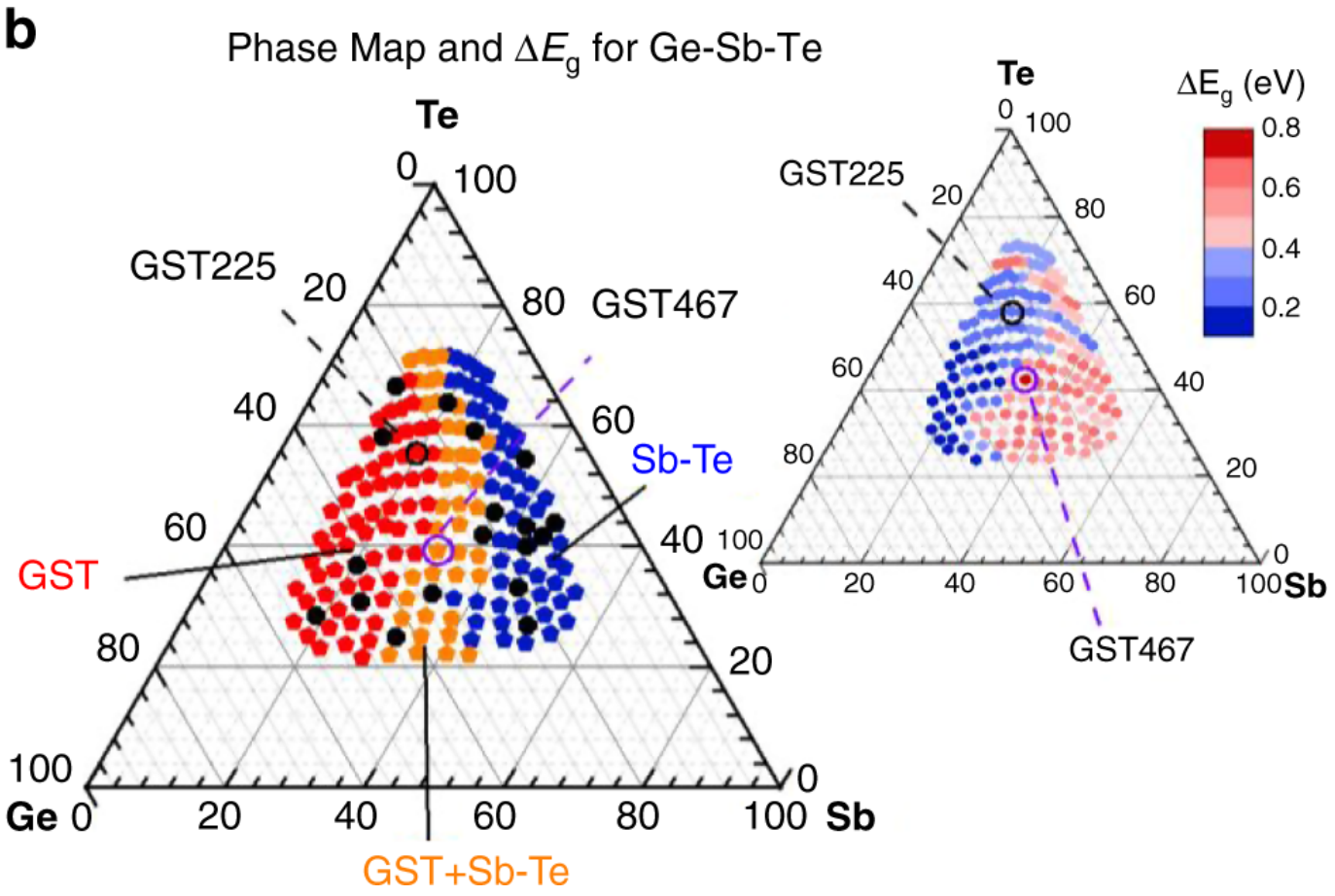


Kusne et al., Nature Communications (2020) https://doi.org/10.1038/s41467-020-19597-w

Smart sampling found the best phase-change alloy in 19 tries

Within 20 iterations CAMEO got within 1% of optimal 31% of the time vs 10% for GP-UCB.

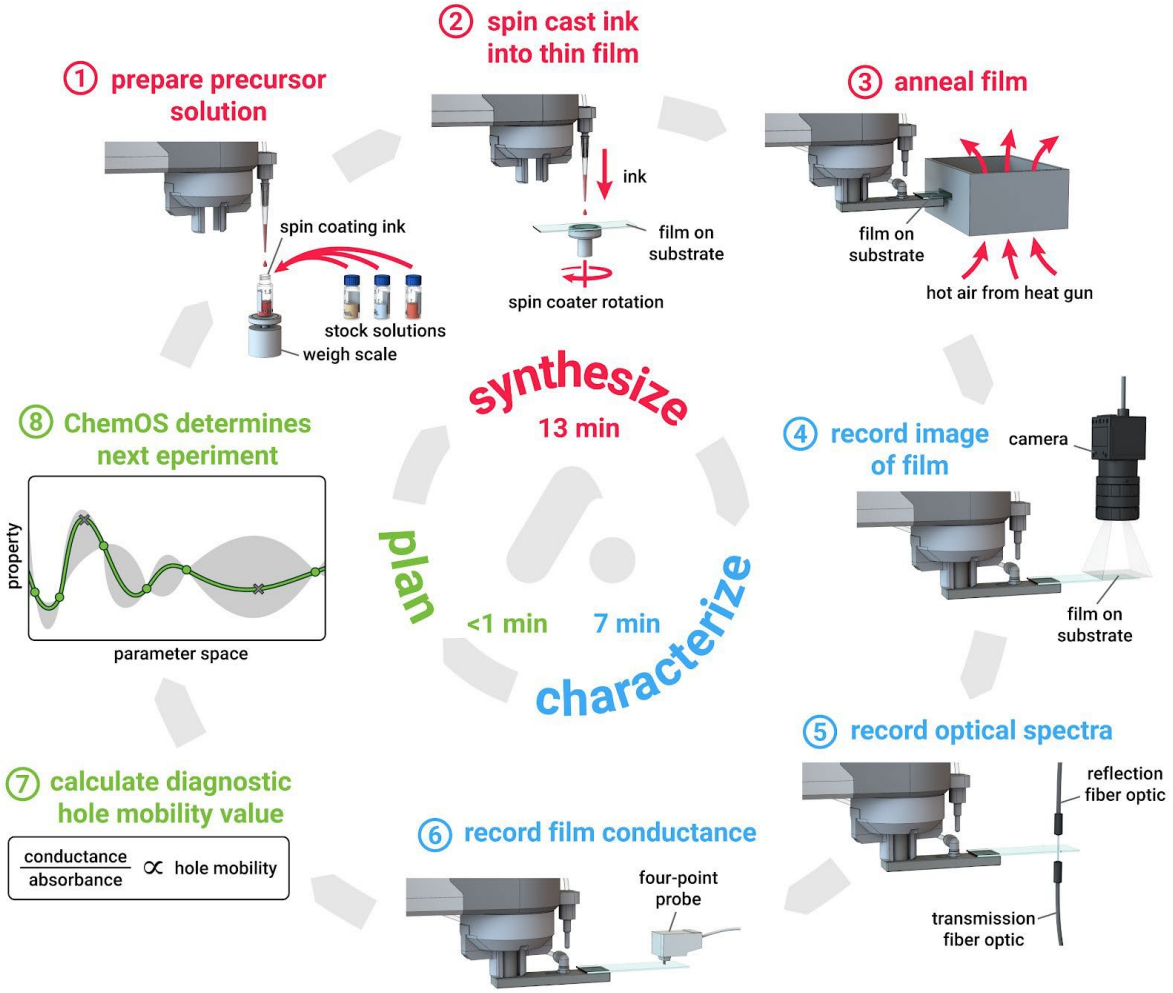




Kusne et al., Nature Communications (2020) https://doi.org/10.1038/s41467-020-19597-w

A self-driving lab makes, measures and plans a film every 20 min

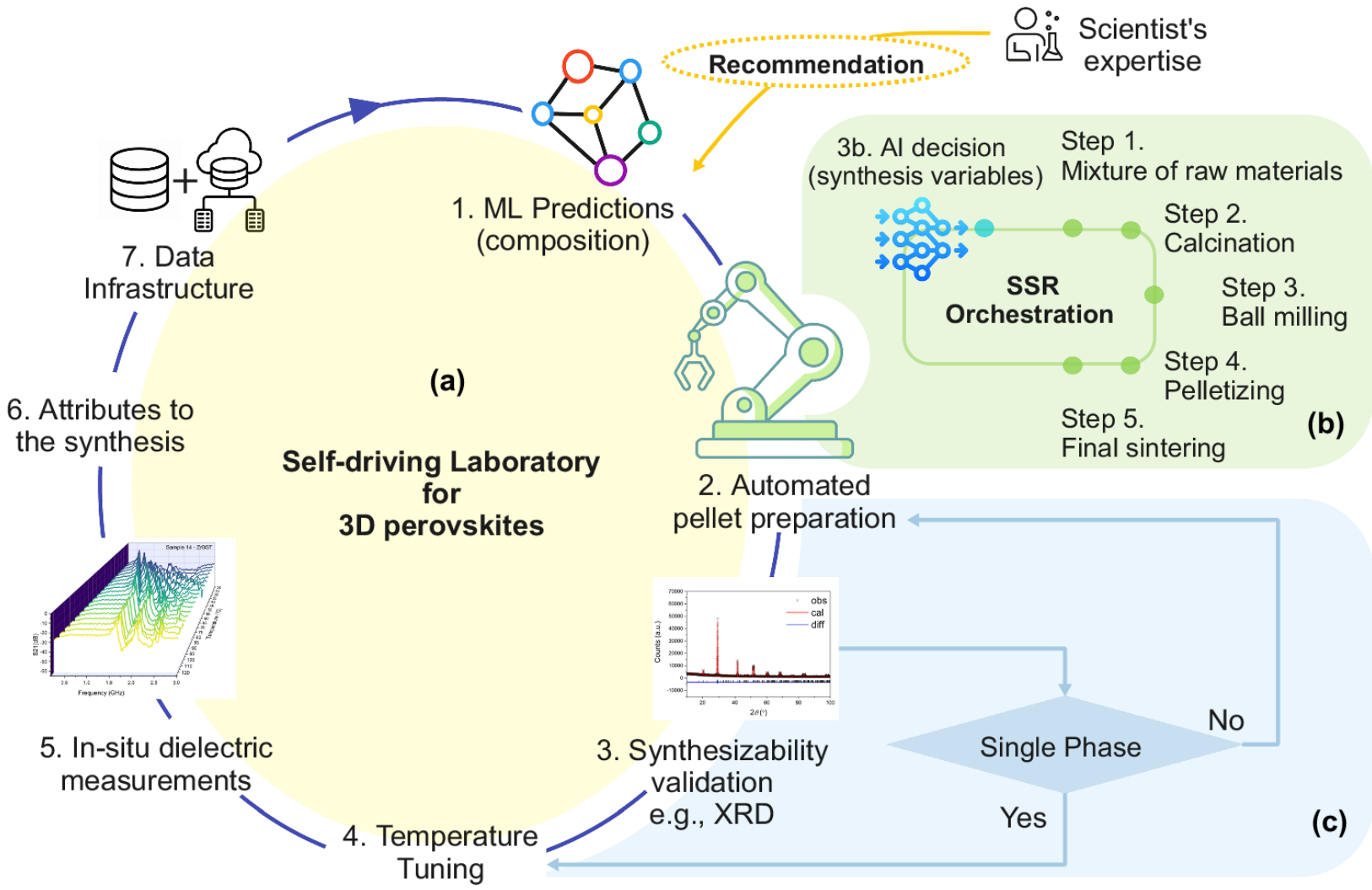
Per cycle: about 13 min to synthesize, 7 min to characterize, under 1 min to plan.



MacLeod et al., Science Advances 6, eaaz8867 (2020) https://doi.org/10.1126/sciadv.aaz8867

The classic solid-state ceramic route can be automated end to end

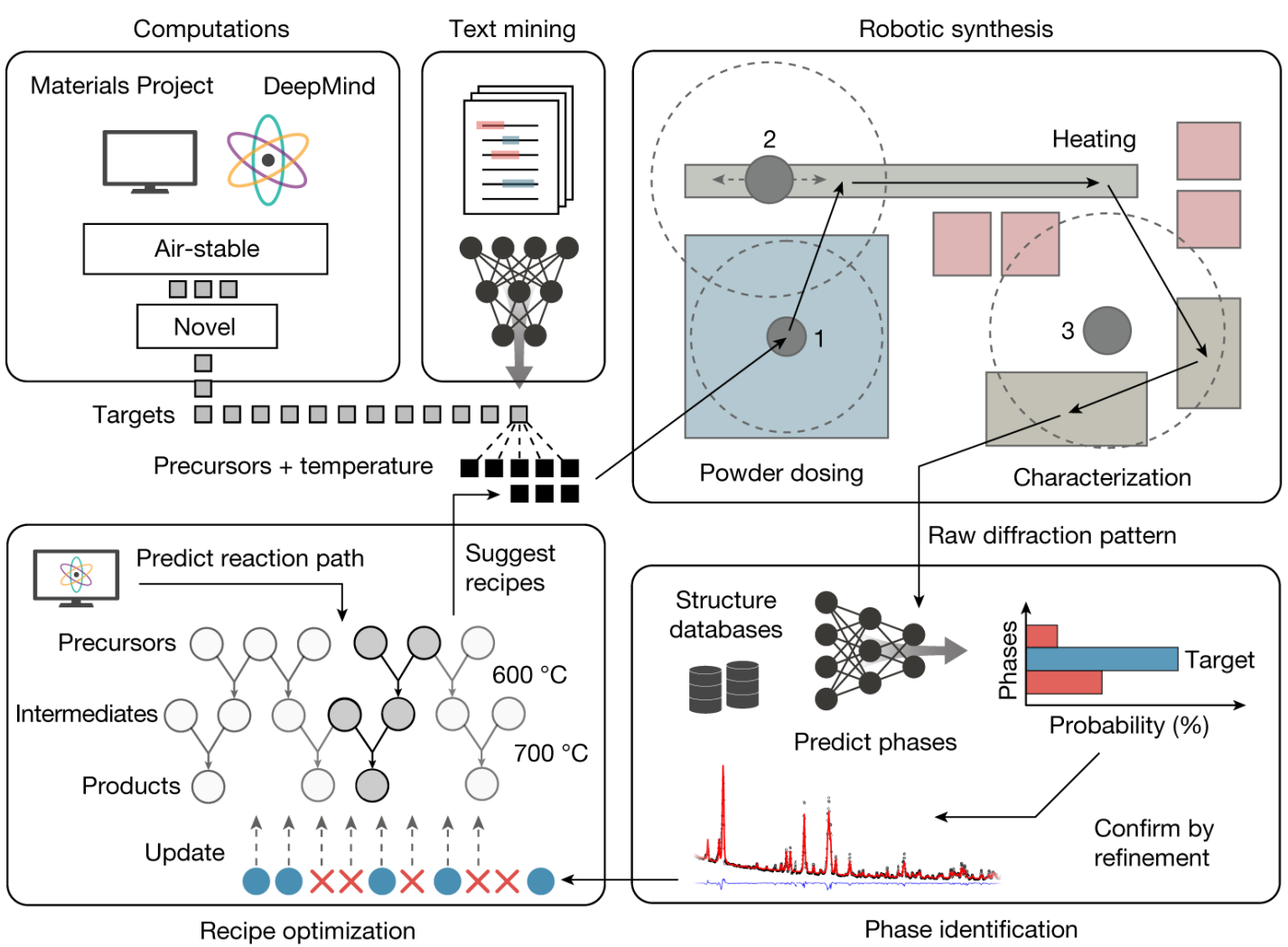
Used to make single-phase (Ba,Sr)CeO_3 solid solutions picked by ML, processing in minutes.



Omidvar et al., Nature Communications (2024) https://doi.org/10.1038/s41467-024-50884-y

Robots plus ML can run solid-state ceramic synthesis end to end

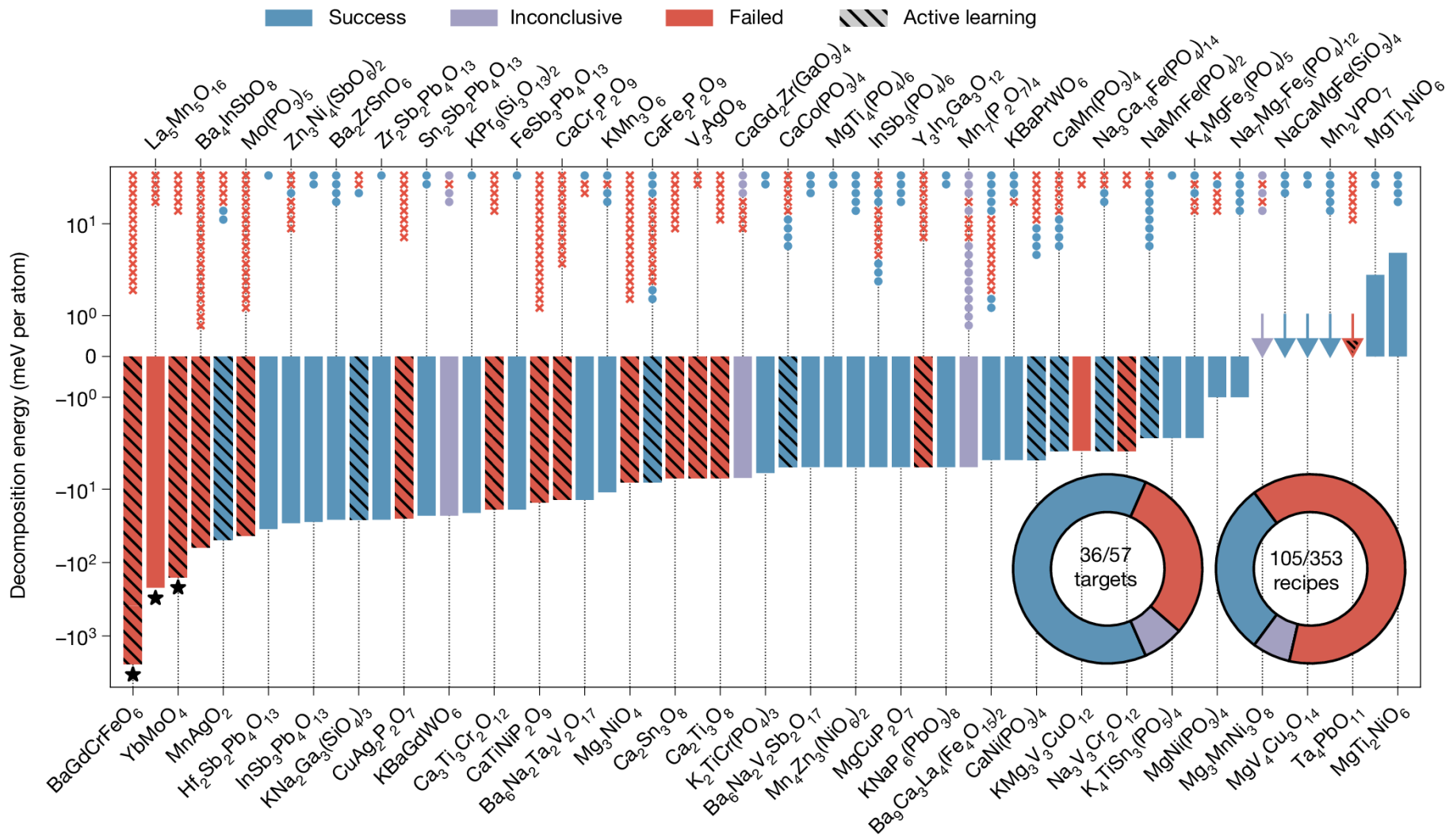
Over 17 days of continuous operation the A-Lab attempted 57 target oxides and phosphates.



Szymanski et al., Nature 624, 86 (2023) https://doi.org/10.1038/s41586-023-06734-w

The A-Lab made 36 of 57 targets, but most recipes still failed

Corrected result: 36 of 57 targets (the paper originally reported 41 of 58)

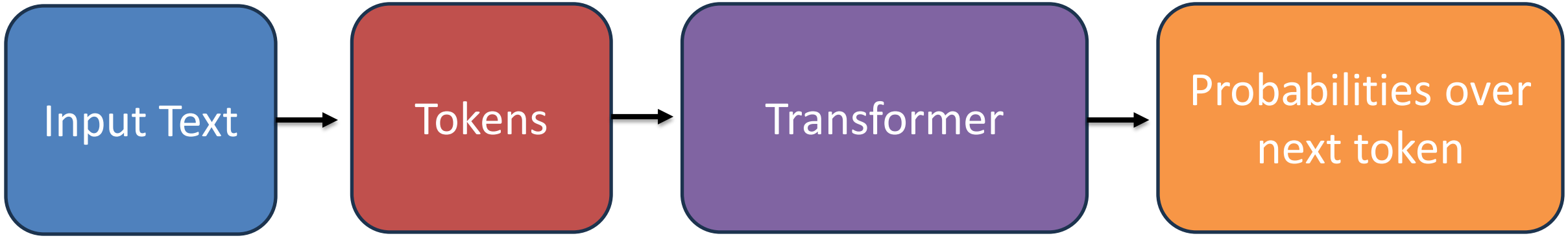


Szymanski et al., Nature 624, 86 (2023) https://doi.org/10.1038/s41586-023-06734-w

LLMs & Agents:
Reading, planning, and calling tools

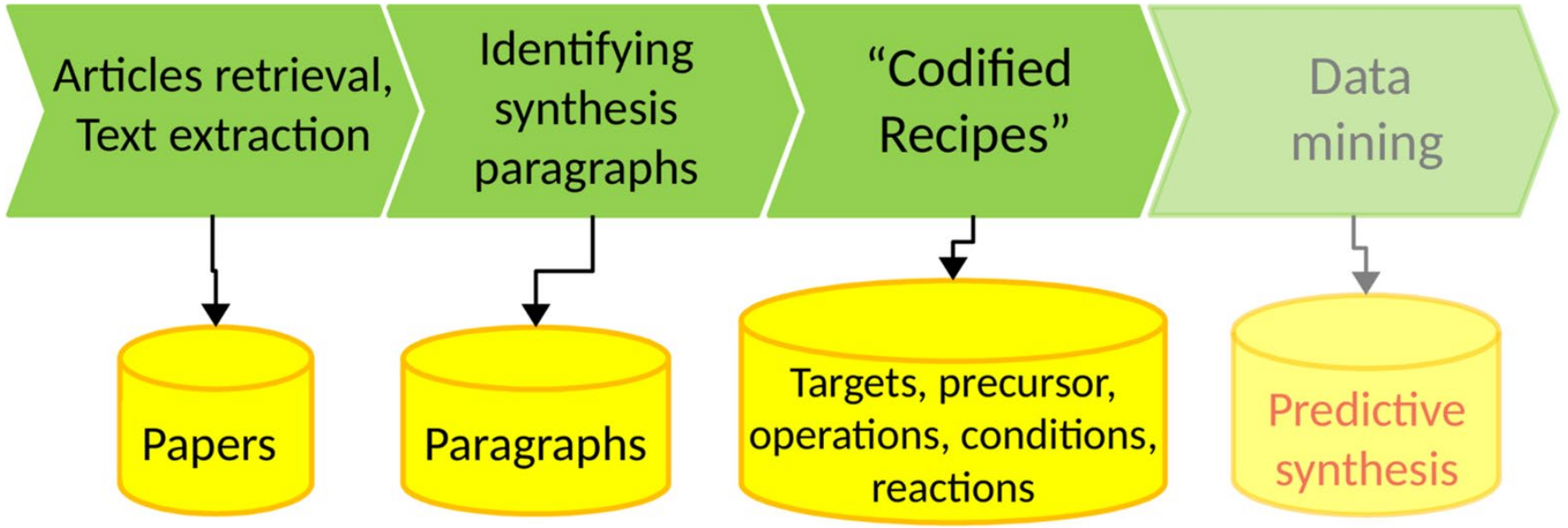
Fill in the blank… at a massive scale

"The band gap of Si is _____ eV."



NLP can turn ceramic synthesis paragraphs into structured recipes

19,488 synthesis entries were extracted from 53,538 solid-state synthesis paragraphs.



Samples of $BaCo_{0.7}Fe_{0.3-x}Nb_xO_{3-\delta}$ were prepared by solid state synthesis. Stoichiometric amounts of $BaCO_3$, Fe_2O_3, Nb_2O_5, $Co(CH_3COO)_2$ were weighted and mixed by ball milling for 6h with agate balls in ethyl alcohol. The powder was dried, calcined at 1000 °C for 10h in air, and finely ground in a mortar for 4h. Finally, the powder was pressed and sintered in air at 1200 °C for 10h.

TARGET: $BaCo_{0.7}Fe_{0.3-x}Nb_xO_{3-\delta}$

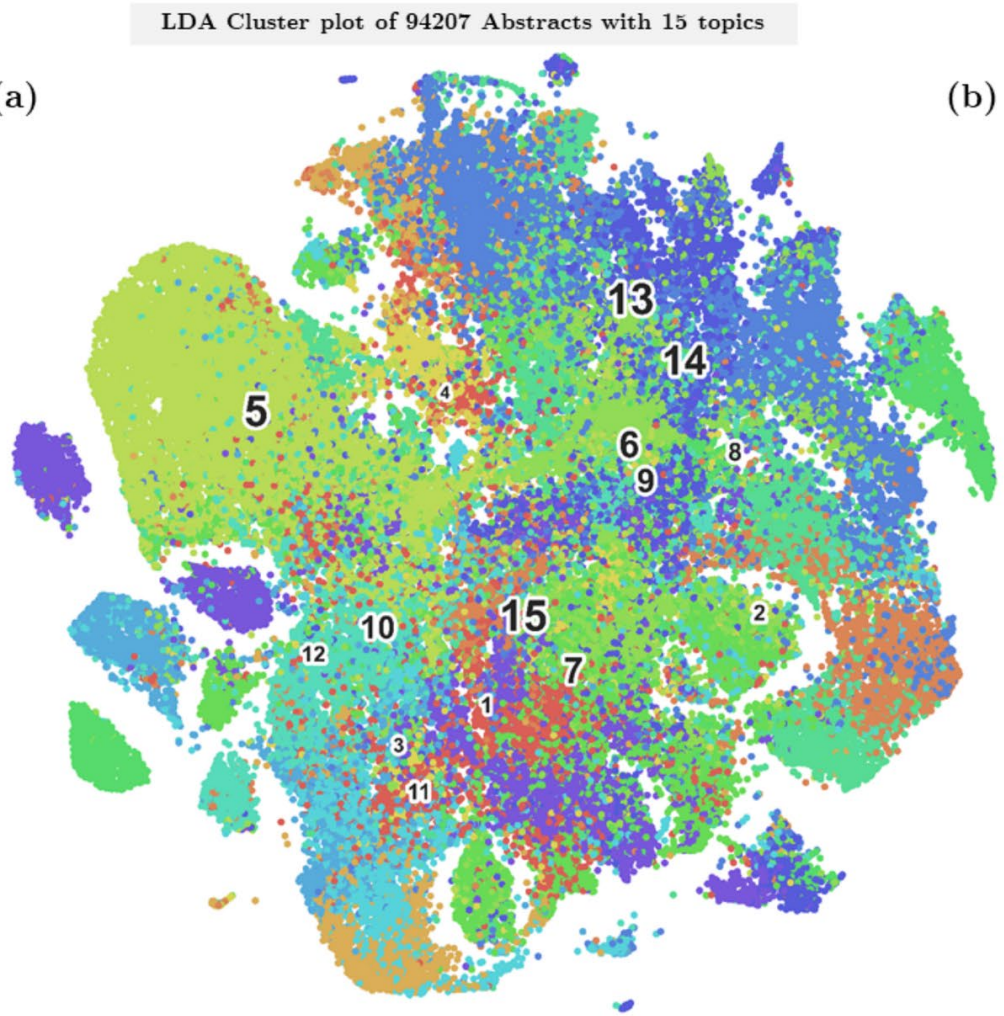
PRECURSORS:	OPERATIONS:	CONDITIONS:
$BaCO_3$	1. mix	6h, ethyl alcohol
Fe_2O_3	2. dry	
Nb_2O_5	3. calcine	1000 °C, 10h, air
$Co(CH_3COO)_2$	4. grind	4h
	5. sinter	1200 °C, 10h, air

Kononova et al., Scientific Data (2019) https://doi.org/10.1038/s41597-019-0224-1

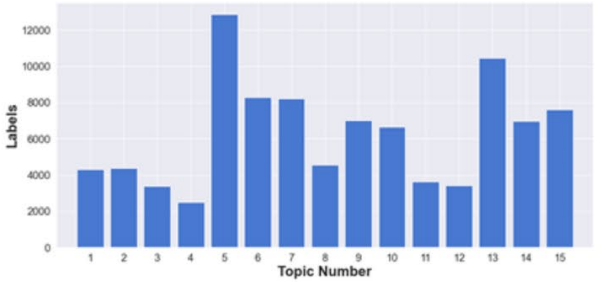
NLP can map the entire glass literature by topic

94,207 abstracts in 15 topics; the corpus also held 106,238 figure captions.

(a)



(b)



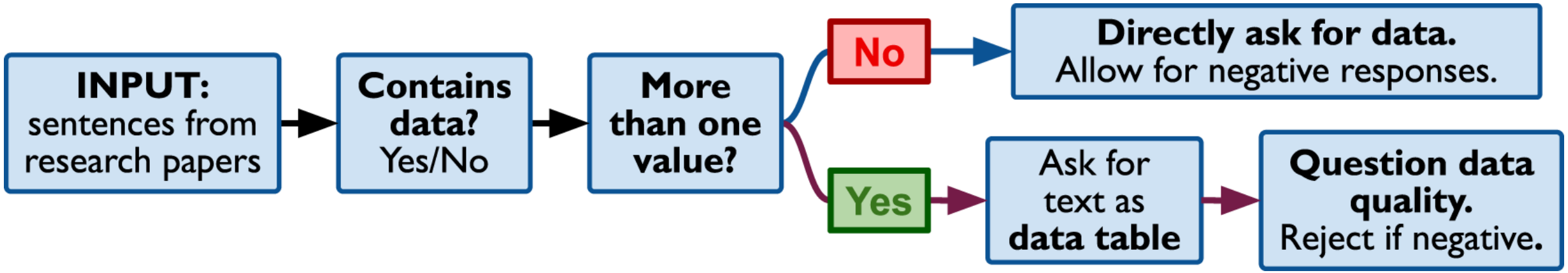
(c)

Topic	Label
1	Luminescence studies
2	Dielectric properties
3	Spectroscopic properties
4	Irradiation studies
5	Thin films
6	Methodological studies
7	Non-linear optical properties
8	Bioactive glasses
9	Electrical properties
10	Glass nanoparticles
11	Rare earth glasses
12	LEDs and Phosphors
13	Glass ceramics
14	Chemical properties
15	Miscellaneous

Venugopal et al. (Krishnan group), Patterns (2021) https://doi.org/10.1016/j.patter.2021.100290

Skeptical follow-up prompts make LLM data extraction accurate

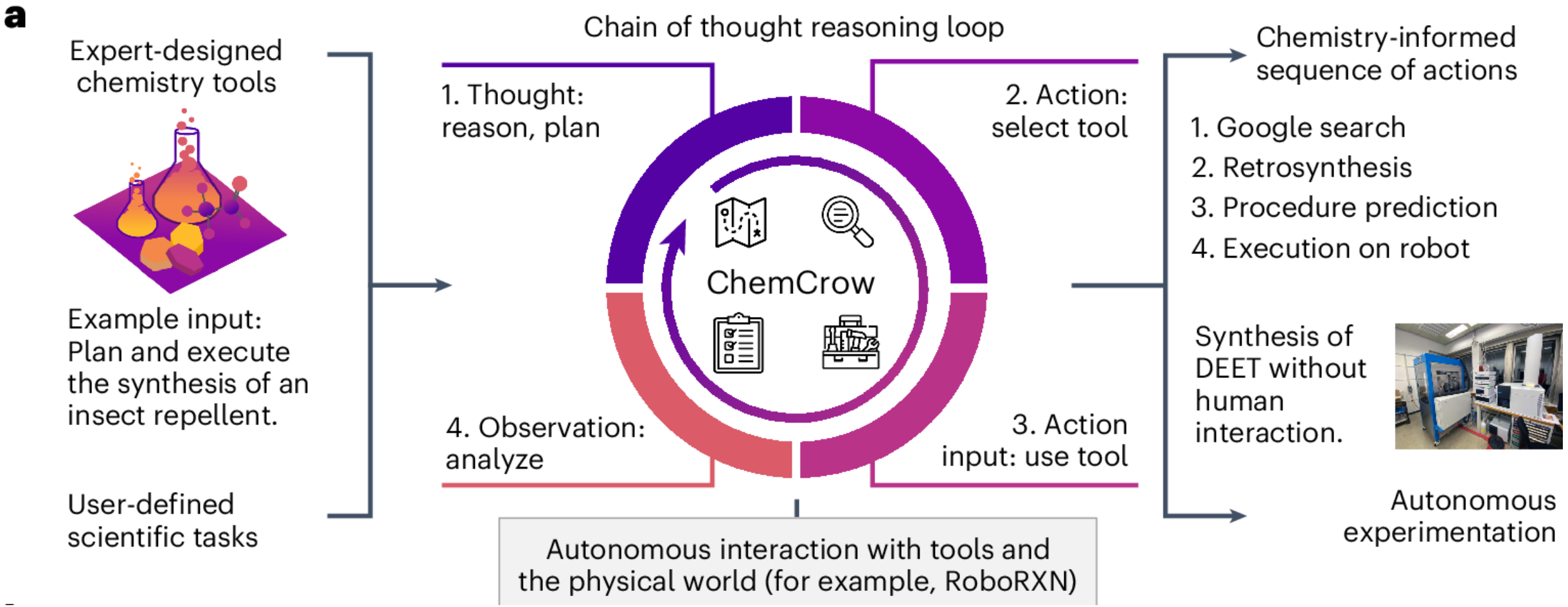
Precision and recall both close to 90% with GPT-4 on materials data.



Polak & Morgan, Nature Communications (2024) https://doi.org/10.1038/s41467-024-45914-8

Agents work by looping: think, pick a tool, act, observe

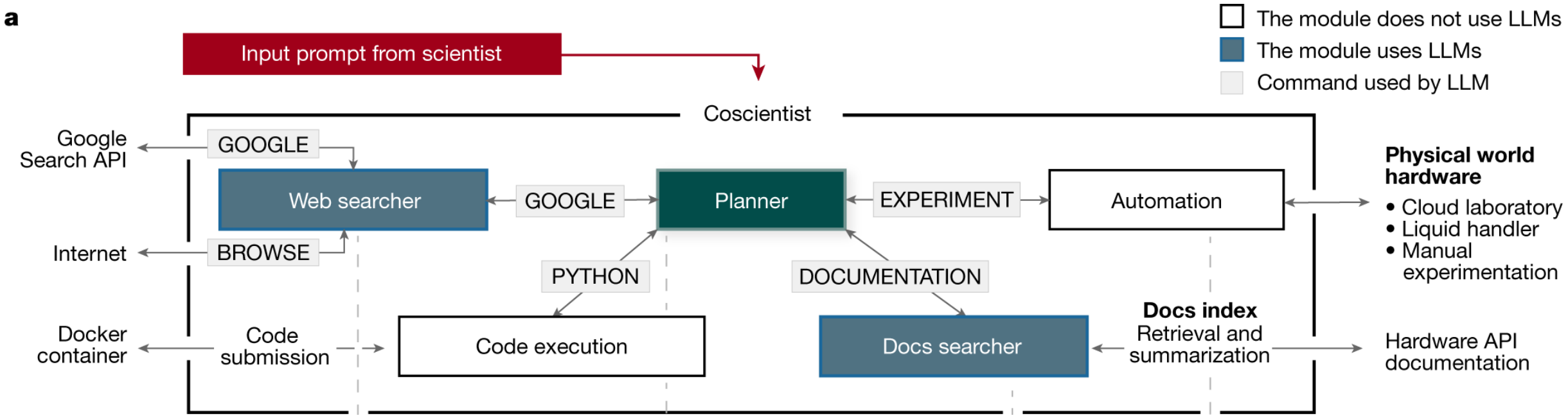
ChemCrow integrates 18 expert-designed tools with GPT-4.



Bran et al., Nature Machine Intelligence (2024) https://doi.org/10.1038/s42256-024-00832-8

An LLM planner can search, write code and drive lab hardware

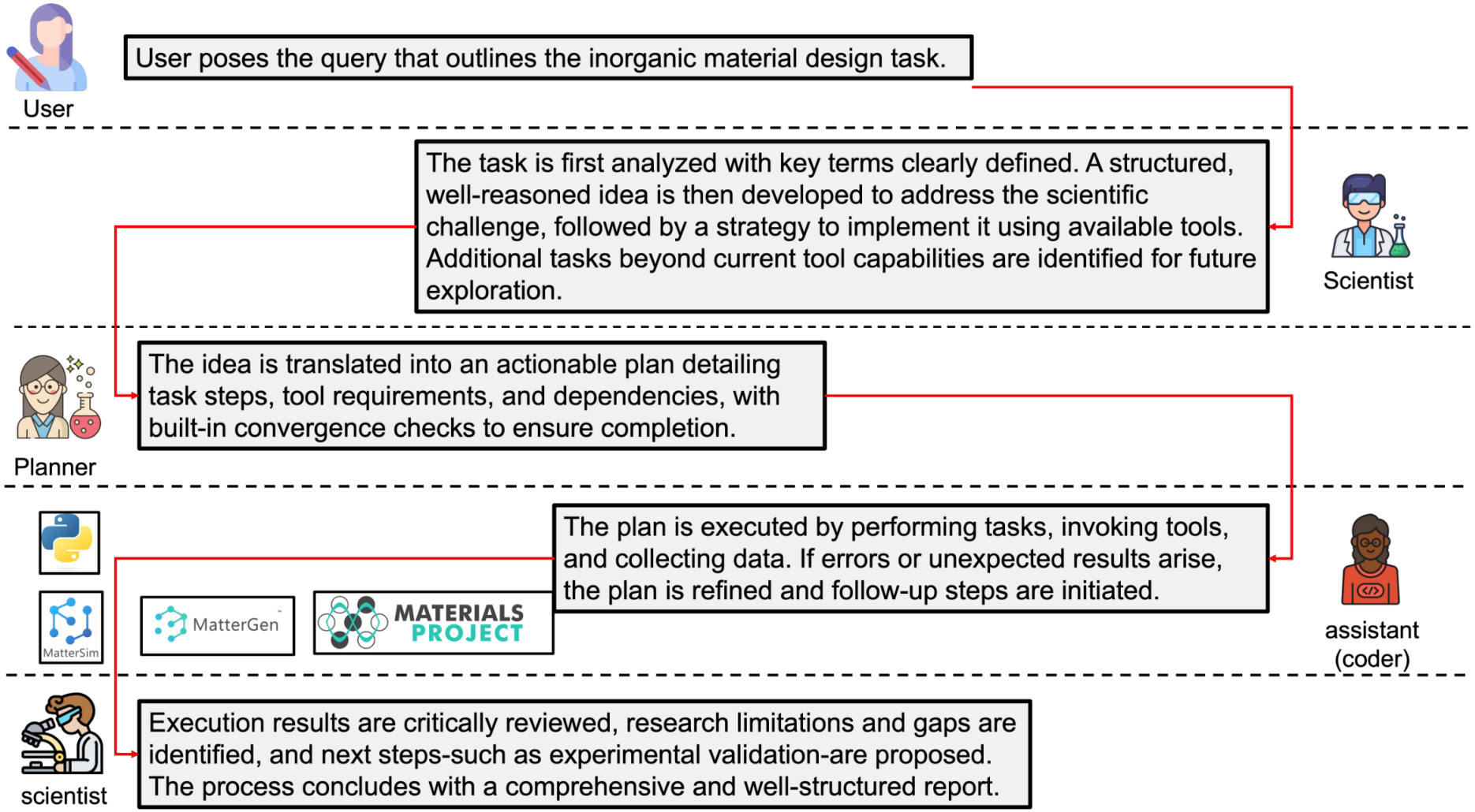
Coscientist planned and ran Pd-catalysed cross-coupling optimizations.



Boiko, MacKnight, Kline & Gomes, Nature (2023) https://doi.org/10.1038/s41586-023-06792-0

Teams of LLM agents can split inorganic design into roles

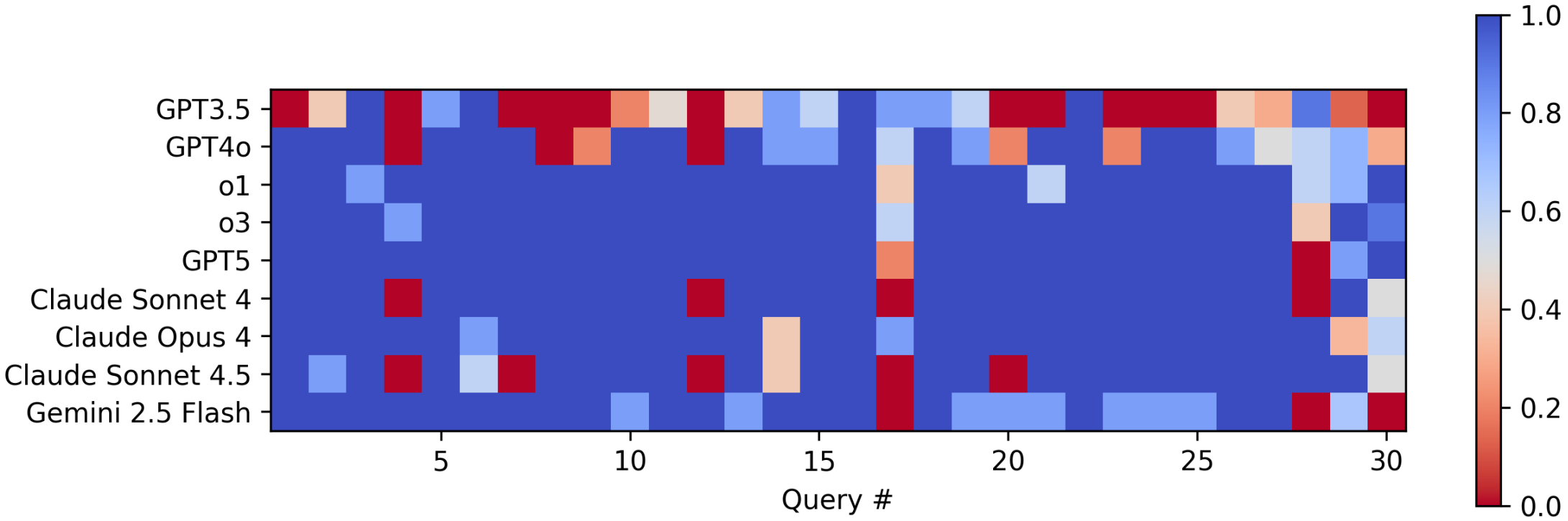
Preprint; tested on thermoelectric, semiconductor and perovskite oxide design tasks



Ghafarollahi & Buehler, arXiv:2508.02956 (2025) https://arxiv.org/abs/2508.02956

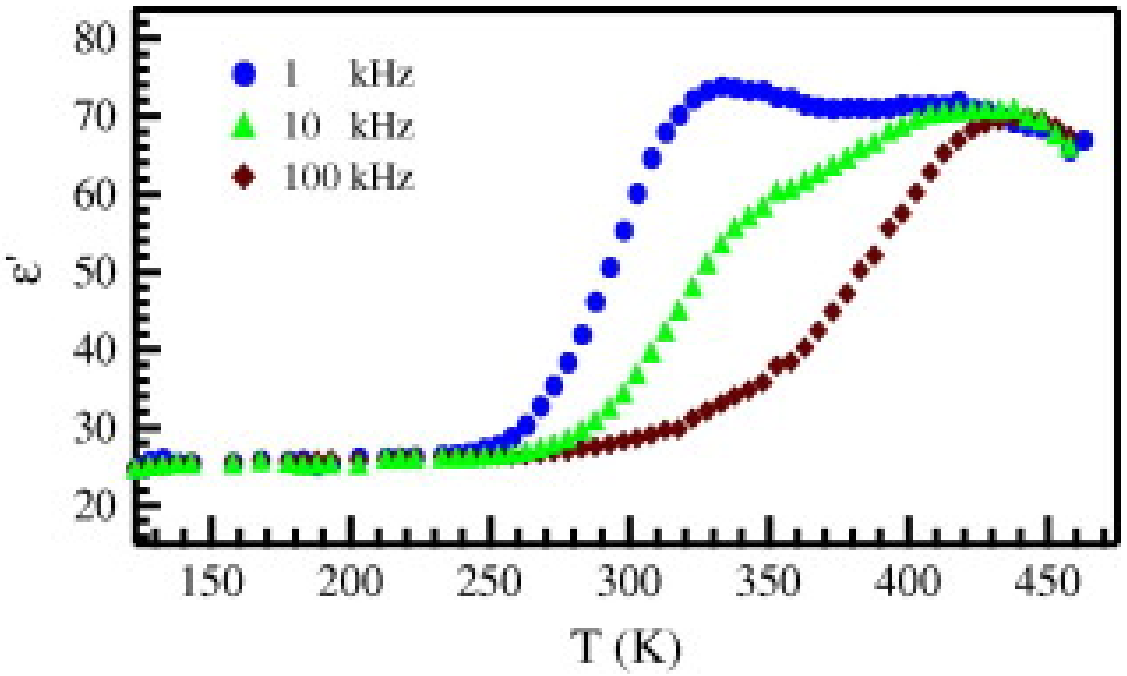
The same agent succeeds or fails depending on the underlying LLM

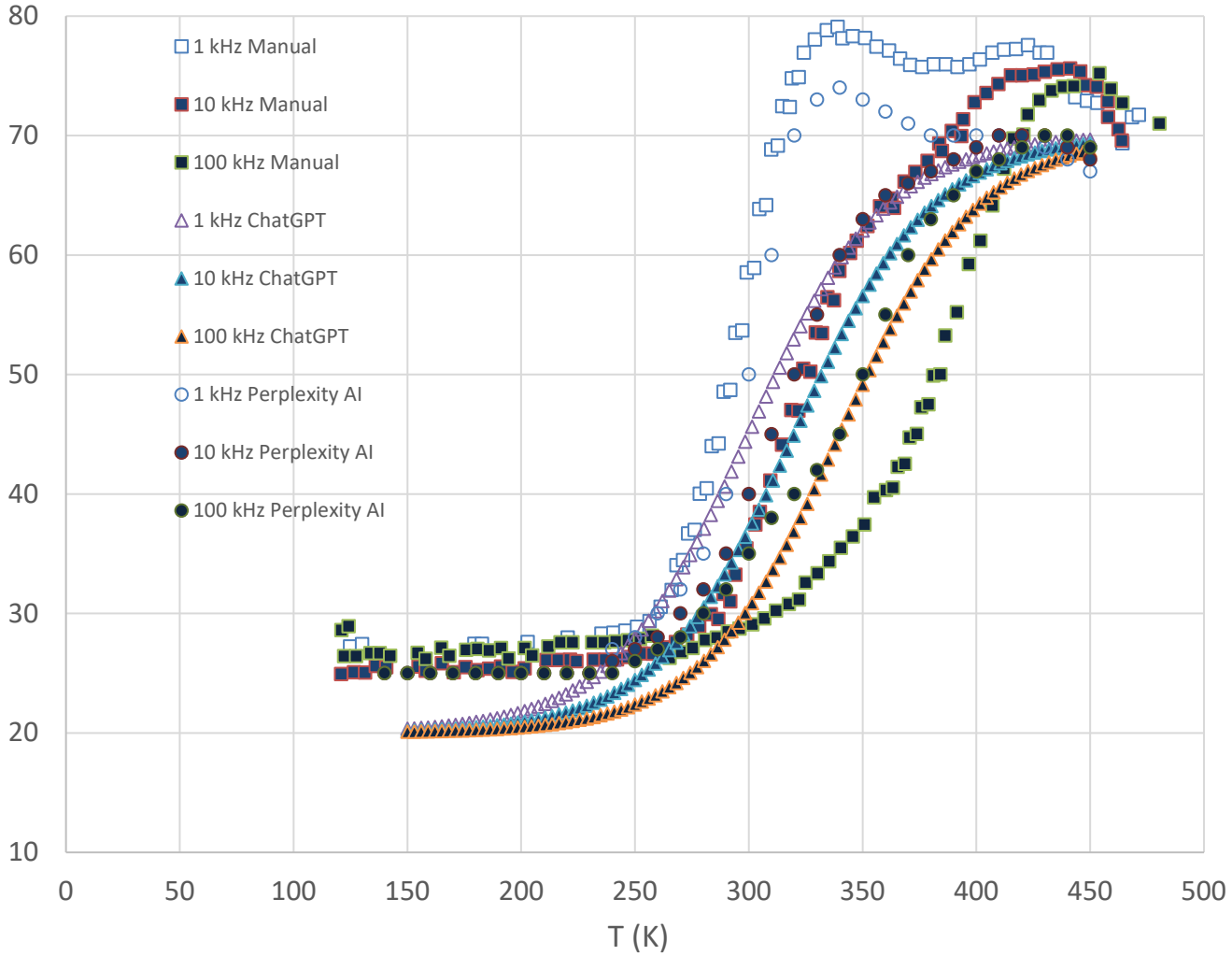
Models handled direct instructions well but struggled on process discovery challenges.



Yanguas-Gil, arXiv:2608.29309 (2026) https://arxiv.org/abs/2608.29309

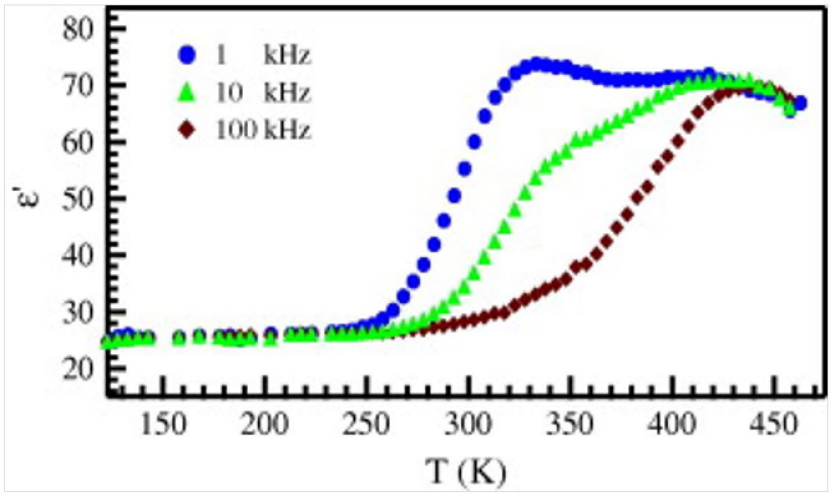
Computer vision is getting much better at data extraction



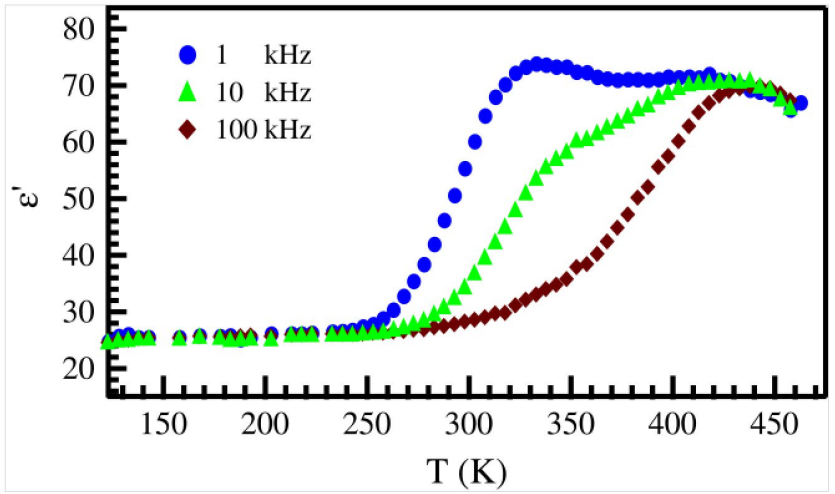


18 months later, Fable 5.1 Max Effort has solved* this problem

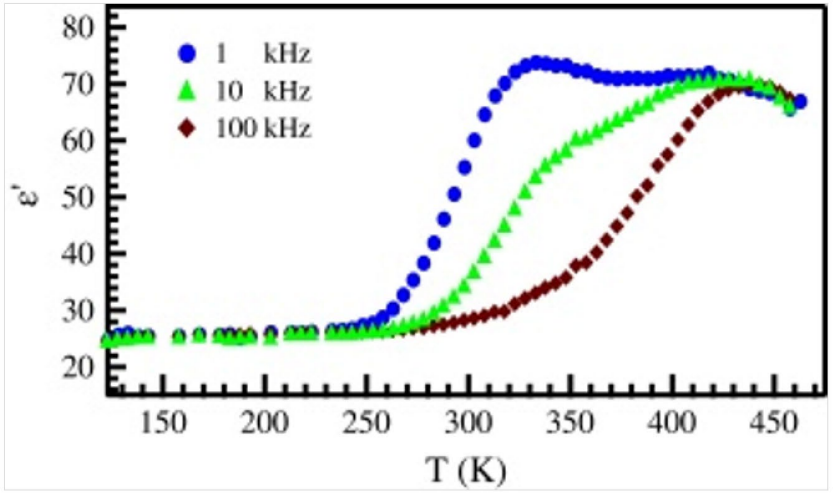
Original the image as supplied, 660 × 390 px



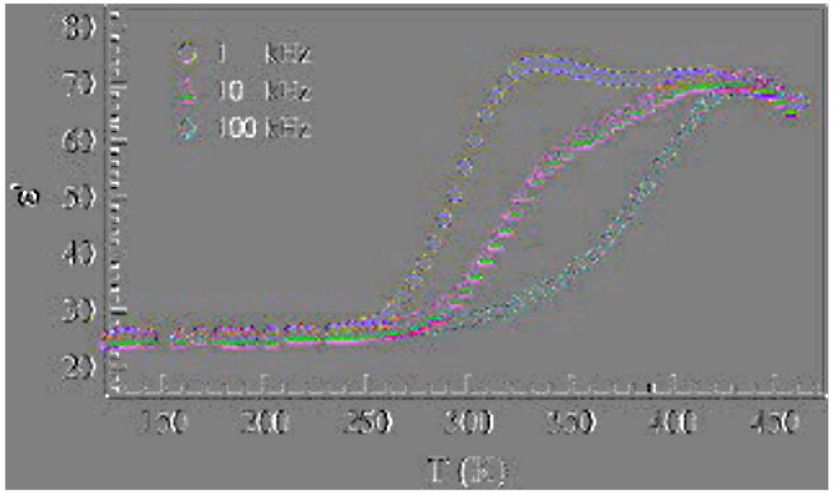
Reconstruction matplotlib replot of the 184 extracted points



Reconstruction at the original's resolution reduced to 319 × 189 px, JPEG, enlarged back



Difference original minus the panel on the left, contrast x3; flat grey = identical



3.5h of wall-clock time

1.3M session tokens:
770k in generation
500k in reviews

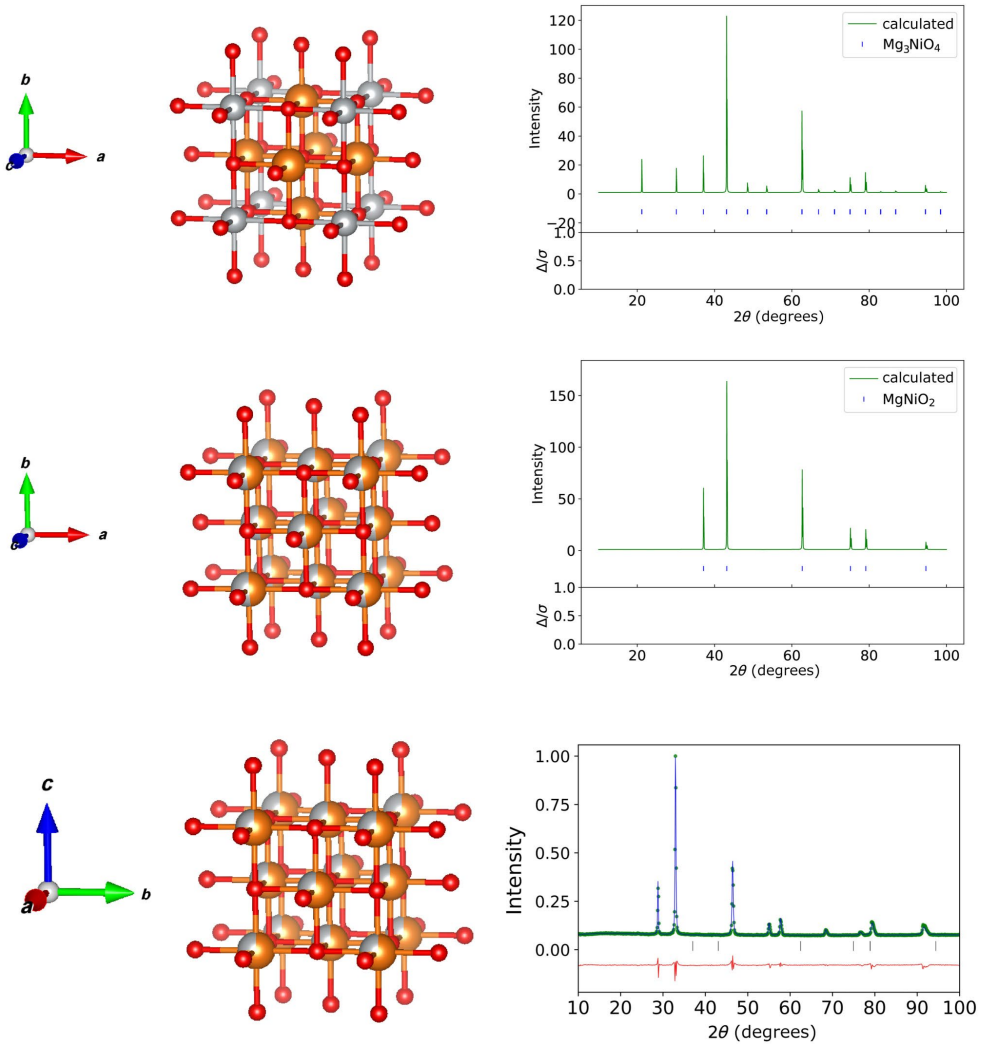
Plus context token: ~10-15M → $50!!

Healthy skepticism

What can go wrong, and how to not fool yourself

A predicted 'new' ordered oxide may just be a known solid solution

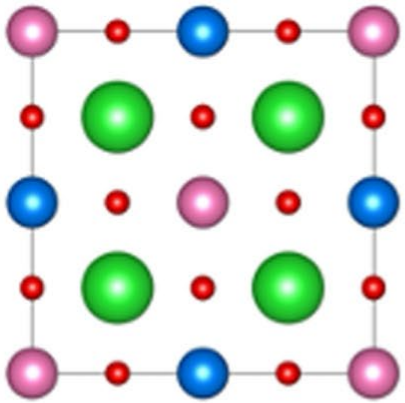
The authors argue about two thirds of the claimed A-Lab successes are likely known compositionally disordered phases.

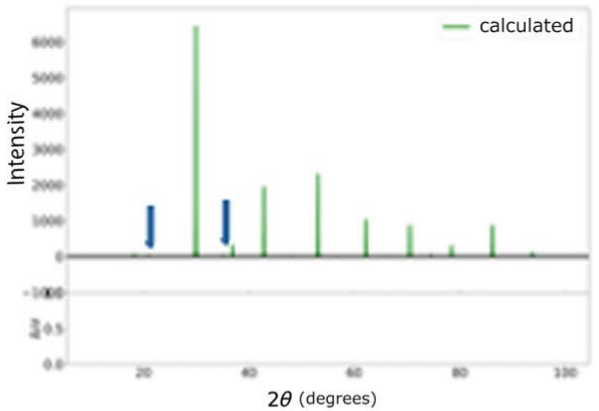


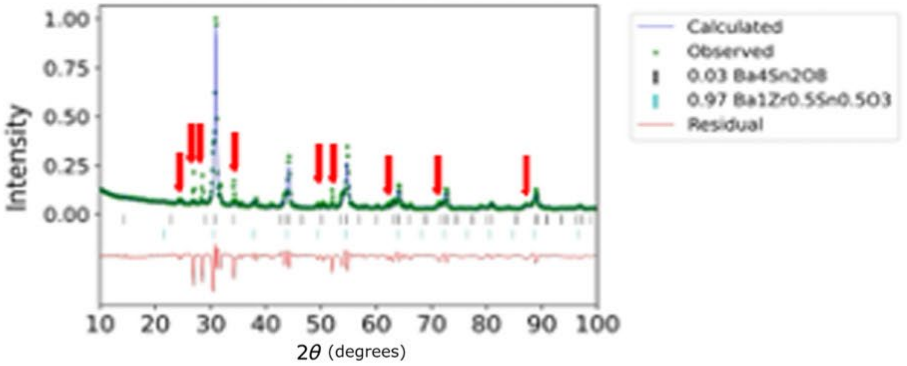
Leeman, ..., Schoop & Palgrave, PRX Energy 3, 011002 (2024) https://doi.org/10.1103/PRXEnergy.3.011002

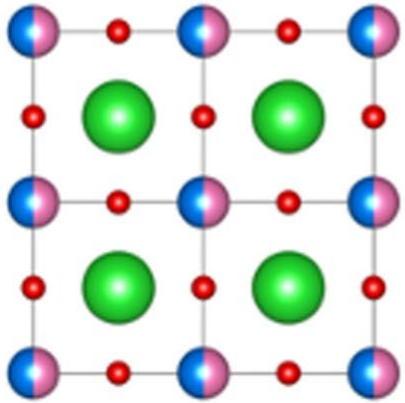
Automated Rietveld fits can overclaim; experts still need to check

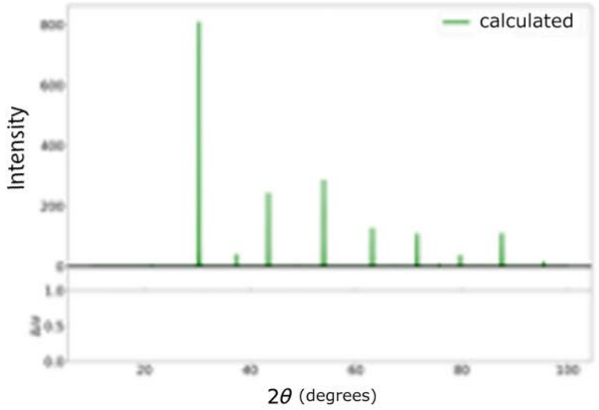
Ordered and disordered versions differ by only two weak XRD peaks.

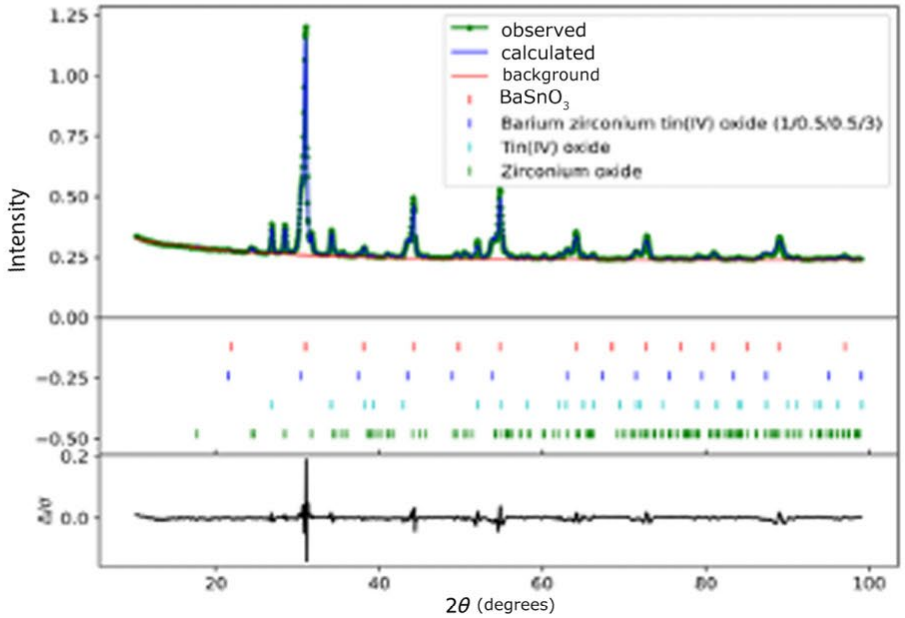






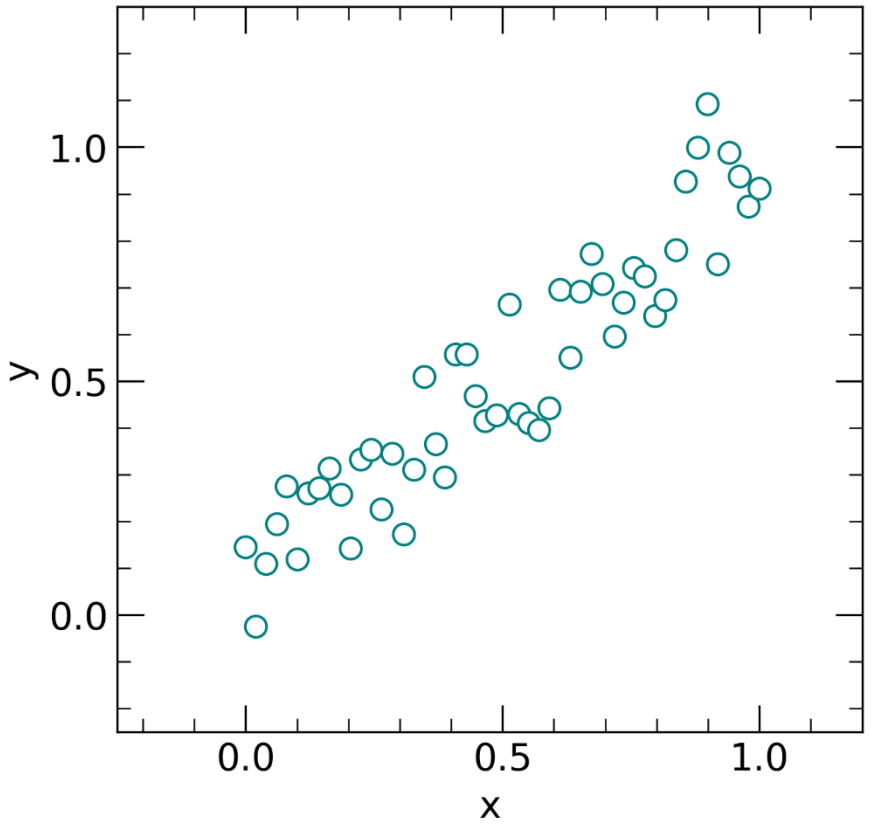


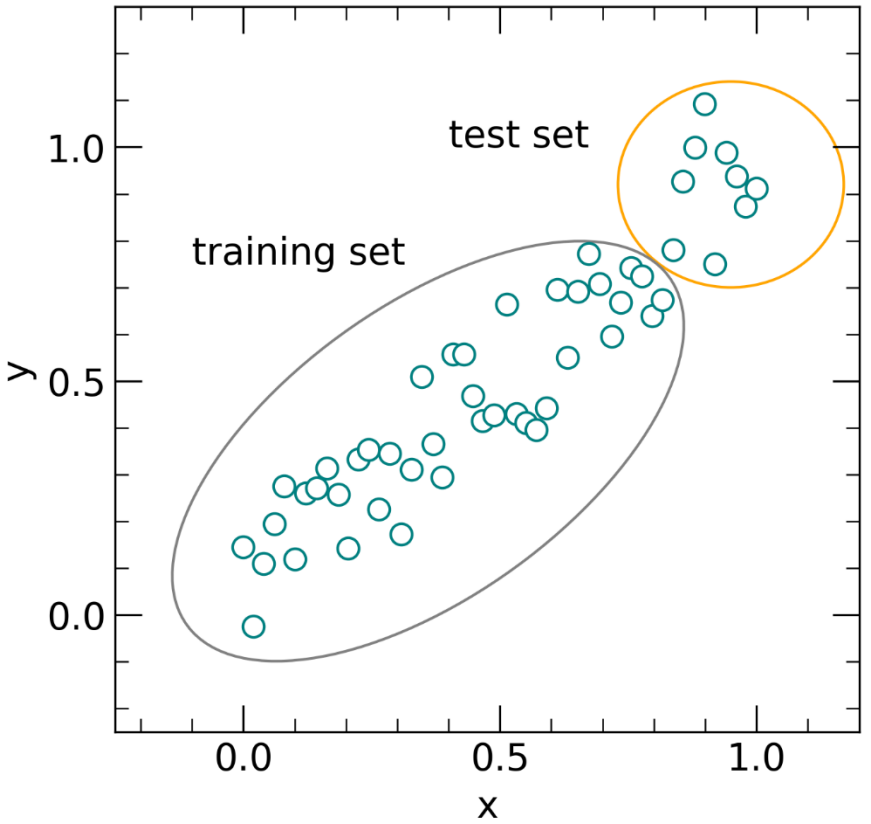




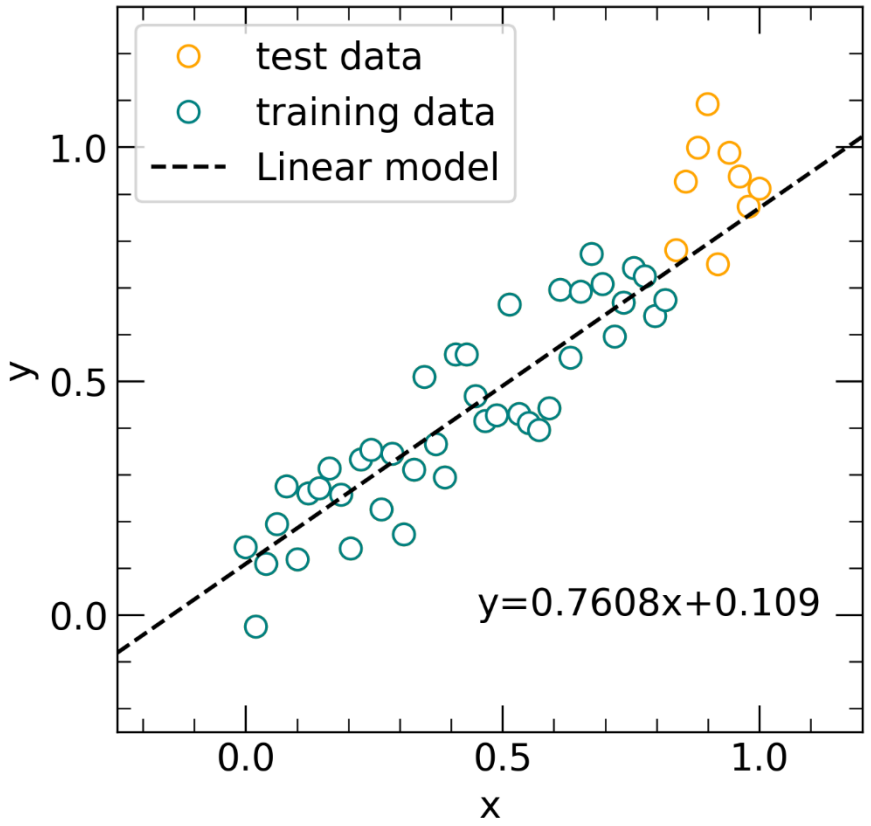
Leeman, ..., Schoop & Palgrave, PRX Energy 3, 011002 (2024) https://doi.org/10.1103/PRXEnergy.3.011002

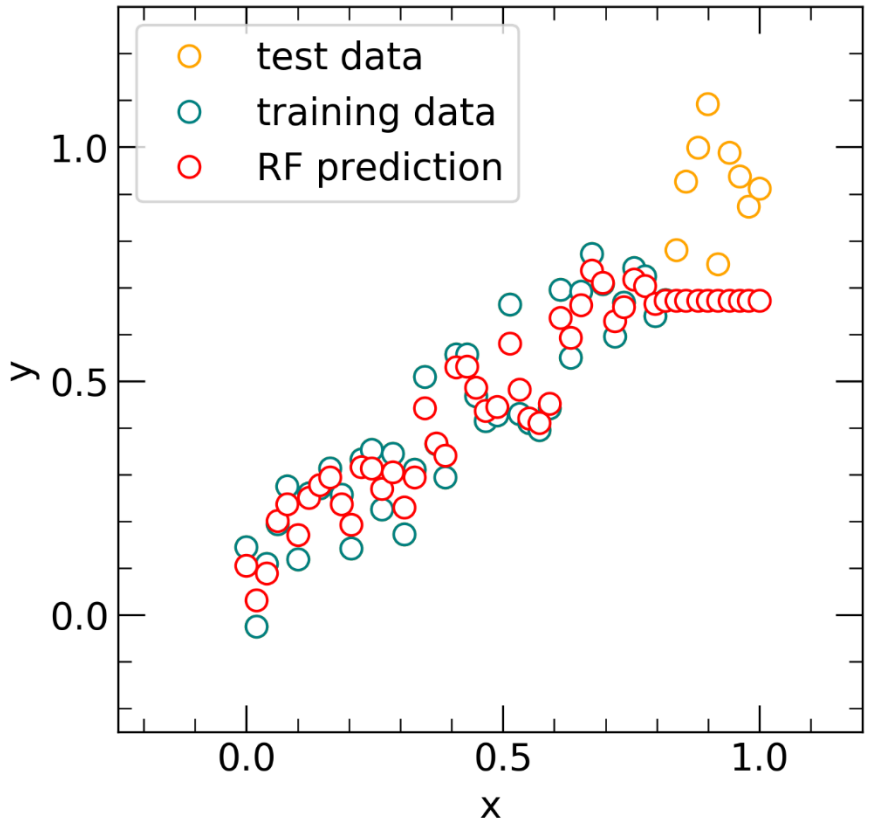
Most machine learning fails at extrapolation



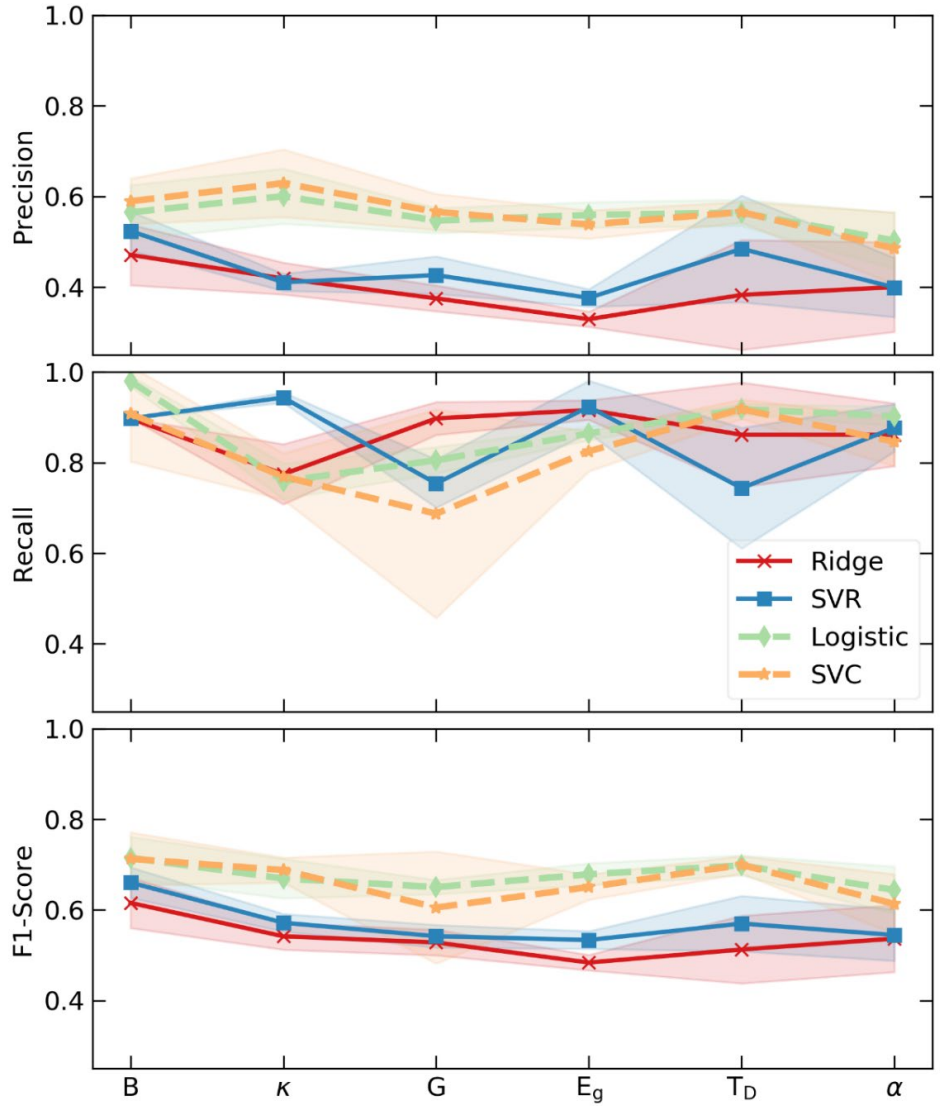


…but it depends on the algorithm!



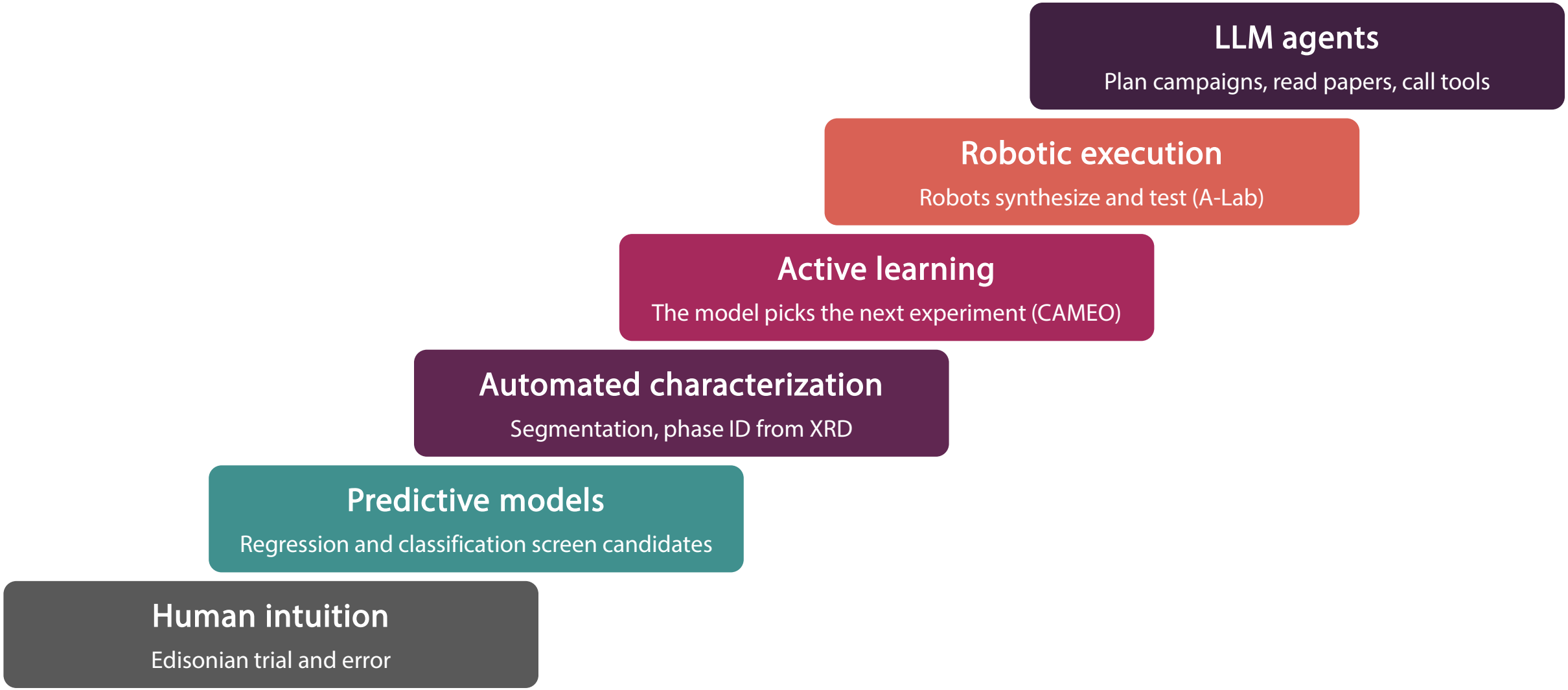


In general, classification and non-linear outperforms



Kauwe, Graser, Murdock, Sparks *Computational Materials Science* (2020)

Each step up the autonomy ladder rests on the steps below it



Four things to remember from today

1. ML is pattern finding

The same structure-processing-property thinking, in many dimensions at once

2. Pick the right task

Regression, classification, clustering, segmentation, generation or automation

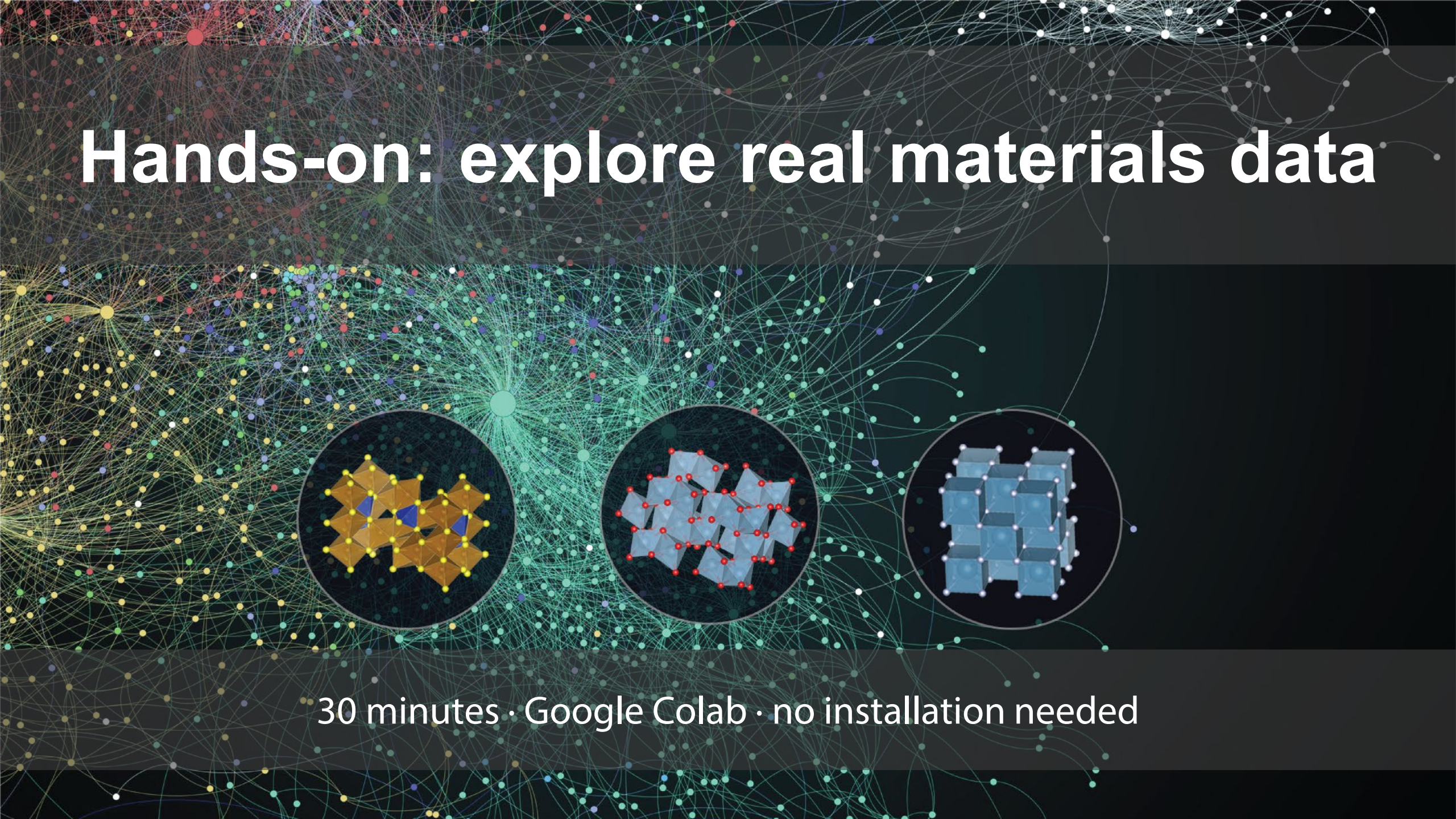
3. Data beats algorithms

Small, messy, biased datasets limit every model; framing matters most

4. Trust, but verify

Extrapolation, leakage and overclaiming are the common failure modes

Hands-on: explore real materials data



30 minutes · Google Colab · no installation needed

Open the Session 1 notebook in Google Colab

1. **Go to the course repo**
   github.com/sp8rks/ACerS_AI_ML_Workshop

2. **Click 'Open in Colab'**
   Session 1: 01_materials_informatics_intro

3. **Runtime → Run all**
   The first cell downloads data and helpers

4. **Work through the questions**
   Pair up; discuss what surprises you

All course datasets load in seconds

Dataset	Rows	What it is
bulk_modulus	4,905	DFT (AFLOW) bulk modulus, GPa
shear_modulus	4,905	DFT (AFLOW) shear modulus, GPa
log10_thermal_conductivity	4,896	DFT (AFLOW) log10 κ at 300 K
bandgap	27,767	Band gap, eV (0 = metal)
heat_capacity_raw / clean	4,583 / 4,547	Cp vs T, raw version has errors on purpose
agnp_flow_synthesis	3,295	Real Ag nanoparticle flow-synthesis experiments
ceramics_text_snippets	10	Synthetic abstracts for LLM extraction

Sources: Wang et al., Chem. Mater. 2020 (Best Practices repo, MIT); Mekki-Berrada et al., npj Comput. Mater. 2021. See data/README.md

Explore with a critical eye before you model anything

Look

What does the bulk modulus distribution look like? Any suspicious values?

Count

Which elements and chemistry families dominate? What is missing?

Sample

Same compound at many temperatures: what counts as one sample?

Frame

Fill in the problem-framing worksheet for a problem from your own work

Most ML projects fail from poor problem framing, not poor algorithms

Is your problem ready for machine learning?

Red flags	Green flags
Fewer than ~50 examples and no physics-informed features	A clear decision and a cheap way to verify predictions
Inputs you won't know when you need the prediction	A loop to add new data when the model is uncertain
Target measured differently across labs or papers	Domain knowledge you can turn into features or constraints
"We want to predict everything"	A measurable baseline the model has to beat